

ONE-BUBBLE NEIGHBORHOODS IN $SO(3)$ -MONOPOLE MODULI SPACES

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ABSTRACT. We prove an oriented neighborhood theorem at the first Uhlenbeck level of $SO(3)$ -monopole moduli spaces, over a compact regular Seiberg–Witten moduli space of arbitrary dimension whose solutions have nonzero spinor. A finite-dimensional obstruction model admits an equivariant compactified embedding that maps the obstruction zero set homeomorphically onto an Uhlenbeck-open neighborhood of the ideal stratum; its restriction to that zero set at positive scale is a diffeomorphism onto its image. The principal analytic step is a strong approximation theorem in critical norms for arbitrary convergent one-bubble monopole sequences. Uniform gauge estimates and summable decay along the neck recover the gluing parameters, including the relative frame, and give reverse uniqueness. A global equivariant stabilization treats the background family without requiring a constant-rank spectral cutoff. Comparison with Donaldson’s instanton-addition map identifies the induced orientation with the standard monopole orientation. Under the additional topological hypotheses of Feehan and Leness, the construction supplies the analytic neighborhood input to their level-one link pairing. We state the resulting polynomial formula and its low-degree consequences for Witten’s conjecture on abundant, effective manifolds of Seiberg–Witten simple type.

1. INTRODUCTION

1.1. The neighborhood problem. Let X be a smooth, closed, connected, oriented four-manifold, and consider the Uhlenbeck compactification of a moduli space of $SO(3)$ monopoles. Its lower levels contain products of Seiberg–Witten moduli spaces with symmetric products of X . The $SO(3)$ -monopole cobordism program compares the link of the top anti-self-dual stratum with the links of these reducible strata. A link calculation has two distinct analytic inputs. One must first construct solutions of a finite-dimensional extension of the monopole equations from splicing data. One must then prove that the resulting zero set is an actual neighborhood of the ideal stratum in the Uhlenbeck topology.

The first input was established in the local gluing theorem of Feehan and Leness [FL99]. That theorem constructs the gluing-data bundle, a smooth solution map for the extended equations, and a finite-rank obstruction section. It gives the inclusion of the obstruction zero set in the monopole moduli space. Section 1.5.1 of the same paper separates from this construction the remaining continuity and embedding statements and the claim of surjectivity onto a Uhlenbeck-open neighborhood; it assigns this package to a companion article. In the one-bubble case, [FL02, Theorem 3.8] records the compactified embedding, the obstruction sections, and the inclusion of their zero set in the monopole moduli space, but it does not assert that this zero set maps onto an open neighborhood of the ideal stratum. That stronger

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conclusion is Property (4) of Hypothesis 7.8.1 in [FL18] and is retained there as part of the local gluing hypothesis.

This paper constructs the neighborhood for one charge-one bubble over a compact regular Seiberg–Witten moduli space of arbitrary dimension. The standard orientation is identified by comparison with instanton addition. At this level there are no collision strata and no overlaps among partitions of a symmetric product. The family problem cannot be reduced to the zero-dimensional case by choosing one spectral cutoff: eigenvalue crossings can change the rank of the corresponding small-eigenvalue spaces. We instead stabilize the normal family globally by a smooth finite-rank equivariant bundle. Local cutoff reductions enter only through a common enlargement, so neither the gluing chart nor its orientation depends on a constant-rank spectral window.

1.2. Relation to Witten’s conjecture. Witten’s conjecture relates the Donaldson and Seiberg–Witten invariants of a four-manifold by an identity of generating series [Wit94]. The neighborhood problem above is an analytic input to the $SO(3)$ -monopole approach to this identity. In the degree range considered in [FL02], the link calculations for strata with nonzero Seiberg–Witten invariant involve only levels zero and one. Theorem B supplies the level-one neighborhood input under the regularity and nonzero-spinor assumptions stated below. Together with the published top-level calculation, it gives the low-degree comparison in Corollary 8.4, retaining abundance, effectiveness, and Seiberg–Witten simple type. Effectiveness remains an assumption on link pairings at every level; higher-level strata need not be absent.

The later results of Feehan and Leness deduce Witten’s conjecture for four-manifolds of Seiberg–Witten simple type with $b_1 = 0$ and odd $b_2^+ \geq 3$, conditional on the general monopole cobordism formula [FL19a, Theorem 1.1], [FL19b, Corollary 1.3]. The gluing hypothesis in that deduction concerns neighborhoods over every relevant symmetric-product stratum [FL19b, Hypothesis 3.1 and Remark 3.4]. The present theorem establishes a one-bubble neighborhood, with its orientation and obstruction properties; it does not establish that all-level hypothesis or enlarge the published low-degree range.

1.3. Results. Let $b_1(X) = 0$ and let $b_2^+(X) \geq 3$ be odd. Fix generic perturbation data, a spin^u structure $\mathfrak{t}'$, and a spin^c structure $\mathfrak{s}$ for which the level-one reducible stratum is

$$(1) \quad M_{\mathfrak{s}} \times X \subset \overline{\mathcal{M}}_{\mathfrak{t}'}$$

Let $\mathfrak{t}$ denote the lower-level spin^u structure containing $M_{\mathfrak{s}}$. In the standard level-one notation,

$$(2) \quad \mathfrak{t} = \mathfrak{s} \oplus \mathfrak{s} \otimes L, \quad c_1(\mathfrak{t}) = c_1(\mathfrak{t}'), \quad p_1(\mathfrak{t}) = p_1(\mathfrak{t}') + 4.$$

Thus $\mathfrak{t}'$ is the ambient structure obtained after splicing one unit of instanton charge into a configuration for $\mathfrak{t}$. Assume that $M_{\mathfrak{s}}$ is a compact regular smooth manifold, every represented Seiberg–Witten pair has nonzero spinor, and

$$(3) \quad \dim M_{\mathfrak{s}} = d_{\mathfrak{s}} = 2d \geq 0.$$

The level-one construction has a compactified virtual gluing space

$$(4) \quad \overline{\mathcal{M}}_{\mathfrak{t}', \mathfrak{s}}^{\text{vir}}(\varepsilon, \delta),$$

an obstruction pseudo-bundle $\overline{\Upsilon}$, and a section χ . Their definitions are recalled in Section 2. We write $\overline{\mathcal{C}}_{\mathfrak{t}'}$ for the space of ideal configuration classes with the Uhlenbeck topology, restricted to levels zero and one. The superscripts $*, 0$ indicate an irreducible connection and a nonzero spinor on the top stratum; a bar includes their ideal limits. An Uhlenbeck-open neighborhood

is open in this ideal configuration space, or in the indicated moduli subspace when intersected with it. The instanton summand of $\overline{\mathbf{Y}}$ has complex rank one at positive scale and collapses at the zero-scale cone point.

Writing $N(\varepsilon)$ for the background virtual-normal disk, the compactified gluing space has the three strata

$$(5) \quad \mathcal{G}^+ \sqcup \underbrace{(N(\varepsilon) - M_{\mathfrak{s}}) \times X}_{\text{middle zero-scale stratum}} \sqcup \underbrace{M_{\mathfrak{s}} \times X}_{\text{lowest stratum}}.$$

At positive scale the obstruction section has background and instanton components $\chi = (\chi_s, \chi_i)$. At zero scale the instanton fiber and χ_i collapse, while χ_s restricts to the background obstruction section $b : N(\varepsilon) \rightarrow \Xi$. Thus b is transverse on the middle stratum and vanishes on the lowest stratum.

Write q_λ for the standard one-bubble pregluing and $J_{s,\lambda}$ for the matched background obstruction embedding. The background monopole map satisfies $\mathfrak{S}(q) = \iota_{s,q} b(q)$. Use $L^{\sharp,2;2}$ for $L^{\sharp,2}$ on the self-dual-curvature component and L^2 on the negative-spinor component; the critical norm is defined in Section 3.

We first state the analytic estimates and embedding used in the neighborhood theorem.

Theorem A (uniform estimates and compactified embedding). *For sufficiently small $\varepsilon, \delta > 0$, the following hold.*

- (1) *The Coulomb operator of the Euclidean charge-one instanton is surjective on its homogeneous first-order domain and has kernel equal to the eight-dimensional framed tangent space. The augmented monopole linearization at q_λ has a complement determined by bounded integrals against background and framed bubble modes. On this complement its inverse, with ordinary L^2 gauge row and mixed critical target, is bounded uniformly into the first-order configuration space together with the finite-dimensional obstruction space.*
- (2) *The inverse and its kernel projection have uniformly bounded first and second derivatives in the graph norms on the fixed four-manifold X defined in (63), with center and scale derivatives multiplied by λ and with ordinary background and frame derivatives.*
- (3) *The pair Laplacian has a uniformly bounded critical Green operator. The full determinant-one gauge action admits a Coulomb slice of uniform radius in the critical configuration norm (108). The Green operator, horizontal projection, and slice maps have two uniformly bounded normalized parameter derivatives.*
- (4) *The pregluing family admits uniform tubular coordinates locally in normalized parameter charts, with complement determined by the configuration retraction R_g of Proposition 3.4.*
- (5) *The projected splicing residual satisfies*

$$(6) \quad \|\mathfrak{S}(q_\lambda) - J_{s,\lambda} b(q)\|_{L^{\sharp,2;2}} \leq C\lambda^{1/3}.$$

There is a unique small correction in this complement, with obstruction coefficients satisfying

$$(7) \quad \|u_\lambda\|_{L_1^2} + \|\eta_{s,\lambda}\| + \|\eta_{i,\lambda}\| \leq C\lambda^{1/3}.$$

The correction is smooth at positive scale, and the corrected map extends continuously at zero scale in the Uhlenbeck topology. The residual and correction also have two normalized parameter derivatives of order $O(\lambda^{1/3})$ in the target and augmented graph norms; the configuration derivatives obey the same bound in the critical configuration norm.

- (6) *The corrected map is an equivariant compactified topological embedding, smooth on the positive-scale stratum. Multiplying its correction by $t \in [0, 1]$ gives an equivariant isotopy through such embeddings from splicing to the corrected map.*

The Euclidean model, complemented inverse, and differentiated bounds are proved in Propositions 3.2, 3.4, and 3.6. Propositions 4.1, 4.2, and 4.3 give the residual and correction estimates, including their derivatives; Propositions 5.2 and 5.4 give the critical gauge estimates and slice. Proposition 5.6 and Lemma 5.7 prove uniform parameter recovery. Lemma 5.8 recovers the relative frame from the critical gauge equation; Corollary 5.10 and (151) give the embedding and isotopy. Proposition 6.1 obtains critical approximation for arbitrary one-bubble sequences by annular gauge patching and summable cylinder estimates. The neighborhood assertion below follows by reverse uniqueness. Proposition 7.4 identifies the standard orientation by instanton addition.

Theorem B (level-one neighborhood; = Theorem 2.1). *For sufficiently small $\varepsilon, \delta > 0$, there is an S^1 -equivariant map*

$$\gamma : \overline{\mathcal{M}}_{\nu, s}^{\text{vir}}(\varepsilon, \delta) \longrightarrow \overline{\mathcal{C}}_{\nu}$$

with the following properties.

- (1) *The map is a topological embedding, is smooth on the positive-scale stratum, and is joined to the splicing map by an S^1 -equivariant isotopy through compactified embeddings. On the zero-scale face it is the canonical background-and-ideal-point map.*
- (2) *There is an Uhlenbeck-open neighborhood $\mathcal{U}$ of $M_s \times X$ for which the restriction*

$$\gamma : \chi^{-1}(0) \cap \gamma^{-1}(\mathcal{U}) \longrightarrow \mathcal{U} \cap \overline{\mathcal{M}}_{\nu}^{*, 0}$$

is a homeomorphism and is a diffeomorphism on the positive-scale stratum.

- (3) *The total obstruction section is transverse on the positive-scale stratum, and its background component is transverse on the middle zero-scale stratum. Its norm is continuous along the compactified section. No transversality is asserted on the lowest stratum.*
- (4) *For a homology orientation of X , the standard determinant-line orientation agrees with the orientation of the transverse zero locus. This orientation descends through the frame and residual-circle quotients and induces the standard boundary orientation used in the level-one link pairing.*

The last qualification in (3) is forced by the geometry. On the lowest stratum the surviving background obstruction section is identically zero. If its rank is positive, its restriction cannot be transverse. This is the statement used in the relative-Euler-class calculation of [FL02, Section 5]; it also agrees with the qualification following Theorem 3.8 there. The all-strata formulation in [FL18, Hypothesis 7.8.1(2)] therefore requires qualification on the fine stratification (5): the transversality supplied here is that of the total section at positive scale and of the background section on the middle stratum. We use these precise properties in the level-one link calculation, rather than a transversality assertion on the lowest stratum.

Under the additional topological hypotheses of the published calculations, Theorem B supplies the deferred analytic input to the level-one link pairing of [FL02, Theorem 6.1]. Section 8 records its polynomial form, including zero Jacobi values, and its zero-dimensional specialization. Corollary 8.4 gives the resulting degree comparison for abundant, effective manifolds of Seiberg–Witten simple type. These application statements retain their stated topological assumptions; no higher-level neighborhood result is used or proved here.

1.4. Proof strategy. The linear construction uses the charge-one instanton on Euclidean $\mathbb{R}^4$, with homogeneous first-order domain and ordinary L^2 target. Proposition 3.2 identifies its Coulomb kernel with the eight-dimensional framed deformation space. Extending the rescaled physical metric on the bubble before constructing its inverse makes the principal symbols agree on the localization region. The remaining errors are zero-order terms and cutoff commutators. Proposition 3.4 gives the resulting complemented inverse in mixed critical norms. The projected residual has norm $O(\lambda^{1/3})$, and contraction gives a correction of the same order. Differentiating the explicit regional residual formulas preserves this rate. The square nonlinear equation obtained by adjoining the configuration retraction then gives two derivatives of the correction with the same bound.

To differentiate the inverse, we identify nearby gluing bundles over the identity of the physical four-manifold. The metric and principal symbols then remain fixed. Differentiated connection coefficients are uniformly bounded critical multipliers, and differentiation of the retraction acts on its smooth dual profiles. The graph domains coincide in these local identifications. Adjoining the retraction makes the operator invertible, so the inverse differentiation formula gives two uniform derivatives in Proposition 3.6. The ordinary compact Coulomb operator on S^4 has only five ASD kernel directions; the Euclidean homogeneous domain accounts for the three additional framing directions. Section 3 also explains why separate compact conformal transports do not have uniform norms.

The nonnegative spinor potential in the pair Laplacian preserves the critical Green estimate. Combined with a uniform spectral gap, it gives a uniform Coulomb slice for the full determinant-one gauge group in Propositions 5.2 and 5.4. The configuration norm controls both the first-order norm and the critical norm of the divergence. To exclude identifications by gauges outside the local exponential chart, we use their first-order equation and a scalar spectral gap to prove that they too have small critical norm. For uniform parameter recovery, we compare pregluing derivatives with explicit Euclidean Coulomb modes. A bounded covariantly harmonic adjoint section supplies the frame modes and cancels the large annular term from differentiating the transition function. The background comparison uses the pair Coulomb connection and its induced connection on relative frames. Differentiating the zero Coulomb row of the kernel lift then controls the moving complement in the critical norm. These comparisons prove the tubular-map bounds and uniform modulation in Proposition 5.6 and Lemma 5.7. To recover the relative frame globally, we compare two constructed pairs in the common exterior trivialization. The gauge equation controls the scalar Laplacian of the comparison gauge in $L^\sharp$. Smooth limits at the two ends allow a matrix-valued extension, to which the scalar critical Green estimate applies. Its bound is independent of neck length and forces the inner constant gauge to agree with the exterior normalization. This proves frame recovery for the corrected family and its interpolation in Lemma 5.8.

For arbitrary one-bubble monopole sequences, local regularity improves small-energy gauges to higher regularity on each normalized annulus. The transition equation then permits patching across the complete neck with uniform local bounds. We impose Coulomb gauge on each round three-sphere and remove the mean temporal gauge term by constant rotations, retaining their inner endpoint as the relative frame. The coexact tangential operator has a spectral gap. Its two-sided estimate bounds the sum of local connection norms by the endpoint errors and a summable curvature source. Scaled weighted estimates and a separate spinor Weitzenböck estimate give critical strong splicing approximation in Proposition 6.1.

With these inputs, contraction, modulation, and reverse uniqueness give the neighborhood theorem. For the orientation, the inverse also acts from $L^{4/3}$ to a physical L^2 space in

which bubble coordinates have weight λ . This estimate identifies the kernel synthesis with orthogonal projection of cutoff modes. We prove that this comparison commutes with finite stabilization and with the homotopy that turns the spinor off. At zero spinor the Dirac blocks are complex linear and the real block is Donaldson's instanton-addition map. Explicit quaternionic modes fix its scale and frame signs. This gives the standard monopole orientation in Proposition 7.4; the horizontal tangent comparison carries it to the corrected chart. We use the residual-circle and outward-radial convention

$$(8) \quad \lambda_{\mathcal{M}} = -v_{S^1} \wedge \vec{r} \wedge \tilde{\lambda}_L,$$

from [FL02, Equation (3.81)].

2. LEVEL-ONE GLUING DATA

2.1. The monopole map and its deformation operator. Let $\mathfrak{t}'$ be represented by a Clifford module $V = V^+ \oplus V^-$ with Clifford multiplication ρ and adjoint bundle $\mathfrak{g}_V$. A configuration is a pair (A, Φ) consisting of a compatible connection and a section of V^+ . We use the perturbed monopole map

$$(9) \quad \mathfrak{S}(A, \Phi) = (F_A^+ - \tau \rho^{-1}(\Phi \otimes \Phi^*)_{00}, (D_A + \rho(\vartheta))\Phi).$$

The perturbations are fixed throughout and belong to a compact generic family. For the determinant-one gauge group, our infinitesimal-action convention is

$$d_{A,\Phi}^0 \zeta = (d_A \zeta, -\zeta \Phi).$$

Thus

$$d_{A,\Phi}^{0,*}(a, \phi) = d_A^* a - (\cdot \Phi)^* \phi,$$

and the derivative of (9) is

$$(10) \quad \begin{aligned} d_{A,\Phi}^1(a, \phi) = & (d_A^+ a - \tau \rho^{-1}(\Phi \otimes \phi^* + \phi \otimes \Phi^*)_{00}, \\ & (D_A + \rho(\vartheta))\phi + \rho(a)\Phi). \end{aligned}$$

The gauge-fixed operator is

$$(11) \quad \mathcal{D}_{A,\Phi} = d_{A,\Phi}^{0,*} \oplus d_{A,\Phi}^1.$$

At an approximate solution $d^1 d^0$ need not vanish. All parametrices below include the gauge row and do not use a deformation complex at the reglued pair.

Equation (2.3) of [FL99] prints a factor $1/2$ in the spinor-bilinear term of d^1 , whereas the literal derivative of the displayed monopole map, Equation (9.3) there, and [FL98, Equation (2.36)] have the normalization in (10). We use (10). Interpolating the discrepant zeroth-order coefficient through $[1/2, 1]$ does not change a determinant-line orientation; the characteristic-class normalization is checked independently in Section 8.

2.2. The background finite-dimensional reduction. The lower-level structure in (2) has the reducible splitting

$$\mathfrak{t} = \mathfrak{s} \oplus \mathfrak{s} \otimes L.$$

Along the reducible embedding of $M_{\mathfrak{s}}$, the gauge-fixed family splits into tangential and normal summands

$$(12) \quad \mathcal{D} = \mathcal{D}^t \oplus \mathcal{D}^n.$$

Regularity of the Seiberg–Witten equations gives

$$(13) \quad \ker \mathcal{D}^t = TM_{\mathfrak{s}}, \quad \text{coker } \mathcal{D}^t = 0.$$

The normal family is complex linear, and the residual circle acts on its domain and target through one complex character.

We use the extrinsic finite-dimensional reduction of [FL01a, Theorem 3.19]. Compactness of M_s gives a finite-rank globally trivial complex bundle $\Xi = M_s \times \mathbb{C}^{r_\Xi}$, equipped with the normal circle character, and an equivariant target embedding ι_Ξ such that

$$(14) \quad \widehat{\mathcal{D}}_p^n = \mathcal{D}_p^n \oplus (-\iota_{\Xi,p}) : \text{Dom } \mathcal{D}_p^n \oplus \Xi_p \longrightarrow \text{Tar } \mathcal{D}_p^n$$

is surjective for every $p \in M_s$. The cited theorem constructs a trivial smooth equivariant target subbundle; this triviality is part of the choice, not a consequence of cokernel spanning alone. Fix a Hermitian trivialization and pull it back over the tubular neighborhood. Surjectivity persists after shrinking that neighborhood. No spectral projection is used.

The augmented kernels form a smooth complex bundle

$$(15) \quad N_{t,s} = \ker \widehat{\mathcal{D}}^n \longrightarrow M_s.$$

Thus the full augmented kernel is $TM_s \oplus N_{t,s}$ and

$$(16) \quad \text{Ind}_{\mathbb{C}}(\mathcal{D}^n) = [N_{t,s}] - [\Xi] \quad \text{in } K^0(M_s).$$

The projected convention of [FL99] is equivalent: if $\Pi_{\Xi^\perp}$ is orthogonal projection, then

$$(17) \quad u \longmapsto (u, \iota_\Xi^{-1} \mathcal{D}^n u)$$

identifies $\ker(\Pi_{\Xi^\perp} \mathcal{D}^n)$ with $\ker \widehat{\mathcal{D}}^n$.

After choosing a sufficiently small lifted disk in $N_{t,s}$, the implicit-function theorem gives a smooth embedding into the thickened configuration quotient and a section

$$(18) \quad b : N_{t,s}(\varepsilon) \longrightarrow \Xi, \quad \mathfrak{S}(q) = \iota_{s,q} b(q).$$

Its zero set is the full nearby lower-level monopole moduli space in this thickening; it contains M_s and can also contain irreducible monopoles. Regularity of the Seiberg–Witten stratum does not exclude those additional zeros. The restriction of b off the zero section is transverse for the generic perturbations under consideration [FL01a, Theorem 3.21].

There is also a distinction between the derivative of this reduction and the augmented operator with its embedding fixed at q . In a smooth local slice and the fixed trivialization of Ξ , differentiation gives

$$(19) \quad D\mathfrak{S}_q(v) - \iota_{s,q} D b_q(v) = (D\iota_s)_q[v] b(q), \quad v \in T_q N(\varepsilon).$$

The right side vanishes on the zero section and is $O(\varepsilon)$ on the fixed disk; it is not an $o(1)$ term as the bubble scale tends to zero. The literal augmented kernels therefore agree with the tangent bundle on the zero section only. Off it, projection of the configuration component of $T_q N(\varepsilon)$ onto the augmented kernel defines a smooth bundle isomorphism A_q after shrinking ε : at the zero section it is the identity, and invertibility is open. All identifications below use this isomorphism, with its uniformly bounded inverse, rather than dropping the right side of (19).

The term *virtual-normal disk* will always mean this disk bundle; it never means an eigenspace selected by a spectral cutoff. Since every pair on the zero section has nonzero spinor, compactness and openness allow the disk to be chosen so that every pair in it has trivial determinant-one infinitesimal stabilizer. Put

$$r_N = \text{rank}_{\mathbb{C}} N_{t,s}, \quad r_\Xi = \text{rank}_{\mathbb{C}} \Xi.$$

The background parameter bundle therefore has real dimension $d_s + 2r_N$, while its obstruction variables have real dimension $2r_\Xi$.

2.3. The charge-one factor and compactification. Let $M_1^{s,\natural}(S^4, \delta)$ be the centered framed charge-one ASD moduli space with scale in $(0, \delta]$. There are natural identifications

$$(20) \quad M_1^{s,\natural}(S^4, \delta) \cong (0, \delta] \times \mathrm{SO}(3), \quad \overline{M}_1^{s,\natural}(S^4, \delta) \cong c(\mathrm{SO}(3)).$$

If I is the standard centered instanton, then

$$(21) \quad \ker D_I = 0, \quad K_I := \mathrm{coker} D_I = \ker D_I^*, \quad \dim_{\mathbb{C}} K_I = 1.$$

The unframed charge-one ASD deformation space on the round S^4 has real dimension five, and the compact Coulomb operator is surjective [AHS78, Sections 6 and 9]. The tangent space of the framed moduli space fits into the exact sequence

$$(22) \quad 0 \longrightarrow \mathfrak{so}(3) \longrightarrow T_{[I]} M_1^{\mathrm{fr}} \longrightarrow H_I^1(S^4) \longrightarrow 0.$$

Thus the framed space has four translation, one dilation, and three relative-frame directions. The last three are additional framing variables, not extra elements of the compact Coulomb kernel. We write $\mathcal{K}_I^{\mathrm{fr}} = T_{[I]} M_1^{\mathrm{fr}}$ for this eight-dimensional tangent space, allowing all centers and scales. We also denote the full framed space with scale in $(0, \delta]$ by $M_1^s(S^4, \delta)$; the superscript $\natural$ imposes the centering condition.

We recall the quotient notation of [FL02, Equations (3.48)–(3.54)]. The plain letter s denotes the marked framing pole on S^4 , using the south-pole convention of that source. In $\mathrm{Fr}(\mathfrak{g}_V|_s)$ the bundle is the adjoint bundle of the charge-one spherical model; the other adjoint frame factor is over X . A tilde denotes the space before the corresponding gauge quotient: $\widetilde{M}_1^{\natural}$ consists of centered instanton connections, and $\widetilde{N}$ is the pullback of the normal disk to Seiberg–Witten representatives. The group $\mathcal{G}_s$ is the Seiberg–Witten circle gauge group, acting on this lift through its inclusion in the determinant-one monopole gauge group.

The pre-splicing factor is

$$(23) \quad \begin{aligned} \widetilde{\mathrm{Gl}}_t(\delta) = & \left(\mathrm{Fr}(\mathfrak{g}_V) \times_X \mathrm{Fr}(T^*X) \times \mathrm{Fr}(\mathfrak{g}_V|_s) \right. \\ & \left. \times \widetilde{M}_1^{\natural}(S^4, \delta) \right) / (\mathrm{SO}(3) \times \mathrm{SO}(4)). \end{aligned}$$

After dividing by the charge-one gauge group, one obtains $\mathrm{Gl}_t(\delta)$. Replacing the instanton factor by the cone in (20) gives $\overline{\mathrm{Gl}}_t(\delta)$ and

$$(24) \quad \overline{\mathcal{M}}_{t,s}^{\mathrm{vir}} = \widetilde{N}_{t,s}(\varepsilon) \times_{\mathcal{G}_s} \overline{\mathrm{Gl}}_t(\delta).$$

At the cone point the relative instanton frame is forgotten. Hence (24) has three strata:

$$(25) \quad \begin{aligned} \overline{\mathcal{M}}_{t,s}^{\mathrm{vir}} = & \mathcal{G}^+ \\ & \sqcup \left((N_{t,s}(\varepsilon) - M_s) \times X \right) \sqcup (M_s \times X), \end{aligned}$$

where $\mathcal{G}^+$ denotes the positive-scale stratum.

Before quotienting, the splicing map is equivariant for the diagonal $\mathcal{G}_s$ and charge-one gauge actions and for the diagonal $\mathrm{SO}(4)$ tangent-frame and $\mathrm{SO}(3)$ adjoint-frame actions. These auxiliary frame actions are divided out; the relative $\mathrm{SO}(3)$ frame in (20) remains.

The residual circle has the two equivalent presentations of [FL02, Lemma 3.6]. It may act by the monopole circle on the normal factor and trivially on the compactified gluing factor, or diagonally with weight two on both. We use the effective circle in the link quotient. Passing from weight two to its effective parametrization does not reverse the orbit orientation.

2.4. The obstruction pseudo-bundle. At positive scale the obstruction space is

$$(26) \quad \Upsilon_g = \Upsilon_g^s \oplus \Upsilon_g^i \cong \Xi_q \oplus K_I.$$

The background summand extends as an ordinary bundle over (25). The instanton summand is the associated line

$$(0, \delta] \times \mathrm{SU}(2) \times_{\{\pm 1\}} \mathbb{C} \longrightarrow (0, \delta] \times \mathrm{SO}(3).$$

Its compactification is the cone map

$$(27) \quad c(\mathrm{SU}(2) \times_{\{\pm 1\}} \mathbb{C}) \longrightarrow c(\mathrm{SO}(3));$$

the fiber collapses at the vertex. This is a rank-dropping pseudo-bundle, not a singular trivialization of a line bundle.

2.5. Strong and critical-norm Sobolev completions. For smooth dependence at a fixed positive scale, choose $k \geq 4$ and use the standard Fredholm map

$$\mathcal{D}_{q_\lambda} : L_k^2(\Lambda^1 \otimes \mathfrak{g}_V \oplus V^+) \longrightarrow L_{k-1}^2(\mathfrak{g}_V \oplus \Lambda^+ \otimes \mathfrak{g}_V \oplus V^-).$$

Uniform estimates as $\lambda \downarrow 0$ use

$$(28) \quad \begin{aligned} \mathbb{X}_{q_\lambda} &= L_{1,A_\lambda}^2, \\ \mathbb{Z}_{q_\lambda} &= L^2(\mathfrak{g}_V), \\ \mathbb{Y}_{q_\lambda} &= L^{\sharp,2}(\Lambda^+ \otimes \mathfrak{g}_V) \oplus L^2(V^-). \end{aligned}$$

Here $L^{\sharp,2}$ is the completion of smooth sections in the sum of the $L^\sharp$ and L^2 norms, following [Fee01, Section 4]. All critical spaces below use this completion convention. Proposition 3.4 constructs an inverse on these spaces, first on smooth target data and then on the graph completion. We do not assert that the full first-order operator maps every arbitrary L_1^2 section into the mixed critical target.

Choose matched finite-dimensional embeddings

$$J_{s,\lambda} : \Xi_q \rightarrow \mathbb{Y}_{q_\lambda}, \quad J_{i,\lambda} : K_I \rightarrow \mathbb{Y}_{q_\lambda}.$$

The augmented linearization is

$$(29) \quad \widehat{\mathcal{D}}_\lambda(u, \xi_s, \xi_i) = (d_{q_\lambda}^{0,*}u, d_{q_\lambda}^1u - J_{s,\lambda}\xi_s - J_{i,\lambda}\xi_i).$$

The nonlinear augmented map is

$$(30) \quad \begin{aligned} \widehat{\mathcal{F}}_\lambda(u, \eta_s, \eta_i) &= (d_{q_\lambda}^{0,*}u, \mathfrak{S}(q_\lambda + u) \\ &\quad - J_{s,\lambda}(b(q) + \eta_s) - J_{i,\lambda}\eta_i). \end{aligned}$$

The retained parameter directions have real dimension $d_s + 2r_N + 8$: the background tangent and normal variables, four translations, dilation, and three relative rotations. The obstruction rank is $2r_\Xi + 2$. Hence the expected dimension of the obstruction zero set is

$$(31) \quad d_s + (2r_N + 8) - (2r_\Xi + 2) = d_s + 2(r_N - r_\Xi) + 6.$$

Theorem 2.1 (level-one neighborhood). *Let $b_1(X) = 0$ and let $b_2^+(X) \geq 3$ be odd. Fix generic perturbation data and a level-one pair $(\mathfrak{t}', \mathfrak{s})$ as in (1)–(2). Suppose that M_s is compact and regular and that every represented Seiberg–Witten pair has nonzero spinor. For sufficiently small $\varepsilon, \delta > 0$, there is an equivariant compactified embedding*

$$\gamma : \overline{\mathcal{M}}_{\mathfrak{t}', \mathfrak{s}}^{\mathrm{vir}}(\varepsilon, \delta) \longrightarrow \overline{\mathcal{C}}_{\mathfrak{t}'}$$

and a continuous compactified obstruction section $\chi = (\chi_s, \chi_i)$ with the following properties.

- (1) *The positive-scale maps are smooth, and γ is equivariantly isotopic through compactified embeddings to splicing.*
- (2) *After restricting to an Uhlenbeck-open neighborhood of $M_{\mathfrak{s}} \times X$, the obstruction zero set maps homeomorphically onto the corresponding ideal monopole neighborhood and diffeomorphically on the positive-scale stratum.*
- (3) *The section χ is transverse on the positive-scale stratum, and χ_s is transverse on the middle zero-scale stratum. Its norm is continuous. No transversality is asserted on the lowest stratum.*
- (4) *The determinant-line orientation induces the standard level-one boundary orientation.*

The proof occupies Sections 3–7. The local pregluing construction and the one-sided inclusion $\gamma(\chi^{-1}(0)) \subset \overline{\mathcal{M}}_t^{*,0}$ are quoted from [FL99]. The Euclidean model, the inverse with two uniform normalized parameter derivatives, and the small correction are proved in Sections 3 and 4. The critical gauge estimates, slice, and uniform modulation, frame recovery, embedding, and isotopy in Section 5 give the compactified embedding. Section 6 proves strong splicing approximation and neighborhood surjectivity. Section 7 compares the kernel map with instanton addition and identifies the orientation of the corrected chart and its link boundary.

3. THE EUCLIDEAN MODEL AND THE AUGMENTED INVERSE

3.1. Critical norms and the square partition. For a bundle-valued section f on X , put

$$\begin{aligned}
 \|f\|_{L^\sharp} &= \sup_{z \in X} \int_X \text{dist}(z, y)^{-2} |f(y)| dV(y), \\
 \|f\|_{L^{2\sharp}} &= \sup_{z \in X} \left(\int_X \text{dist}(z, y)^{-2} |f(y)|^2 dV(y) \right)^{1/2}, \\
 \|f\|_{L^{\sharp,2}} &= \|f\|_{L^\sharp} + \|f\|_{L^2}.
 \end{aligned}
 \tag{32}$$

The critical multiplication and Sobolev estimates of [Fee01, Lemmas 4.1 and 4.3] give

$$L_1^2 \hookrightarrow L^{2\sharp}, \quad L^{2\sharp} \cdot L^{2\sharp} \longrightarrow L^\sharp, \tag{33}$$

uniformly for the compact family of background connections under consideration. In particular,

$$\|v\|_{L^4} + \|v\|_{L^{2\sharp}} \leq C \|v\|_{L_{1,A}^2}. \tag{34}$$

Fix $N > 4$. The logarithmic cutoff of [FL99, Lemma 5.4], pulled back by geodesic normal coordinates at x , may be chosen to satisfy

$$\begin{aligned}
 \beta &= 0 & \text{if } r &\leq N^{-1}\sqrt{\lambda}, & \beta &= 1 & \text{if } r &\geq \tfrac{1}{2}\sqrt{\lambda},
 \end{aligned}
 \tag{35}$$

$$\|d\beta\|_{L^4} \leq C(\log N)^{-3/4}, \quad \|d\beta\|_{L^{2\sharp}} \leq C(\log N)^{-1/2}, \quad \|d\beta\|_{L^2} \leq C(\log N)^{-1}\sqrt{\lambda}.$$

Define

$$\beta_X = \sin(\pi\beta/2), \quad \beta_I = \cos(\pi\beta/2). \tag{36}$$

Then $\beta_X^2 + \beta_I^2 = 1$, and both derivatives obey the bounds in (35). These estimates are uniform in x : near x this follows from normal-coordinate comparison, and for a base point outside a fixed larger chart the last, scale-decaying bound in (35) controls the supremum in (32).

Lemma 3.1 (cutoff commutator). *For $j = X, I$ and smooth $u = (a, \phi)$,*

$$(37) \quad [\mathcal{D}_{A_\lambda, \Phi_\lambda}, \beta_j]u = (-\iota_{\nabla \beta_j} a, (d\beta_j \wedge a)^+, \rho(d\beta_j)\phi),$$

and

$$(38) \quad \|[\mathcal{D}_{A_\lambda, \Phi_\lambda}, \beta_j]u\|_{\mathbb{Z} \oplus \mathbb{Y}} \leq C(\log N)^{-1/2} \|u\|_{\mathbb{X}_{q_\lambda}}.$$

The estimate extends by density to the weak domain.

Proof. All zero-order terms commute with a scalar cutoff. The three principal parts give (37). The gauge row, the L^2 part of the curvature row, and the Dirac row are bounded using $L^4 \cdot L^4 \subset L^2$ and the first estimate in (35). For the critical part of the curvature row, (33) gives

$$\|(d\beta_j \wedge a)^+\|_{L^\#} \leq C\|d\beta_j\|_{L^{2\#}}\|a\|_{L^{2\#}} \leq C(\log N)^{-1/2}\|a\|_{L^2_{1, A_\lambda}}.$$

Adding the rows proves (38). Notice that no derivative of $d\beta_j$ occurs: the augmented operator is first order. $\square$

3.2. Framed bubble spaces and norm transport. For the round compact model, the stabilized differential expression is

$$(39) \quad (a, \phi, \kappa) \longmapsto (d_I^* a, d_I^+ a, D_I \phi - \iota_I \kappa).$$

On the ordinary $L^2_1(S^4)$ domain its kernel has real dimension five. Adding the spinor cokernel restores surjectivity of the Dirac block but does not add the three framing variables in (22).

There is a separate issue with conformal norm transport. Write $y = z/\lambda$, $\Omega(y) = 2/(1+|y|^2)$ and $g_{S^4} = \Omega^2 \delta$. A conformal spinor transport with scalar normalization a_λ has the domain and target weights

$$(40) \quad T_\lambda^+ \phi(z) = a_\lambda \lambda^{-1} \Omega(y)^{3/2} \phi(y), \quad T_\lambda^- \psi(z) = a_\lambda \lambda^{-2} \Omega(y)^{5/2} \psi(y).$$

Direct change of variables gives

$$(41) \quad \|T_\lambda^- \psi\|_{L^2(\mathbb{R}^4)}^2 = |a_\lambda|^2 \int_{S^4} \Omega |\psi|^2 dV_{S^4}.$$

For target sections supported on a collar $|y| \asymp \lambda^{-1/2}$, where the bubble cutoff is bounded below, the inverse target transport has norm of order $|a_\lambda|^{-1} \lambda^{-1/2}$. On fixed profiles near $y = 0$, the homogeneous first-order domain norm of the transported section is of order $|a_\lambda|$. No scalar normalization makes both bounds uniform. This calculation concerns the separate transport maps; it does not rule out a bounded composite parametrix.

We instead use the Euclidean instanton with the homogeneous space

$$(42) \quad \dot{H}_I^1 = \overline{C}_c^\infty \|\nabla_I(\cdot)\|_{L^2} \hookrightarrow L^4(\mathbb{R}^4).$$

The pure dilations $u(z) = \lambda^{-1} v(z/\lambda)$ and $f(z) = \lambda^{-2} h(z/\lambda)$ preserve respectively the homogeneous first-order norm and the L^2 target norm. We first establish the Fredholm assertion for this domain, then localize the resulting inverse.

Proposition 3.2 (Euclidean framed inverse). *Let I be the centered, unit-scale charge-one ASD connection on the $SU(2)$ bundle E_I over S^4 , restricted to the complement of its framing pole with the Euclidean metric. The operator*

$$Q_I = d_I^* \oplus d_I^+ : \dot{H}_I^1(\Lambda^1 \otimes \text{ad } E_I) \longrightarrow L^2((\Lambda^0 \oplus \Lambda^+) \otimes \text{ad } E_I)$$

is surjective. Its kernel identifies with $\mathcal{K}_I^{\text{fr}}$ and has real dimension eight. The operator $D_I^+ : \dot{H}_I^1(S^+ \otimes E_I) \rightarrow L^2(S^- \otimes E_I)$ is injective, has closed range, and has complex one-dimensional cokernel K_I .

Choose a smooth compactly supported embedding $\kappa_0 : K_I \rightarrow C_c^\infty(S^- \otimes E_I)$ whose pairing with the cokernel is nonsingular. Then

$$(43) \quad \widehat{B}_0(a, \phi, c) = (d_I^* a, d_I^+ a, D_I^+ \phi - \kappa_0 c)$$

is surjective, has kernel $\mathcal{K}_I^{\text{fr}} \times \{0\} \times \{0\}$, and has a bounded inverse P_0 on the homogeneous orthogonal complement of that kernel. Kernel elements satisfy $|a(y)| = O(|y|^{-3})$ and $|\nabla_I a(y)| = O(|y|^{-4})$ in the exterior framing.

Proof. We use the Dirac operator twisted by either $F = E_I$ or $F = \text{End}_0 E_I$. In the exterior frame obtained from radial gauge at the pole, $I = O(|y|^{-3})$. Choose R so large that its L^4 norm outside B_R is small. For sections v supported there, the flat Fourier identity and the scalar Sobolev inequality give

$$\|\partial v\|_2 = \|D_0^\pm v\|_2 \leq \|D_{I,F}^\pm v\|_2 + C\|I\|_{L^4(\mathbb{R}^4 \setminus B_R)}\|v\|_4.$$

Absorbing the last term controls $\nabla_I v$ as well. An exterior cutoff and interior elliptic estimates yield

$$(44) \quad \|v\|_{\dot{H}_I^1} \leq C(\|D_{I,F}^\pm v\|_2 + \|v\|_{L^2(B_{2R})}).$$

Kato's inequality gives $\dot{H}_I^1 \hookrightarrow L^4$ and hence compact restriction to $L^2(B_{2R})$ by local Rellich compactness. Estimate (44) therefore gives finite kernel and closed range. Indeed, a unit sequence orthogonal to the kernel whose images tend to zero has a locally L^2 convergent subsequence; the estimate applied to differences makes it converge in $\dot{H}_I^1$, contradicting orthogonality. The orthogonal complement of the range consists of L^2 distributional solutions of the opposite-chirality equation.

Both the homogeneous kernel and this L^2 harmonic space can be computed on S^4 . Under the isometric spin-bundle identification β_{sp} for $g_{S^4} = \Omega^2 \delta$, conformal covariance gives

$$(45) \quad D_{I,F,g_{S^4}}(\Omega^{-3/2} \beta_{\text{sp}} \psi) = \Omega^{-5/2} \beta_{\text{sp}} D_{I,F,\delta} \psi.$$

This is the ordinary spinor covariance law [Fis13, Section 3, after Proposition 3.3]; the twisting term transforms by the same rule because Clifford multiplication by a one-form acquires the factor Ω^{-1} . For a Euclidean harmonic spinor $\psi \in L^4$, its transform $\psi_S = \Omega^{-3/2} \beta_{\text{sp}} \psi$ satisfies

$$\int_{S^4} |\psi_S|^{4/3} = \int_{\mathbb{R}^4} \Omega^2 |\psi|^{4/3} \leq \left(\int |\psi|^4 \right)^{1/3} \left(\int \Omega^3 \right)^{2/3} < \infty.$$

For $\psi \in L^2$, the same conclusion follows with upper bound $(\int |\psi|^2)^{2/3} (\int \Omega^6)^{1/3}$. To remove the pole, choose χ_ϵ vanishing within distance ϵ and equal to one beyond 2ϵ , with $\|d\chi_\epsilon\|_4 \leq C$. Against a smooth test spinor the commutator error is bounded by

$$C\|\psi_S\|_{L^{4/3}(B_{2\epsilon} \setminus B_\epsilon)} \|d\chi_\epsilon\|_4 \rightarrow 0.$$

Thus the compact equation holds distributionally across the pole, and elliptic regularity makes ψ_S smooth. Conversely, a smooth compact harmonic spinor pulls back with value $O(|y|^{-3})$ and covariant derivative $O(|y|^{-4})$. It belongs to L^2 and L^4 ; cutoffs at increasing radii put it in the homogeneous completion.

For $F_I^+ = 0$, the positive scalar curvature of the round metric and the Lichnerowicz formula give $\ker D_{I,F}^+ = 0$. The index calculation is

$$(46) \quad \begin{array}{c|cc} F & \text{ind}_{\mathbb{C}} D_{I,F}^+ & \dim_{\mathbb{C}} \ker D_{I,F}^- \\ \hline E_I & -1 & 1 \\ \text{End}_0 E_I & -4 & 4 \end{array}$$

since $\text{ch}_2(E_I) = -c_2(E_I)$, $\text{ch}_2(\text{End}_0 E_I) = -4c_2(E_I)$, and $p_1(TS^4) = 0$. This is also the twisted calculation in [AHS78, Section 10, pp. 459–460], with chirality reversed to match our ASD convention.

Choose parallel positive spinors s_1, s_2 forming a complex basis on $\mathbb{R}^4$. The constant bundle isomorphisms

$$a \longmapsto (\rho(a)s_1, \rho(a)s_2), \quad (f, v) \longmapsto ((f + \rho(v))s_1, (f + \rho(v))s_2)$$

identify the complexification of Q_I with two copies of $D_{I, \text{End}_0 E_I}^-$. Indeed, the Clifford relation gives

$$(47) \quad D_{I, \text{End}_0 E_I}^-(\rho(a)s_j) = (d_I^*a + \rho(d_I^+a))s_j.$$

Anti-self-dual two-forms act trivially on S^+ . The isomorphisms preserve the relevant norms up to fixed constants. Consequently Q_I is surjective and has real kernel dimension $2 \cdot 4 = 8$; its kernel has the asserted decay.

It remains to identify this kernel with the framed tangent space. Represent a framed infinitesimal ASD deformation b by a family in radial gauge at the pole. Then $b_{\text{rad}} = 0$ there, and in the exterior frame $b = O(|y|^{-3})$. In particular $b \in L^2 \cap \dot{H}_I^1$. Lax–Milgram on the scalar homogeneous space gives a unique $u \in \dot{H}_I^1(\text{ad } E_I) \subset L^4$ such that

$$(48) \quad \int \langle d_I u, d_I v \rangle = \int \langle b, d_I v \rangle \quad (v \in \dot{H}_I^1(\text{ad } E_I)).$$

We have $d_I u \in L^2$ and $Q_I(d_I u) = (d_I^* b, 0)$. Local regularity and (44), transferred to Q_I and applied to expanding cutoffs of $d_I u$, give $d_I u \in \dot{H}_I^1$: the additional cutoff term is at most $CR^{-1}\|d_I u\|_2$ and the interior remainder is uniformly bounded. Hence $b - d_I u \in \ker Q_I$. Two radial-gauge representatives differ by a based infinitesimal gauge vanishing near the pole, which (48) removes. If $b = d_I u$, the radial components of I and b vanish on exterior rays, so u is constant along each such ray. Its L^4 condition forces these constants to vanish. It therefore extends smoothly by zero at the pole and is a based infinitesimal gauge. The framed tangent map is injective; the dimension $5 + 3 = 8$ from (22) makes it an isomorphism.

Finally, pairing the Dirac equation with its cokernel first determines c , after which closed range gives a unique ϕ . Together with the orthogonal complement for Q_I , this proves the assertion about P_0 . A radial cutoff of the harmonic cokernel line provides κ_0 equivariantly; invariant inner products define the complements. $\square$

3.3. Matching the metric on the bubble. For use in the elementary coefficient estimates, normal coordinates satisfy

$$(49) \quad g_{ij}(z) = \delta_{ij} + O(|z|^2), \quad \partial g_{ij}(z) = O(|z|), \quad \partial^2 g_{ij}(z) = O(1).$$

In the exterior framing of the Euclidean instanton, $|I(y)| \leq C|y|^{-3}$ and $|\nabla I(y)| \leq C|y|^{-4}$. For its truncation outside radius $R \asymp \lambda^{-1/2}$, radial integration gives

$$(50) \quad \|I^{\text{tr}} - I\|_{L^4} + \|\nabla(I^{\text{tr}} - I)\|_{L^2} \leq C\lambda, \quad \|I^{\text{tr}} - I\|_{L^2} + \|I^{\text{tr}} - I\|_{L^{2\sharp}} \leq C\sqrt{\lambda}.$$

Let ϑ_{met} be smooth, equal to one on $[0, 1/2]$, and zero on $[1, \infty)$. Extend the rescaled physical metric to $\mathbb{R}^4$ by

$$(51) \quad h_{x,\lambda}(y) = \delta + \vartheta_{\text{met}}(\sqrt{\lambda}|y|)(g_x(\lambda y) - \delta).$$

It agrees with the rescaled physical metric wherever β_I is supported. The normal-coordinate bounds give

$$(52) \quad \|h_{x,\lambda} - \delta\|_\infty + \|\partial h_{x,\lambda}\|_4 \leq C\lambda.$$

Indeed, the first term is $O(\lambda^2|y|^2)$ on a ball of radius $\lambda^{-1/2}$. The derivative is $O(\lambda^{3/2})$ on that ball, whose volume to the power $1/4$ is $O(\lambda^{-1/2})$.

Identify forms and spinors for h and δ by the positive square root of the metric, preserving the three target summands. Use the corresponding identification for κ_0 . On the fixed Euclidean spaces, the resulting operator $\widehat{B}_h$ satisfies

$$(53) \quad \|\widehat{B}_h - \widehat{B}_0\|_{\dot{H}_I^1 \oplus K_I \rightarrow L^2} \leq C(\|h - \delta\|_\infty + \|\partial h\|_4) \leq C\lambda.$$

The principal coefficient multiplies $\nabla_I u$, and the remaining coefficient is bounded by $|\partial h|$ and multiplies $u \in L^4$. Consequently

$$(54) \quad P_h = P_0(1 + (\widehat{B}_h - \widehat{B}_0)P_0)^{-1}$$

is a uniformly bounded right inverse with range in the fixed complement used for P_0 . This perturbation argument does not require I to be ASD for h .

3.4. Localized coefficients and matched obstruction spaces. The remaining comparison terms are zero-order. We repeatedly use the following two consequences of (32). If $f = (v, \psi)$ is a target section supported in $B(x, r)$ and is pointwise bounded by M , then

$$(55) \quad \|v\|_{L^\sharp, 2} + \|\psi\|_{L^2} \leq CMr^2.$$

Indeed, the L^2 bounds follow from volume, while the integral of $\text{dist}(z, \cdot)^{-2}$ over a four-ball is $O(r^2)$, uniformly in z .

If instead f is a bounded coefficient supported in the same ball, then

$$(56) \quad \|f\|_{L^4} + \|f\|_{L^{2^\sharp}} \leq Cr\|f\|_{L^\infty}, \quad \|fu\|_{L^2} + \|fu\|_{L^\sharp} \leq Cr\|f\|_{L^\infty}\|u\|_{L^2}.$$

The first inequality follows by volume comparison and by splitting the supremum in (32) according as its base point lies within $2r$ of x ; the second follows from (33) and $L^4 \cdot L^4 \subset L^2$.

The pregluing formulas have the exact support properties

$$(57) \quad \begin{aligned} \Phi_\lambda &= \Phi_0 & \text{outside } B(x, 8\lambda^{1/3}), & \quad \Phi_\lambda = 0 & \text{inside } B(x, 4\lambda^{1/3}), \\ A_\lambda &= A_0 & \text{outside } B(x, 4\sqrt{\lambda}). \end{aligned}$$

Set $s_\lambda = \Phi_\lambda - \Phi_0$. The compact background family and (56) give

$$(58) \quad \|s_\lambda\|_{L^4} + \|s_\lambda\|_{L^{2^\sharp}} \leq C\lambda^{1/3}.$$

This controls the spinor coefficients in the gauge, curvature, and Dirac rows, using L^4 for the ordinary rows and $L^{2^\sharp}$ for the critical curvature row.

In radial gauge, $|A_0 - \Gamma| \leq C|z|$. On $\text{supp } \beta_X$, where $|z| \geq N^{-1}\sqrt{\lambda}$, the south-pole instanton decay gives

$$|I_\lambda(z)| \leq C\lambda^2|z|^{-3} \leq CN^3\sqrt{\lambda}.$$

Thus $c_{x,\lambda} = \beta_X(A_\lambda - A_0)$ satisfies

$$(59) \quad \|c_{x,\lambda}\|_{L^\infty} \leq CN\sqrt{\lambda}, \quad \|c_{x,\lambda}\|_{L^4} + \|c_{x,\lambda}\|_{L^{2^\sharp}} \leq CN\lambda.$$

It contributes $O_N(\sqrt{\lambda})$ to the three rows, including the Coulomb row. Finally, the fixed perturbing one-form in the bubble Dirac row is supported by β_I in a ball of radius $\frac{1}{2}\sqrt{\lambda}$; ordinary Hölder gives an $O(\sqrt{\lambda})$ multiplier bound. Consequently the total localized coefficient error is

$$(60) \quad \|E_{\text{spin}} + E_{\text{conn}}^X + E_{\text{gauge}}^X\| \leq C_N \lambda^{1/3}.$$

Choose the Hermitian metric on Ξ so that $\iota_{s,q}$ is an isometry in the ordinary target L^2 norm. Choose a smooth compactly supported negative-spinor profile κ_0 on the Euclidean charge-one model with L^2 norm one and nonzero pairing with its Dirac cokernel. The line of profiles varies equivariantly with the frames and represents the same complex obstruction line K_I . In oriented normal frames define

$$(61) \quad J_{s,\lambda}\xi = \beta_X T_{X,\lambda}^Y \iota_{s,q}\xi, \quad J_{i,\lambda}c(z) = c\lambda^{-2}\kappa_0(z/\lambda),$$

where $T_{X,\lambda}^Y$ is the background bundle identification. For sufficiently small scale the second support is inside the bubble core. The metric identifications used in (53) are included in the second formula. All formulas involving K_I are interpreted equivariantly on local framed covers, so no global trivialization of the framing line is required.

Lemma 3.3 (obstruction embeddings). *The map $J_{s,\lambda} \oplus J_{i,\lambda}$ is injective with a uniformly bounded inverse on its image. Its ordinary L^2 projection is uniformly bounded in the mixed target norm.*

Proof. The removed background profile is smooth on a shrinking ball, and its mixed norm tends to zero by (55). Pure dilation preserves the Euclidean L^2 norm of κ_0 , while the physical metric changes that norm by $o(1)$. The two supports are disjoint for small scale. The Gram matrix therefore tends to the identity. The background profiles have bounded mixed norm; the bubble profile is entirely in the L^2 Dirac row. The projection formula and Cauchy–Schwarz prove the second assertion. $\square$

3.5. The complemented inverse. Let $\mathcal{K}_X(q)$ be the kernel of the literal augmented background operator with $\iota_{s,q}$ fixed. It has dimension $d_s + 2r_N$ on a sufficiently small disk, by openness of surjectivity. The isomorphism A_q after (19) identifies the background parameter space with $\mathcal{K}_X(q)$.

Normalize the parameter metric by

$$(62) \quad g_{\text{par},\lambda} = g_{\overline{N}} + \lambda^{-2}g_X + (d\lambda/\lambda)^2 + g_{\text{SO}(3)}.$$

The normalized bubble directions are $\lambda\partial_{x^j}$, $\lambda\partial_\lambda$, and ∂_{θ^k} . For smooth corrections and auxiliary coefficients use the graph norm

$$(63) \quad \|(u, \eta_s, \eta_i)\|_{\text{gr}} = \|u\|_{L_1^2} + \|\eta_s\| + \|\eta_i\| + \|\widehat{\mathcal{D}}_\lambda(u, \eta_s, \eta_i)\|_{\mathbb{Z} \oplus \mathbb{Y}}.$$

Write $\mathbb{X}_{q,\lambda}^{\text{gr}}$ for the completion. This is the domain on which the mixed-target operator is bounded; it is smaller than the space of arbitrary L_1^2 corrections.

Proposition 3.4 (uniform complemented inverse). *For N sufficiently large and then λ sufficiently small, there are smooth equivariant synthesis and retraction maps*

$$(64) \quad S_g : \mathcal{K}_X(q) \oplus \mathcal{K}_I^{\text{fr}} \longrightarrow \mathbb{X}_{q,\lambda}^{\text{gr}}, \quad R_g : \mathbb{X}_{q,\lambda} \longrightarrow \mathcal{K}_X(q) \oplus \mathcal{K}_I^{\text{fr}},$$

with uniformly bounded norms and

$$(65) \quad M_g = R_g \text{pr}_{\mathbb{X}} S_g = I + o(1), \quad \Pi_g = S_g M_g^{-1} R_g \text{pr}_{\mathbb{X}}.$$

Here S_g lifts the model kernel elements, while R_g extracts their parameter coordinates and ignores auxiliary obstruction coefficients. With

$$(66) \quad \mathbb{H}_g^{\text{cmp}} = \ker(R_g \text{pr}_{\mathbb{X}}),$$

the augmented operator in (29) restricts to an isomorphism on the graph completion and on each fixed positive-scale strong Fredholm level. Its inverse P_g satisfies

$$(67) \quad \|P_g y\|_{\text{gr}} \leq C \|y\|_{\mathbb{Z} \oplus \mathbb{Y}}.$$

The image of S_g is the full augmented kernel. Smoothness is asserted at positive scale on strong Sobolev levels.

Proof. Write $T_\lambda v(z) = \lambda^{-1} v(z/\lambda)$ and $U_\lambda f(z) = \lambda^{-2} f(z/\lambda)$, with the metric and bundle identifications above understood. The map $U_\lambda^{-1} \beta_I$ is uniformly bounded in ordinary L^2 . Domain synthesis $v \mapsto \beta_I T_\lambda v$ is bounded from $\dot{H}_I^1$ to physical L_{1,A_λ}^2 . For its differentiated cutoff term use $\|d\beta_I\|_4 \|T_\lambda v\|_4$; its undifferentiated L^2 term is bounded by $C\sqrt{\lambda} \|v\|_4$. The truncated-connection difference is a small L^4 multiplier. The critical estimate needed below also holds on the Euclidean model: the translated Hardy inequality and Kato's inequality give

$$(68) \quad \sup_z \int_{\mathbb{R}^4} |z - y|^{-2} |v(y)|^2 dy \leq C \|\nabla_I v\|_2^2.$$

It is preserved by T_λ .

Let P_X be the ordinary L^2 inverse of the literal augmented background operator on a smooth complement. It is uniformly bounded over the small compact background disk by openness of surjectivity. For target y , set

$$\begin{aligned} (u_X, \xi_X) &= P_X (T_{X,\lambda}^Y)^{-1} (\beta_X y), \\ (v_I, c_I) &= P_h U_\lambda^{-1} (\beta_I y), \end{aligned}$$

and define

$$(69) \quad \mathcal{Q}_\lambda y = (\beta_X T_{X,\lambda} u_X + \beta_I T_\lambda v_I, \xi_X, c_I).$$

Restrictions are extended by zero across regions on which the cutoff vanishes. Applying the augmented operator cancels the obstruction terms exactly: the background embedding contains β_X , and the bubble profile is supported where $\beta_I = 1$. The principal terms sum to $\beta_X^2 y + \beta_I^2 y = y$. There is no principal-symbol remainder on the bubble because h agrees there with the rescaled physical metric.

For the curvature commutator, use (68) and the $L^{2\sharp}$ bound for $d\beta_I$. Ordinary rows use the L^4 bound for $d\beta_I$. Lemma 3.1 gives the same estimates for the background term. The comparison coefficients on the background are bounded by (60). On the bubble support the spinor is zero. The difference between the full and truncated instanton is supported at physical radius comparable to $\sqrt{\lambda}$ and is $O(\sqrt{\lambda})$ there; its L^4 and $L^{2\sharp}$ norms are $O(\lambda)$. The fixed smooth Dirac coefficients contribute $O(\sqrt{\lambda})$. Thus all remaining errors are zero-order, and

$$(70) \quad \widehat{\mathcal{D}}_\lambda \mathcal{Q}_\lambda = 1 + E_\lambda, \quad \|E_\lambda\|_{\mathbb{Z} \oplus \mathbb{Y} \rightarrow \mathbb{Z} \oplus \mathbb{Y}} \leq C(\log N)^{-1/2} + C_N \lambda^{1/3}.$$

Fixing N large and then λ small gives a right inverse $P_\lambda^r = \mathcal{Q}_\lambda (1 + E_\lambda)^{-1}$. Its ordinary domain bound and the equation $\widehat{\mathcal{D}}_\lambda P_\lambda^r = 1$ give the uniform graph bound. Smooth target data give smooth solutions by fixed-scale elliptic regularity. Density therefore extends the construction to the target and graph completions specified above.

We next choose the complement using bounded integrals of configurations. The configuration projection of $\mathcal{K}_X(q)$ is injective, since $\iota_{s,q}$ is injective. Its configuration L^2 Gram matrix defines a left inverse r_X ; put $R_X u = r_X(\beta_X T_{X,\lambda}^{-1} u)$. For $K = \ker Q_I$, choose a smooth compactly supported radial cutoff χ_0 so that

$$G(k, l) = \int_{\mathbb{R}^4} \chi_0 \langle k, l \rangle$$

is positive definite on K . Such a cutoff exists because $K \subset L^2$ is finite dimensional. Define $R_I u$ by pairing $T_\lambda^{-1} u$ with $\chi_0 K$ and applying G^{-1} . The compact dual profiles belong to $L^{4/3}$, so Hölder gives $\|R_I u\| \leq C \|u\|_4$, uniformly in scale. Thus $R_g = (R_X, R_I)$ is bounded on the configuration domain and ignores all auxiliary coefficients.

Define preliminary synthesis S_g^0 by $(v, \xi) \mapsto (\beta_X T_{X,\lambda} v, \xi, 0)$ on $\mathcal{K}_X(q)$ and $k \mapsto (\beta_I T_\lambda k, 0, 0)$ on K . The background diagonal block of $R_g \text{pr}_{\mathbb{X}} S_g^0$ tends to the identity, while the bubble block is exactly the identity once the dual support lies where $\beta_I = 1$. The off-diagonal blocks tend to zero: a dilated bubble mode has physical L^2 norm $O(\lambda)$, and the rescaled restriction of a smooth background mode on the fixed dual support is $O(\lambda)$. Consequently

$$(71) \quad R_g \text{pr}_{\mathbb{X}} S_g^0 = 1 + o(1).$$

Also $\widehat{\mathcal{D}}_\lambda S_g^0 = o(1)$ in the mixed target for fixed N . For background modes, the cutoff commutator is $O_N(\sqrt{\lambda})$ by smoothness and shrinking support, and the comparison coefficients tend to zero. For bubble modes the commutator is supported at rescaled radii comparable to $\lambda^{-1/2}$. Their value and derivative decay in Proposition 3.2 give an $O_N(\lambda)$ bound in both target norms. The metric principal error is bounded by $C\lambda \|\nabla_I k\|_{L^2 \cap L^\sharp}$, which is finite by that decay. The remaining metric coefficients have L^4 and $L^{2^\sharp}$ norms $O(\lambda)$. These critical derivative estimates are used only on the finite-dimensional smooth mode space.

Set

$$(72) \quad S_g = (1 - P_\lambda^r \widehat{\mathcal{D}}_\lambda) S_g^0.$$

Then $\widehat{\mathcal{D}}_\lambda S_g = 0$, and $M_g = R_g \text{pr}_{\mathbb{X}} S_g = 1 + o(1)$. To see that S_g spans the entire kernel, use the compact elliptic index at any fixed positive scale. Adding one instanton charge changes the real ASD index by +8 and the real twisted Dirac index by -2. The added complex stabilization contributes +2. The augmented index therefore increases by eight, and surjectivity identifies it with the kernel dimension. It follows that

$$(73) \quad P_g = (1 - S_g M_g^{-1} R_g \text{pr}_{\mathbb{X}}) P_\lambda^r$$

is a right inverse with image in $\mathbb{H}_g^{\text{cmp}}$. Every kernel vector is $S_g v$ and has retraction $M_g v$; this proves injectivity on that complement and the uniform bound.

Radial cutoffs and invariant metrics make the choices equivariant under the compact frame and residual-circle actions. At positive scale the smooth elliptic family argument gives smooth dependence on strong Sobolev levels. This does not give bounds for derivatives as λ tends to zero. $\square$

3.6. Uniform parameter derivatives. We first specify local identifications independently of the inverse. Near $g_0 = (q_0, x_0, \lambda_0, \rho_0)$ use ordinary background coordinates and

$$(74) \quad x = \exp_{x_0}(\lambda_0 \xi), \quad \lambda = \lambda_0 e^s, \quad \rho = \rho_0 \exp \theta.$$

Restrict these coordinates to a sufficiently small ball of fixed radius. They are uniformly comparable with (62), with bounded connection coefficients for second derivatives. In the

following lemma, ∂_p^α denotes a mixed coordinate derivative in these charts. Norms on inter-sections are sum norms.

Lemma 3.5 (physical bundle identifications). *There are smooth unitary identifications of the gluing bundles over each chart (74), covering the identity of X , such that, for $1 \leq |\alpha| \leq 2$,*

$$(75) \quad \|\partial_p^\alpha A_g\|_{L^4 \cap L^{2\sharp}} + \|\partial_p^\alpha \Phi_g\|_{L^4 \cap L^{2\sharp}} \leq C.$$

In these identifications the first two derivatives of $J_{s,g}$ and $J_{i,g}$ are uniformly bounded into the mixed target, and those of R_g are uniformly bounded from L^2_{1,A_g} to $\mathcal{K}_X(q) \oplus \mathcal{K}_I^{\text{fr}}$. All constants are independent of g_0 after fixing N .

Proof. Use a regular inner frame for the instanton and the exterior frame from its framing pole. The relative frame parameter is placed in their transition function, so the regular instanton expression on the core is independent of that parameter. Local $\text{SU}(2)$ lifts give the corresponding spinor identifications. Choose a fixed overlap annulus about x_0 of radius $r_0 = \sqrt{\lambda_0}$ and fixed radius ratio; its constants may depend on N . The original gluing formulas can be expressed on this cover because their connection cutoff radii are comparable throughout the parameter chart.

Let t_g be the transition function. On this annulus its quotient with t_{g_0} is uniformly close to the identity when the chart radius is small. The angular instanton transition is scale independent, normalized center displacement divided by annulus radius is $O(\lambda_0/r_0)$, and the remaining frames vary over compact smooth families. Take the Lie-group logarithm of this quotient. Extend it to a slightly larger annulus, multiply it by a fixed radial cutoff equal to one on the overlap, and exponentiate. The resulting change of inner frame makes the transition equal to t_{g_0} and is the identity toward the core. Leave the outer frame fixed. This identifies the bundles over id_X . The logarithm, its first two parameter derivatives, and their first spatial derivatives are bounded respectively by C and C/r_0 . Thus extra connection coefficients and two parameter derivatives are $O(r_0^{-1})$ on an annulus of fixed radius ratio. Their $L^4 \cap L^{2\sharp}$ norms are bounded by (56). These estimates also hold for the transition multipliers between overlapping parameter charts, so the local identifications are compatible with the configuration and target norms.

For completeness, the regular charge-one potential has components

$$(76) \quad (I_\lambda)_\mu(z) = \sum_\nu C_{\mu\nu} \frac{z^\nu}{\lambda^2 + |z|^2},$$

where the fixed Lie-algebra coefficients $C_{\mu\nu}$ encode the chosen chirality and normalization. Its leading angular pure-gauge term is scale independent. Direct differentiation gives

$$(77) \quad |\partial_p^\alpha I_\lambda(z)| \leq C \frac{\lambda}{(\lambda + |z|)^2}, \quad 1 \leq |\alpha| \leq 2,$$

for center and scale derivatives in flat coordinates. For example, $\lambda \partial_\lambda$ replaces the numerator in (76) by $-2\lambda^2 C_{\mu\nu} z^\nu$ and squares the denominator; $\lambda \partial_{x^j} = -\lambda \partial_{z^j}$. Repeated derivatives satisfy the same bound. The profile on the right has bounded $L^4 \cap L^{2\sharp}$ norm: after dilation the weighted integral is

$$\sup_z \int_{\mathbb{R}^4} |z - y|^{-2} (1 + |y|)^{-4} dy < \infty.$$

This follows by splitting into the regions near 0, near z , and their complement, or by radial monotonicity and evaluation at $z = 0$.

Derivatives of the normal-coordinate and orthonormal-frame coefficients gain a factor λ . Multiplied by the regular potential $O((\lambda + |z|)^{-1})$ on a fixed ball of radius at most one, they are also bounded by (77). Derivatives of connection cutoffs give $O(r_0^{-1})$ terms on annuli of fixed radius ratio. Exterior instanton coefficients have the better decay $O(\lambda^2|z|^{-3})$. The background family is smooth and compact. These observations prove the connection part of (75), including mixed derivatives.

The spinor is zero on the connection-gluing annulus. At its outer cutoff radius $\lambda^{1/3}$, normalized scale derivatives of the cutoff are bounded pointwise, while center derivatives gain $\lambda/\lambda^{1/3}$. The smooth background and frame coefficients therefore give uniformly bounded spinor derivatives. The same reasoning applies to the smooth background obstruction profiles. Derivatives of the bubble embedding are λ^{-2} times smooth compactly supported profiles in z/λ , with bounded L^2 norm. They remain entirely in the negative-spinor row, since these bundle identifications preserve the fixed physical metric and the target splitting.

Finally, write the compact dual profiles defining R_I as w_a , including the inverse of their Gram matrix. In flat physical coordinates its coefficients are

$$(78) \quad (R_I u)_a = \int \langle u(z), \lambda^{-3} w_a((z-x)/\lambda) \rangle dz.$$

Center and scale differentiation act on the dual profile by

$$(79) \quad w \mapsto -\partial_j w, \quad w \mapsto -3w - y \cdot \partial_y w,$$

respectively. After two derivatives the weights still have bounded $L^{4/3}$ norm, so Hölder gives the asserted bound without differentiating u . The metric, volume-density, and bundle factors in the actual formula have two bounded normalized derivatives on the fixed rescaled support. For R_X , use the compact smooth background Gram inverse and its L^2 dual profiles; the normalized derivatives of β_X are pointwise bounded. This proves the lemma. $\square$

Put $\tilde{S}_g = S_g M_g^{-1}$; this is the unique kernel lift whose configuration retraction is the identity.

Proposition 3.6 (uniform differentiated inverse). *In the physical bundle identifications of Lemma 3.5, the graph completions form locally a common Banach space with uniformly equivalent norms. The operators P_g , R_g , $\tilde{S}_g$, and Π_g have uniformly bounded first and second derivatives with respect to (62). For maps with values in the augmented configuration space these are graph-norm bounds. The operators remain equivariant and are smooth on strong Sobolev levels at positive scale.*

Proof. Abbreviate $\widehat{\mathcal{D}}_{q_g}$ by B_g and write $W = \mathbb{Z} \oplus \mathbb{Y}$ and $\mathcal{K} = \mathcal{K}_X(q) \oplus \mathcal{K}_I^{\text{fr}}$. Use smooth local orthonormal frames for the finite-dimensional bundles. Because the metric and principal symbols are fixed, every parameter derivative of B_g is zero-order in the configuration variable. Lemma 3.5 and the critical product estimates give

$$(80) \quad \|(\partial_p^\alpha B_g)(u, \eta)\|_W \leq C(\|u\|_{L_{1,A_g}^2} + |\eta|), \quad 1 \leq |\alpha| \leq 2.$$

Use $L^4 \cdot L^4 \subset L^2$ in the ordinary rows and $L^{2^\#} \cdot L^{2^\#} \subset L^\#$ in the curvature row. No $L^\#$ bound for an arbitrary ∇u is required.

Integrating (75) along a parameter segment bounds $\|A_g - A_{g_0}\|_4$ by $C|p - p_0|$. Kato's inequality and Sobolev embedding then compare L_{1,A_g}^2 and $L_{1,A_{g_0}}^2$ uniformly in both directions. Integrating (80) similarly bounds $B_g - B_{g_0}$ from the first-order augmented space to W . The graph norms are therefore uniformly equivalent on their common smooth domain.

Taking smooth completions identifies them with one Banach space D ; this identification was obtained without using an inverse.

Consider the square operator

$$(81) \quad F_g : D \longrightarrow W \oplus \mathcal{K}, \quad F_g h = (B_g h, R_g \operatorname{pr}_{\mathbb{X}} h).$$

Proposition 3.4 gives its uniformly bounded inverse

$$(82) \quad L_g(y, k) = P_g y + \tilde{S}_g k.$$

The coefficient and dual-profile formulas show that F_g is C^2 in operator norm on these fixed Banach spaces, with two uniform derivatives. The inverse identity gives

$$(83) \quad \begin{aligned} \partial_i L &= -L(\partial_i F)L, \\ \partial_j \partial_i L &= L(\partial_j F)L(\partial_i F)L + L(\partial_i F)L(\partial_j F)L - L(\partial_j \partial_i F)L. \end{aligned}$$

All factors are uniformly bounded between the specified spaces. This proves the bounds for P_g and $\tilde{S}_g$, and hence for $\Pi_g = \tilde{S}_g R_g \operatorname{pr}_{\mathbb{X}}$. The bounds for R_g were proved in the lemma. Conversion to covariant second derivatives uses the uniformly bounded connection coefficients of (74). Equivariance follows from the uniqueness characterizations and the equivariant undifferentiated construction. Positive-scale strong smoothness is the ordinary elliptic family assertion already established above. $\square$

The normalized synthesis has the same image as S_g and changes its finite-dimensional basis by $M_g^{-1} = I + o(1)$, of positive determinant. The original synthesis remains useful for comparison with the ordered model modes. No derivative bound for its particular raw-parametrix formula is needed. The estimates above concern the first-order graph norms; they do not give uniform bounds in unrescaled higher Sobolev norms.

3.7. Comparison with orthogonal projection. The orientation argument will use orthogonal projection of cutoff kernel fields. Bubble fields have physical L^2 norm of order λ ; comparison in the unweighted configuration norm would not suffice. Write $T_g = S_g^0$ for the preliminary synthesis above, and put

$$(84) \quad \begin{aligned} \|(u, \xi_s, \xi_i)\|_{\mathcal{H}_\lambda}^2 &= \|u\|_2^2 + |\xi_s|^2 + \lambda^2 |\xi_i|^2, \\ |(v, k)|_\lambda^2 &= |v|^2 + \lambda^2 |k|^2 \quad (v \in \mathcal{K}_X(q), k \in \mathcal{K}_I^{\text{fr}}). \end{aligned}$$

Here $|v|$ includes the auxiliary coordinates of the background kernel.

Proposition 3.7 (weak inverse and projected kernel). *With N fixed as in Proposition 3.4, the inverse P_g extends to $L^{4/3}$ in all three target rows and satisfies*

$$(85) \quad \|P_g f\|_{\mathcal{H}_\lambda} + \|\nabla_{A_g} \operatorname{pr}_{\mathbb{X}} P_g f\|_{4/3} \leq C_N \|f\|_{4/3}.$$

Let π_g be orthogonal projection onto $\ker B_g$ in $\mathcal{H}_\lambda$, and set $\mathcal{L}_g = (1 - P_g B_g)T_g$. Then

$$(86) \quad \mathcal{L}_g^{-1} \pi_g T_g = 1 + O_N(\sqrt{\lambda})$$

in the weighted parameter norm. In particular $\pi_g T_g$, $\mathcal{L}_g$, and S_g are kernel isomorphisms with the same orientation. The conclusion also holds for $\pi_g T'_g$ if $\|T'_g - T_g\|_{|\cdot|_\lambda \rightarrow \mathcal{H}_\lambda} = o(1)$.

Proof. First consider the Euclidean completion

$$\dot{W}_I^{1,4/3} = \overline{C_c^\infty}^{\|\nabla_I(\cdot)\|_{4/3}} \hookrightarrow L^2.$$

The embedding follows from Kato's inequality and scalar Sobolev. The exterior argument in Proposition 3.2 applies at this exponent: the flat Calderón–Zygmund estimate bounds

$\|\partial v\|_{4/3}$ by $C\|D_0 v\|_{4/3}$, and the term $\|I\|_4\|v\|_2$ is absorbed outside a fixed large ball. Interior elliptic estimates give a compact remainder $\|v\|_{L^{4/3}(B_{2R})}$. Thus the kernel is finite dimensional and the range is closed. The kernel consists of L^2 harmonic fields, and the annihilator of the range consists of harmonic fields in the dual space L^4 . Both spaces compactify and extend across the pole by the conformal and cutoff calculation in that proposition. The same compact kernel computation therefore applies. In particular $\widehat{B}_0$ is surjective at exponent $4/3$, with the same real eight-dimensional kernel.

One can use the same complement at both exponents. For each kernel field k its homogeneous orthogonality functional is

$$\int \langle \nabla_I v, \nabla_I k \rangle = \int \langle v, \nabla_I^* \nabla_I k \rangle.$$

The coefficient on the right is smooth and in L^2 : the Dirac-square identity makes it curvature times k , of order r^{-7} at infinity. The functionals extend to L^2 and their matrix on the kernel is its positive homogeneous Gram matrix. For compactly supported smooth target data, either inverse is harmonic off a compact set. Conformal removability gives r^{-3} decay there, so each solution belongs to both homogeneous domains. Complemented uniqueness makes the inverses equal. Density in $L^2 \cap L^{4/3}$ extends this equality to common data. The metric perturbation in (53) has a principal coefficient of size $O(\lambda)$ in L^∞ and a remaining coefficient of size $O(\lambda)$ in L^4 . They act on $\nabla v \in L^{4/3}$ and $v \in L^2$, respectively. Its Neumann inverse consequently has the same agreement and bound at exponent $4/3$.

The physical scaling is

$$(87) \quad \|T_\lambda v\|_2 = \lambda\|v\|_2, \quad \|\nabla T_\lambda v\|_{4/3} = \lambda\|\nabla v\|_{4/3}, \quad \|U_\lambda f\|_{4/3} = \lambda\|f\|_{4/3}.$$

Smooth finite-dimensional complements also give a background inverse from $L^{4/3}$ to $W^{1,4/3}$ plus its auxiliary space. Repeat (69) in $L^{4/3}$ in every target row. The principal symbols still match, and the finite columns cancel exactly. Cutoff derivatives and connection errors are L^4 multipliers on L^2 configurations, so the error is bounded by

$$C\|d\beta_X\|_4 + C\|d\beta_I\|_4 + C_N\lambda^{1/3} < \frac{1}{2}.$$

This gives a raw inverse P_g^r with the bound in (85). The weight on the bubble auxiliary coordinate follows from (87).

The preliminary modes satisfy the sharper error estimates

$$(88) \quad \|B_g T_g(v, 0)\|_{4/3} \leq C_N \lambda |v|, \quad \|B_g T_g(0, k)\|_{4/3} \leq C_N \lambda^{3/2} |k|.$$

For the first estimate, a smooth background mode times a cutoff derivative on radius $r \asymp \sqrt{\lambda}$ has norm $O(r^2)$. The removed bounded spinor coefficients have support radius $\lambda^{1/3}$ and contribute $O((\lambda^{1/3})^3) = O(\lambda)$. The finite column cancels the cut-off background equation. For the second estimate, a bubble mode is $O(\lambda^2 r^{-3})$ on the cutoff annulus. Its product with $d\beta_I$ is bounded on volume $O_N(\lambda^2)$, giving $O_N(\lambda^{3/2})$. The principal metric error contributes $O(\lambda)O(\lambda)$ by (87); the remaining connection and metric coefficients have L^4 norm $O(\lambda)$ and multiply a field of L^2 norm $O(\lambda)$. The bubble modes have no spinor component, and the preglued spinor vanishes on their support.

The map T_g has uniform upper and lower bounds from $|\cdot|_\lambda$ to $\mathcal{H}_\lambda$. Its background Gram matrix tends to the positive augmented background Gram matrix. After dividing bubble fields by λ , their Gram matrix tends to the positive L^2 Gram matrix of the Euclidean kernel. The cross terms tend to zero: the L^1 norm of $\lambda^{-1}T_\lambda k$ on the support is $O_N(\lambda^{3/2})$, apart from the smaller $O(\lambda^2)$ core contribution.

The retraction R_g is bounded from configuration L^2 to the weighted parameter norm. Its bubble dual has L^2 norm $O(\lambda^{-1})$, canceled by the weight on that row. Moreover $M_g^0 = R_g \text{pr}_{\mathbb{X}} T_g = 1 + o(1)$ also in this norm. The background diagonal error is $O_N(\lambda^2)$; the bubble diagonal is the identity; the bubble row on a background column vanishes because $\beta_X = 0$ on the compact dual support. The remaining cross term is $O_N(\lambda^{5/2})$, by integrating $\lambda^2 r^{-3}$ on the overlap.

Put $S_g^r = (1 - P_g^r B_g) T_g$. The weak row inverse and (88) give

$$(89) \quad \|(S_g^r - T_g)(v, k)\|_{\mathcal{H}_\lambda} \leq C_N(\lambda|v| + \lambda^{3/2}|k|).$$

Its retraction matrix is invertible. At fixed positive scale, weak kernel elements are smooth by elliptic regularity; surjectivity and index therefore imply that S_g^r spans the full kernel. Subtract its kernel projection from P_g^r to obtain a right inverse in $\ker(R_g \text{pr}_{\mathbb{X}})$. It retains the weak bound, including the derivative bound since the cutoff kernel modes have uniformly bounded $W^{1,4/3}$ norm relative to $|\cdot|_\lambda$. On smooth data this is P_g by uniqueness in the stated complement. This proves (85).

The same estimate (89) holds for $\mathcal{L}_g$, and $R_g \text{pr}_{\mathbb{X}} \mathcal{L}_g = M_g^0$ exactly. Its orientation agrees with S_g , since their retraction matrices both have positive determinant. Finally $\pi_g \mathcal{L}_g = \mathcal{L}_g$ and $\|\pi_g\| = 1$, whence

$$\|(\pi_g T_g - \mathcal{L}_g)(v, k)\|_{\mathcal{H}_\lambda} \leq C_N \sqrt{\lambda} |(v, k)|_\lambda.$$

The lower bound for T_g and the lift error give a uniform lower bound for $\mathcal{L}_g$. This proves (86) and its assertion for T'_g . $\square$

In the remaining sections write $\mathbb{H}_{\lambda,N}^{\text{cmp}} = \mathbb{H}_g^{\text{cmp}}$ and $P_{\lambda,N} = P_g$ after fixing the cutoff parameter N .

In particular, by Proposition 3.4, a unit vector in the complement cannot have image tending to zero in the mixed target. This conclusion follows from (67) without a separate assumption excluding neck-supported approximate kernels.

4. PROJECTED RESIDUAL AND NONLINEAR CORRECTION

4.1. The projected splicing error. Put $r = \sqrt{\lambda}$ and $R = \lambda^{1/3}$. For small λ the pregluing formula divides X into the core $B(x, r/4)$; the inner, flat, and middle annuli with radii $(r/4, r/2)$, $(r/2, 2r)$, and $(2r, 4r)$; the zero-spinor plateau $(4r, 4R)$; the outer spinor-cutoff annulus $(4R, 8R)$; and the exterior of $B(x, 8R)$. The logarithmic parametrix cutoffs lie inside the r -scale annuli but play no role in this residual decomposition.

Proposition 4.1 (projected residual). *Uniformly in the compact background family, the center, and the frames,*

$$(90) \quad \|\mathfrak{S}(q_\lambda) - J_{s,\lambda} b(q)\|_{L^{\sharp,2;2}} \leq C \lambda^{1/3}.$$

Here $L^{\sharp,2;2}$ means $L^{\sharp,2}$ in the curvature component and L^2 in the negative-spinor component.

Proof. On the exterior, the preglued pair equals the background pair and $J_{s,\lambda}$ equals the background obstruction embedding; the residual therefore vanishes identically.

Write $\mathfrak{S}(q) = (v_0, \psi_0)$ and let χ_{out} be the outer spinor cutoff. On $4R < d(x, \cdot) < 8R$ the pair is $(A_0, \chi_{\text{out}} \Phi_0)$ and $J_{s,\lambda} b(q) = \mathfrak{S}(q)$. If $\mathcal{Q}(\Phi)$ denotes the quadratic spinor term in the curvature equation, the residual is

$$(91) \quad (\mathcal{Q}(\Phi_0) - \mathcal{Q}(\chi_{\text{out}} \Phi_0), (\chi_{\text{out}} - 1)\psi_0 + \rho(d\chi_{\text{out}})\Phi_0).$$

The first component and the first term in the second component are uniformly bounded on a ball of radius $8R$, and hence have mixed-target norm $O(R^2)$ by (55). Direct radial integration gives $\|d\chi_{\text{out}}\|_{L^2} \leq CR$; consequently the last term in (91) is $O(R) = O(\lambda^{1/3})$.

On the plateau $4r < d(x, \cdot) < 4R$ the pair is $(A_0, 0)$, so its Dirac component is zero. Subtracting the matched background residual gives

$$(92) \quad \mathfrak{S}(A_0, 0) - \mathfrak{S}(A_0, \Phi_0) = (\mathcal{Q}(\Phi_0), -\psi_0),$$

with the sign in $\mathcal{Q}$ fixed by the convention in (91). Both entries are uniformly bounded and supported in $B(x, 4R)$, so their mixed-target norm is $O(R^2) = O(\lambda^{2/3})$.

On the middle annulus $2r < d(x, \cdot) < 4r$, write the connection in radial gauge as $\Gamma + \chi_{\text{mid}}(A_0 - \Gamma)$. Then

$$(93) \quad F = \chi_{\text{mid}}F_{A_0} + d\chi_{\text{mid}} \wedge (A_0 - \Gamma) + (\chi_{\text{mid}}^2 - \chi_{\text{mid}})(A_0 - \Gamma) \wedge (A_0 - \Gamma).$$

The bounds $|A_0 - \Gamma| \leq Cr$ and $|d\chi_{\text{mid}}| \leq C/r$ show that (93) is uniformly bounded. The spinor is zero and the matched background target is uniformly bounded, so (55) gives an $O(r^2) = O(\lambda)$ contribution. On the flat annulus $r/2 < d(x, \cdot) < 2r$ the preglued pair is $(\Gamma, 0)$; the only term after subtraction is again a uniformly bounded matched background target, and its contribution is $O(\lambda)$.

On the inner annulus $r/4 < d(x, \cdot) < r/2$, the instanton decay gives

$$(94) \quad |F_I| \leq C\lambda^2 d(x, \cdot)^{-4} \leq C, \quad |A_I| \leq C\lambda^2 d(x, \cdot)^{-3} \leq C\sqrt{\lambda}.$$

Since the cutoff derivative is $O(\lambda^{-1/2})$, every curvature term in the truncated connection and every subtracted background term is uniformly bounded. This annulus therefore also contributes $O(\lambda)$.

Finally, on $B(x, r/4)$ the model instanton is anti-self-dual for the Euclidean metric δ , but not identically for the physical metric g . Its self-dual curvature is

$$(95) \quad F_I^{+,g} = \frac{1}{2}(*_g - *_\delta)F_I.$$

The normal-coordinate estimate $|g - \delta| \leq Cd(x, \cdot)$, conformal invariance of the four-dimensional curvature norm, and radial integration of the standard instanton give

$$(96) \quad \|F_I^{+,g}\|_{L^{\sharp,2}(B(x,r/4))} \leq C\sqrt{\lambda}.$$

The matched background target on the core contributes $O(r^2) = O(\lambda)$. Adding the seven disjoint regional estimates gives

$$C(\lambda^{1/3} + \lambda^{2/3} + \lambda + \lambda^{1/2}) \leq C\lambda^{1/3},$$

which proves (90). $\square$

4.2. Solving the augmented equation. Write $h = (u, \eta_s, \eta_i)$ and let P_λ be the inverse in Proposition 3.4. Expanding (30) gives

$$(97) \quad \widehat{\mathcal{F}}_\lambda(h) = r_\lambda + \widehat{\mathcal{D}}_\lambda h + Q_\lambda(u, u),$$

where r_λ is the residual in Proposition 4.1. Sobolev multiplication gives the scale-independent estimates

$$(98) \quad \begin{aligned} \|Q_\lambda(u, u)\|_{\mathbb{Z} \oplus \mathbb{Y}} &\leq C\|u\|_{L_1^2}^2, \\ \|Q_\lambda(u, u) - Q_\lambda(v, v)\|_{\mathbb{Z} \oplus \mathbb{Y}} &\leq C(\|u\|_{L_1^2} + \|v\|_{L_1^2})\|u - v\|_{L_1^2}. \end{aligned}$$

Proposition 4.2 (small correction). *For sufficiently small λ , there is a unique $h_\lambda = (u_\lambda, \eta_{s,\lambda}, \eta_{i,\lambda})$ in a fixed small ball in $\mathbb{H}_{\lambda,N}^{\text{cmp}}$ satisfying $\hat{\mathcal{F}}_\lambda(h_\lambda) = 0$. It obeys*

$$(99) \quad \|u_\lambda\|_{L_1^2} + \|\eta_{s,\lambda}\| + \|\eta_{i,\lambda}\| \leq C\lambda^{1/3}.$$

It is smooth in all positive-scale gluing parameters.

Proof. On the ball of radius $2C\|r_\lambda\|$, use the map

$$h \longmapsto -P_\lambda(r_\lambda + Q_\lambda(u, u)).$$

Propositions 3.4 and 4.1, together with (98) show that it preserves this ball and has Lipschitz constant less than $1/2$ after decreasing the scale. Banach's fixed-point theorem gives existence, uniqueness, and (99). Any solution in a fixed sufficiently small graph-norm ball satisfies $\|h\|_{\text{gr}} \leq C\|r_\lambda\| + C\|h\|_{\text{gr}}^2$ and therefore lies in the contraction ball. This proves the stated uniqueness. At a fixed positive scale, local elliptic regularity makes the solution smooth. For the initial critical step, take small coordinate balls on which the L^4 norm of u is small, absorb its multiplication operator in the local first-order $W^{1,p}$ estimate for some $p > 2$, and then bootstrap the quadratic equation. All other coefficients and the finite-dimensional source profiles are smooth at this fixed scale. The linearized equation is invertible on the complement by the same small-operator estimate as in the contraction; elliptic regularity gives its inverse on the strong levels. The implicit-function theorem now gives smooth dependence for $\lambda > 0$. $\square$

Define the corrected gluing map and obstruction section by

$$(100) \quad \begin{aligned} \gamma(g) &= [q_\lambda + u_\lambda], \\ \chi(g) &= (b(q) + \eta_{s,\lambda}, \eta_{i,\lambda}). \end{aligned}$$

Then $\gamma(g)$ is a genuine monopole exactly when $\chi(g) = 0$.

4.3. Continuity at zero scale. The constants above are independent of a moving center. If $g_n \rightarrow g_0$ with $\lambda_n \rightarrow 0$, then (99) gives $u_{\lambda_n} \rightarrow 0$ in the global L_1^2 norm. On compact subsets away from the limiting center, the preglued pairs converge in every Sobolev norm to the background. After rescaling at the centers, their connections converge to the framed instanton while their spinors converge to zero. For the corrected pairs, local elliptic estimates upgrade the L_1^2 convergence away from the center and on fixed rescaled balls: in these regions the coefficients are uniformly smooth, and the difference of the augmented source from the limiting source tends to zero smoothly. Moreover, writing $u_\lambda = (a_\lambda, \phi_\lambda)$, we have

$$\|F_{A_\lambda + a_\lambda} - F_{A_\lambda}\|_2 \leq C(\|a_\lambda\|_{L_{1,A_\lambda}^2} + \|a_\lambda\|_{L_{1,A_\lambda}^2}^2) \longrightarrow 0.$$

Thus the correction does not change the limiting curvature measure. These properties are precisely Uhlenbeck convergence to (q, x) . Thus γ extends continuously by

$$(101) \quad \gamma(q, x, 0) = (q, x).$$

The same estimates and the definitions in (61) show that the norm of χ is continuous when the instanton line is collapsed at the cone vertex.

4.4. Derivatives of the residual and correction.

Proposition 4.3 (differentiated correction). *Use the physical bundle identifications from Lemma 3.5. For $1 \leq |\alpha| \leq 2$, the residual and correction satisfy*

$$(102) \quad \|\partial_p^\alpha r_g\|_{\mathbb{Z} \oplus \mathbb{Y}} \leq C\lambda^{1/3},$$

$$(103) \quad \|\partial_p^\alpha h_g\|_{\text{gr},g} + \|\partial_p^\alpha u_g\|_{\text{crit},g} \leq C\lambda^{1/3}.$$

Here the graph norm is (63), and the critical configuration norm is defined in (108). The constants are uniform over the compact parameter family.

Proof. We differentiate the regional formulas in the proof of Proposition 4.1. For a radial cutoff at radius $s = \lambda^\nu$, where $\nu = 1/2$ or $1/3$, its first two normalized parameter derivatives have bounded amplitude, and their first spatial derivatives have amplitude at most C/s . Logarithmic scale derivatives differentiate a fixed profile. A normalized center derivative gains $\lambda/s \leq 1$. Use slightly enlarged annuli for these bounds; no characteristic function of a moving region is differentiated. All background coefficients and their two parameter derivatives are bounded on the compact smooth family.

In (91), the differentiated Dirac cutoff term has amplitude C/R on a set of volume $O(R^4)$, where $R = \lambda^{1/3}$. Its L^2 norm is therefore $O(R)$. Every other outer or plateau term, including its two derivatives, is bounded on $B(x, 8R)$ and contributes $O(R^2)$ to the mixed target. The exterior contribution remains zero: differentiate the exact identity $\mathfrak{S}(q) = \iota_q b(q)$, retaining all derivatives of the embedding and of b .

Put $r_0 = \sqrt{\lambda}$. On the middle annulus, the radial background potential and its normalized derivatives are $O(r_0)$; center derivatives may add an $O(\lambda)$ term, which is smaller. The cutoff differentials are $O(r_0^{-1})$. Thus (93) and its two derivatives have bounded coefficients. On the inner annulus, normalized derivatives of the exterior instanton potential have the same $O(\lambda^2 r_0^{-3}) = O(r_0)$ bound, and the curvature and cutoff curvature terms have bound $O(\lambda^2 r_0^{-4}) = O(1)$. The matched background target and its derivatives are also bounded, including derivatives of its cutoff. These annuli and the flat annulus therefore contribute $O(r_0^2)$.

On the core write (95) as $\mathfrak{m}_g F_{I_\lambda}$, where $\mathfrak{m}_g = \frac{1}{2}(*_g - *_\delta)$ in the chosen identifications. The normal-coordinate Taylor formulas, expressed in the fixed physical target, give

$$|\partial_p^\alpha \mathfrak{m}_g| \leq Cr_0 \quad \text{on } B(x, Cr_0), \quad 0 \leq |\alpha| \leq 2.$$

Indeed the undifferentiated metric discrepancy vanishes at the center; normalized center derivatives add a factor λ , and changes of normal frames have bounded coefficients. The regular-gauge rational curvature formula gives

$$|\partial_p^\alpha F_{I_\lambda}(z)| \leq C \frac{\lambda^2}{(\lambda + |z|)^4}, \quad |\alpha| \leq 2.$$

This envelope has uniformly bounded L^2 and $L^\sharp$ norms. Leibniz's rule thus gives $O(r_0)$ for the differentiated core error. On the bundle-transition annulus, first use gauge covariance: the residual transforms as a target section by a unitary multiplier with bounded first two parameter derivatives. Its potentially large spatial derivatives cancel before any estimate. The spinor is zero there. Summing $O(R) + O(R^2) + O(r_0^2) + O(r_0)$ proves (102).

Use the common graph domain and square operator F_g from (81). The correction satisfies

$$(104) \quad F_g h_g + (Q_g(u_g, u_g), 0) + (r_g, 0) = 0.$$

The quadratic maps Q_g have uniformly bounded bilinear norms $L_1^2 \times L_1^2 \rightarrow \mathbb{Z} \oplus \mathbb{Y}$, including two parameter derivatives: their coefficients are smooth metric and Clifford contractions in the fixed physical bundle charts. The derivative of (104) in h is a small perturbation of F_g , with uniformly bounded inverse into the graph domain. Differentiate once. The residual derivative is $O(\lambda^{1/3})$, and the other inhomogeneous terms contain h_g or two factors of u_g . The inverse gives the same bound for $\partial_p h_g$. Differentiating a second time uses the two bounded operator derivatives and the first correction derivative just obtained, so all inhomogeneous terms are again $O(\lambda^{1/3})$.

For the critical upgrade, differentiate the exact zero Coulomb row:

$$(105) \quad \begin{aligned} d_g^{0,*} \partial_i u_g &= -(\partial_i d_g^{0,*}) u_g, \\ d_g^{0,*} \partial_{ij} u_g &= -(\partial_{ij} d_g^{0,*}) u_g - (\partial_i d_g^{0,*}) \partial_j u_g - (\partial_j d_g^{0,*}) \partial_i u_g. \end{aligned}$$

The differentiated gauge rows are zero-order critical multipliers by (75). The right sides therefore have $L^{\sharp,2}$ norm $O(\lambda^{1/3})$. Difference quotients converge in the first-order space and their divergences converge in $L^{\sharp,2}$, which proves the assertion in the smooth critical completion. $\square$

5. PARAMETER RECOVERY AND THE COMPACTIFIED EMBEDDING

5.1. Critical gauge estimates. Use the full determinant-one gauge group of pairs. At a preglued pair $q_g = (A_g, \Phi_g)$ impose the Coulomb condition

$$(106) \quad d_g^{0,*}((A, \Phi) - q_g) = 0.$$

No value at a point is prescribed for this ambient gauge transformation. The relative instanton frame remains a separate gluing parameter. Put $\mathfrak{r}_{\Phi_g} \zeta = \zeta \Phi_g$ and

$$(107) \quad \Delta_g^0 = d_g^{0,*} d_g^0 = d_{A_g}^* d_{A_g} + \mathfrak{r}_{\Phi_g}^* \mathfrak{r}_{\Phi_g}.$$

For configurations and infinitesimal gauges, respectively, use

$$(108) \quad \|c\|_{\text{crit},g} = \|c\|_{L_1^2} + \|d_g^{0,*} c\|_{L^{\sharp,2}},$$

$$(109) \quad \|\zeta\|_{2,\sharp,g} = \|\Delta_g^0 \zeta\|_{L^{\sharp,2}} + \|\zeta\|_{L^2}.$$

Denote these smooth completions by $\mathbb{E}_g^1$ and $\mathbb{E}_g^0$, respectively. On the horizontal slice the configuration norm is exactly its L_1^2 norm. For arbitrary corrections, L_1^2 control only puts the divergence in L^2 , which does not suffice for the critical Green-operator argument. The critical configuration norm is the pair analogue of the norm in [Fee01, Theorem 8.2].

Lemma 5.1 (gauge and scalar spectral gaps). *For the compact preglued family with sufficiently small background disk and scale bound, there is a uniform C such that*

$$\|\zeta\|_{L^2}^2 \leq C(\|d_{A_g} \zeta\|_{L^2}^2 + \|\zeta \Phi_g\|_{L^2}^2).$$

The same uniform estimate holds for real functions f , in the form

$$(110) \quad \|f\|_2^2 \leq C(\|df\|_2^2 + \|\Phi_g f\|_2^2).$$

Proof. The compact background family has a positive lower bound for its spinor L^2 norm and a finite upper bound for its C^0 norm. The spinor cutoff removes a term of L^2 norm $O(\lambda^{2/3})$, so, for small scale,

$$\inf_g \|\Phi_g\|_2 = m_\Phi > 0, \quad \sup_g \|\Phi_g\|_\infty = M_\Phi < \infty.$$

For a real function f , let $\bar{f}$ be its average. The scalar Poincaré inequality gives

$$m_\Phi |\bar{f}| \leq \| \Phi_g |f| \|_2 + M_\Phi \|f - \bar{f}\|_2 \leq \| \Phi_g |f| \|_2 + C \|df\|_2.$$

Together with $\|f\|_2 \leq \|f - \bar{f}\|_2 + \text{vol}(X)^{1/2} |\bar{f}|$, this proves (110). For a skew-Hermitian trace-free matrix in the rank-two factor, $\zeta^* \zeta = \frac{1}{2} |\zeta|_{\text{HS}}^2 1$. Thus $|\zeta \Phi_g|^2 = \frac{1}{2} |\zeta|_{\text{HS}}^2 |\Phi_g|^2$. Apply the scalar estimate to $|\zeta|_{\text{HS}}$ and use Kato's inequality to obtain the gauge estimate. All bundle norm conventions differ from this one by fixed constants. $\square$

Let $G_g = (\Delta_g^0)^{-1}$, initially on smooth sections, and set

$$(111) \quad H_g = 1 - d_g^0 G_g d_g^{0,*}, \quad \mathcal{C}_g^{\text{sl}} = \ker R_g \cap \ker d_g^{0,*}.$$

Proposition 5.2 (critical pair Green estimates). *The operator G_g extends from $L^{\sharp,2}$ to $\mathbb{E}_g^0$ and satisfies the uniform estimate*

$$(112) \quad \|G_g f\|_{\mathbb{E}_g^0} + \|G_g f\|_{C^0} + \|G_g f\|_{L_{2,A_g}^2} + \|\nabla_{A_g} G_g f\|_{L^4 \cap L^{2\sharp}} \leq C \|f\|_{L^{\sharp,2}}.$$

The map H_g is a uniformly bounded projection on $\mathbb{E}_g^1$ with range $\ker d_g^{0,*}$. In the physical bundle identifications of Lemma 3.5, both graph completions have uniformly equivalent local norms, and G_g and H_g have uniformly bounded first and second normalized parameter derivatives between these spaces.

Proof. We first record the coefficient estimates used in this proof. The fixed instanton energy, compact background family, and cutoff formulas give

$$(113) \quad \|F_{A_g}\|_2 + \|\Phi_g\|_\infty + \|\nabla_{A_g} \Phi_g\|_2 \leq C.$$

For $a_\alpha = \partial_p^\alpha A_g$ and $\phi_\alpha = \partial_p^\alpha \Phi_g$ in the charts of (74), one also has

$$(114) \quad \|\nabla_{A_g} a_\alpha\|_{L^{\sharp,2}} + \|\phi_\alpha\|_\infty + \|\nabla_{A_g} \phi_\alpha\|_2 \leq C, \quad 1 \leq |\alpha| \leq 2.$$

Indeed, one spatial derivative of the rational formula (76) bounds $\nabla_{A_g} a_\alpha$ by $C\lambda/(\lambda + |z|)^3$. This profile has bounded L^2 and $L^\sharp$ norms; the corresponding radial integrals are $\int_0^\infty t^3(1+t)^{-6} dt = 1/20$ and $\int_0^\infty t(1+t)^{-3} dt = 1/2$. On the fixed-ratio transition annulus, the logarithms used in Lemma 3.5 have two spatial derivatives bounded by C/r_0^2 , also after two parameter derivatives. The resulting connection terms are $O(r_0^{-2})$ on that annulus. Their $L^{\sharp,2}$ norms are bounded by (55). The additional commutators use the critical product estimates. Finally, the spinor vanishes on the connection-gluing region, and its outer cutoff and two normalized derivatives have spatial derivative $O(\lambda^{-1/3})$ on a ball of radius $C\lambda^{1/3}$. This proves (113)–(114).

Let $\zeta = G_g f$ for smooth f . The pointwise identity is

$$(115) \quad \frac{1}{2} d^* d |\zeta|^2 + |\nabla_{A_g} \zeta|^2 + |\zeta \Phi_g|^2 = \langle f, \zeta \rangle.$$

The positive Green kernel of $d^* d + 1$ is uniformly comparable with $\text{dist}(x, y)^{-2}$ [Fee01, Lemma 5.1]. Apply it to (115) and discard the nonnegative spinor term. The argument of [Fee01, Lemma 5.3] gives

$$\|\nabla_{A_g} \zeta\|_{L^{2\sharp}}^2 + \|\zeta\|_\infty^2 \leq C \|\zeta\|_\infty (\|f\|_{L^\sharp} + \|\zeta\|_{L^\sharp}).$$

Using $\|\zeta\|_{L^\sharp} \leq C \|\zeta\|_4 \leq C \|\zeta\|_2^{1/2} \|\zeta\|_\infty^{1/2}$ and Young's inequality, then Lemma 5.1, yields

$$(116) \quad \|\zeta\|_\infty + \|\nabla_{A_g} \zeta\|_{L^{2\sharp}} + \|\zeta\|_2 \leq C \|f\|_{L^{\sharp,2}}.$$

Put $V_g = \mathfrak{r}_{\Phi_g}^* \mathfrak{r}_{\Phi_g}$. Since V_g is uniformly bounded, the equation $\nabla_{A_g}^* \nabla_{A_g} \zeta = f - V_g \zeta$ has right side bounded in $L^{\sharp,2}$ by (116). The estimates of [Fee01, Lemmas 5.4–5.5], whose constants

depend on the curvature only through its L^2 norm, give the L^2_{2,A_g} and L^4 derivative bounds. The gauge graph bound follows from the equation and the spectral gap. Smooth approximation proves (112) on the stated completions, and also gives these embeddings for every $\zeta \in \mathbb{E}_g^0$.

In the fixed physical bundle charts, a parameter derivative of Δ_g^0 is a sum of contractions of $a_\alpha \nabla_{A_g} \zeta$, $(\nabla_{A_g} a_\alpha) \zeta$, and $(\partial_p^\alpha V_g) \zeta$. Second derivatives also contain products $a_\alpha a_\beta \zeta$. Equations (75), (114), and (112) therefore imply

$$(117) \quad \|(\partial_p^\alpha \Delta_g^0) \zeta\|_{L^{\sharp,2}} \leq C \|\zeta\|_{\mathbb{E}_g^0}, \quad 1 \leq |\alpha| \leq 2.$$

Expand $\Delta_g^0 - \Delta_{g_0}^0$ using $\nabla_{A_{g_0}}$. Integrating the coefficient bounds controls this difference by $C|p - p_0|$ from $\mathbb{E}_{g_0}^0$ to $L^{\sharp,2}$. For a sufficiently small fixed chart radius the smooth gauge graph norms are thus equivalent in both directions. The configuration graph norms are likewise equivalent, since $d_g^{0,*} - d_{g_0}^{0,*}$ is zero-order with bounded critical multiplier coefficients. Taking smooth completions gives common local Banach spaces.

We may now apply the inverse differentiation identities (83) to Δ_g^0 and G_g . They give the two uniform derivatives of G_g . Moreover,

$$d_g^0 : \mathbb{E}_g^0 \longrightarrow \mathbb{E}_g^1, \quad d_g^{0,*} : \mathbb{E}_g^1 \longrightarrow L^{\sharp,2}$$

are uniformly bounded with two parameter derivatives. For the first map use $\nabla_{A_g}^2 \zeta$ in L^2 , ζ in C^0 , and $\nabla_{A_g} \Phi_g$ in L^2 ; its divergence is $\Delta_g^0 \zeta$. Derivatives of this map use (114) and the same products, including in the critical divergence. Derivatives of the second map are zero-order. The formula (111) now proves the claims for H_g . $\square$

Lemma 5.3 (stability of the horizontal projection). *Let $c = (a, \phi)$ have sufficiently small critical norm $\epsilon = \|c\|_{\text{crit},g}$ at q_g . The pair Laplacian of $q_g + c$ is invertible on the same gauge graph completion, with uniform bound. Its horizontal projection H_{q_g+c} satisfies*

$$(118) \quad \|H_{q_g+c} - H_g\|_{\mathbb{E}_g^1 \rightarrow \mathbb{E}_g^1} \leq C\epsilon.$$

Proof. The difference of the pair Laplacians is a sum of contractions of

$$a \nabla_{A_g} \zeta, \quad (d_{A_g}^* a) \zeta, \quad a^2 \zeta, \quad \Phi_g \phi \zeta, \quad \phi^2 \zeta.$$

Here $d_{A_g}^* a = d_g^{0,*} c + \mathfrak{r}_{\Phi_g}^* \phi$ has $L^{\sharp,2}$ norm at most $C\epsilon$. Use ζ in C^0 , $\nabla_{A_g} \zeta$ in $L^4 \cap L^{2\sharp}$, and the critical product bounds. This gives

$$\|\Delta_{q_g+c}^0 - \Delta_g^0\|_{\mathbb{E}_g^0 \rightarrow L^{\sharp,2}} \leq C(\epsilon + \epsilon^2).$$

A Neumann series therefore gives G_{q_g+c} on the same smooth graph completion, with $G_{q_g+c} - G_g = O(\epsilon)$. Also $d_{q_g+c}^{0,*} - d_g^{0,*}$ is a zero-order operator of norm $O(\epsilon)$ from $\mathbb{E}_g^1$ to $L^{\sharp,2}$. For $d_{q_g+c}^0 - d_g^0$, the first-order configuration norm uses $\nabla_{A_g} a, \nabla_{A_g} \phi$ in L^2 and ζ in C^0 . Its critical divergence uses $d_{A_g}^* a, a \nabla_{A_g} \zeta$, and $\Phi_g \phi \zeta$, so this difference has norm $O(\epsilon)$ from $\mathbb{E}_g^0$ to $\mathbb{E}_g^1$. Insert these bounds in $H_{q_g+c} = 1 - d_{q_g+c}^0 G_{q_g+c} d_{q_g+c}^{0,*}$ to obtain (118). All estimates pass to the specified smooth completions. In particular no pointwise bound for the perturbed spinor is required. $\square$

5.2. A uniform slice for the full gauge group.

Proposition 5.4 (uniform critical pair slice). *There are positive radii $\epsilon_{\text{sl}}, r_{\text{sl}}$, independent of small scale and datum, such that every $q_g + c$ with $\|c\|_{\mathbb{E}_g^1} < \epsilon_{\text{sl}}$ is gauge equivalent to a unique $q_g + h$ with*

$$d_g^{0,*}h = 0, \quad \|h\|_{\mathbb{E}_g^1} < r_{\text{sl}}.$$

The gauge transformation is unique for the full determinant-one gauge group. Its logarithm and the slice representative depend smoothly on c , with uniformly bounded first two derivatives; this includes two normalized parameter derivatives in the physical bundle charts. The slice is a local chart for the critical configuration quotient. At positive scale the maps are smooth on the corresponding strong Sobolev levels.

Proof. For endomorphisms of the rank-two factor, use the auxiliary graph norm

$$(119) \quad \|v\|_{\text{end},g} = \|\nabla_{A_g}^* \nabla_{A_g} v\|_{L^{\sharp,2}} + \|v\|_2.$$

Its smooth completion embeds in $C^0 \cap L_{2,A_g}^2$ and has $\nabla_{A_g} v \in L^4 \cap L^{2\sharp}$, with uniform bounds, by (113) and [Fee01, Lemmas 5.3–5.5]. The Laplacian product identity and the critical product inequalities make this a unital Banach algebra with a uniform multiplication constant. Its determinant-one unitary elements form the critical gauge group used below. On skew-Hermitian trace-free sections its norm is uniformly equivalent to $\mathbb{E}_g^0$, because V_g is bounded. The exponential and local logarithm are therefore smooth, with uniform derivative bounds on fixed small balls; compare [Fee01, Lemmas 7.2–7.3].

Use the action convention

$$(120) \quad u \cdot (A + a, \Phi + \phi) = (A + u^{-1}au + u^{-1}d_A u, u^{-1}(\Phi + \phi)).$$

For $h \in \ker d_g^{0,*}$ define

$$(121) \quad \Psi_g(\zeta, h) = e^\zeta \cdot (q_g + h) - q_g.$$

Its derivative at the origin is $(\zeta, h) \mapsto d_g^0 \zeta + h$, with uniformly bounded inverse

$$(122) \quad c \mapsto (G_g d_g^{0,*} c, H_g c).$$

We verify the nonlinear estimates in these same norms. Spatial differentiation of the connection terms in (120) uses the L_2^2 bounds on $u = e^\zeta$, the L^4 bounds on $d_{A_g} u$ and a , and the L_1^2 bound on a . Their divergence consists of conjugates of $d_{A_g}^* a$ and $\nabla_{A_g}^* \nabla_{A_g} u$, and contractions $d_{A_g} u \cdot a$ and $d_{A_g} u \cdot d_{A_g} u$. Here $d_{A_g}^* a = d_g^{0,*} h + \mathfrak{r}_{\Phi_g}^* \phi$. Every term is bounded in $L^{\sharp,2}$ by the critical products. The spinor terms use the bounded background spinor and its L^2 derivative; their contribution to the divergence is zero-order. Differentiating these multiplication maps gives a uniform C^2 bound on a fixed small ball and

$$\|D\Psi_g(\zeta, h) - D\Psi_g(0, 0)\| \leq C(\|\zeta\|_{\mathbb{E}_g^0} + \|h\|_{\mathbb{E}_g^1}).$$

Parameter derivatives differentiate the coefficients d_{A_g} , Φ_g , and $d_g^{0,*}$ and are controlled by (114). The horizontal domains have compatible local identifications: $H_g H_{g_0} + (1 - H_g)(1 - H_{g_0})$ is invertible on a small fixed parameter chart and maps $\text{ran } H_{g_0}$ to $\text{ran } H_g$. The quantitative inverse-function theorem applied to (122) gives a uniform local diffeomorphism, whose image contains a fixed ball in $\mathbb{E}_g^1$. The inverse and its first two derivatives, including normalized parameter derivatives, are uniformly bounded.

It remains to exclude identifications by gauges outside this exponential chart. Suppose $q_g + c_i$, $c_i = (a_i, \phi_i)$, have $\|c_i\|_{\mathbb{E}_g^1} < r$ and are related by any determinant-one gauge transformation

u . Equation (120) gives

$$(123) \quad d_{A_g} u = u a_2 - a_1 u, \quad (u - 1) \Phi_g = \phi_1 - u \phi_2.$$

Unitarity gives

$$\|d_{A_g} u\|_{L^4 \cap L^{2\sharp}} + \|d_{A_g} u\|_2 + \|(u - 1) \Phi_g\|_2 \leq C r.$$

For a determinant-one unitary matrix of rank two,

$$(124) \quad (u - 1)^* (u - 1) = \frac{1}{2} |u - 1|_{\text{HS}}^2 1.$$

Apply the scalar gap (110) to $|u - 1|_{\text{HS}}$ and use Kato's inequality. The spinor term is a fixed multiple of $\|(u - 1) \Phi_g\|_2$, so $\|u - 1\|_2 \leq C r$. Differentiating the first equation in (123) expresses the covariant Laplacian of u as $u d_{A_g}^* a_2 - (d_{A_g}^* a_1) u$ and contractions of $d_{A_g} u$ with a_i . The critical configuration norms control $d_{A_g}^* a_i$. Consequently

$$(125) \quad \|u - 1\|_{\text{end}, g} \leq C(r + r^2).$$

Thus $u - 1$ is small in C^0 and in the Banach algebra norm. Its logarithm satisfies $\|\log u\|_{\mathbb{E}_g^0} \leq C r$. The argument holds first for strong gauges and then, by the same weak product identities, for the critical gauge group.

If both c_i are horizontal, choose r_{sl} so small that $(\log u, c_1)$ and $(0, c_2)$ lie in the injectivity domain of Ψ_g . Their images agree; hence $u = 1$ and $c_1 = c_2$. Choose ϵ_{sl} smaller so that the local inverse sends its ball into this slice ball. This proves full-group uniqueness and freeness there. The group translates of the local product chart give the asserted quotient chart. For strong inputs at positive scale, regularity follows by the gauge-equation bootstrap of [Fee01, Lemma 8.1]: the additional spinor terms are smooth zeroth-order terms, and the small L^4 norm of $d_{A_g} u$ permits the same elliptic absorption. The strong smoothness assertion then follows from the inverse-function theorem at those levels. $\square$

5.3. Framed Coulomb modes and pregluing derivatives. We first make the Euclidean parameter modes explicit. Identify $\mathbb{R}^4$ with $\mathbb{H}$, put $r = |y|$ and $a(y) = y/r$, and use imaginary quaternion units with $[\tau_i, \tau_j] = 2\epsilon_{ijk} \tau_k$. The regular and exterior expressions for the unit ASD instanton are

$$(126) \quad I_{\text{reg}} = \frac{\text{Im}(\bar{y} dy)}{1 + r^2}, \quad I_{\text{ext}} = a I_{\text{reg}} a^{-1} - da a^{-1} = -\frac{\text{Im}(dy \bar{y})}{r^2(1 + r^2)}.$$

Both expressions are in radial gauge. Choose the relative-frame convention in which ρ acts on the exterior expression by $\rho^{-1} I_{\text{ext}} \rho$.

Lemma 5.5 (explicit framed Coulomb modes). *For $\theta \in \text{Im } \mathbb{H}$, define the adjoint section*

$$(127) \quad h_{\theta}^{\text{ext}} = \frac{r^2}{1 + r^2} \theta, \quad h_{\theta}^{\text{reg}} = \frac{\bar{y} \theta y}{1 + r^2}.$$

It is smooth, covariantly harmonic, and tends to θ in the exterior frame. The normalized translation, dilation, and relative-frame directions have Coulomb representatives

$$(128) \quad k_j = -\iota_{e_j} F_I, \quad k_s = -\iota_y F_I, \quad k_{\theta} = d_I h_{\theta}.$$

These modes form a basis of $\mathcal{K}_I^{\text{fr}}$. Their values are $O(r^{-3})$ and their covariant derivatives are $O(r^{-4})$ in the exterior frame.

Proof. The regular expression in (127) is smooth at zero. The exterior connection satisfies $\sum y^\mu I_\mu = 0$ and $\sum \partial_\mu I_\mu = 0$. The imaginary component rows of $dy \bar{y}$ are orthogonal with squared length r^2 . Since $\sum_i [\tau_i, [\tau_i, \theta]] = -8\theta$, it follows that

$$\sum_\mu [I_\mu, [I_\mu, \theta]] = -\frac{8}{r^2(1+r^2)^2} \theta.$$

For a radial scalar function f this gives

$$(129) \quad \nabla_I^* \nabla_I(f\theta) = -\left(f'' + \frac{3}{r}f' - \frac{8f}{r^2(1+r^2)^2}\right)\theta.$$

Taking $f = r^2/(1+r^2)$, both terms in parentheses are $8/(1+r^2)^3$, so the section is harmonic. Its exterior difference from θ is $O(r^{-2})$, giving the stated derivative decay.

Cartan's covariant formula and radial gauge give

$$(130) \quad -\partial_j I = k_j + d_I(-I_j), \quad -I - y \cdot \partial I = k_s.$$

The Yang–Mills equation and antisymmetry of F_I imply $d_I^* k_j = d_I^* k_s = 0$. Their self-dual derivatives vanish because translation and dilation preserve the ASD equation. For the frame modes, $d_I^* k_\theta = 0$ follows from harmonicity, and $d_I^+ k_\theta = [F_I^+, h_\theta] = 0$. To identify the frame sign, use a cutoff χ equal to zero toward the origin and one on an inner–outer overlap. On a fixed bundle the derivative of the framed family is $d_I \sigma$, where $\sigma_{\text{reg}} = \chi a^{-1} \theta a$ and $\sigma_{\text{ext}} = \theta$. Then $d_I \sigma - k_\theta = d_I(\sigma - h_\theta)$. The section $\sigma - h_\theta$ is smooth, is $O(r^{-2})$ at infinity, and has derivative $O(r^{-3})$. It belongs to the based homogeneous scalar space in (48), so this is exactly the framed Coulomb representative.

The curvature density is $C(1+r^2)^{-4}$, with $C > 0$. Its derivative under translation v and logarithmic dilation s is

$$\frac{8C\langle v, y \rangle + 4Cs(r^2 - 1)}{(1+r^2)^5}.$$

Gauge variations do not change it, so a relation among the eight modes first forces $v = s = 0$. If $d_I h_\theta = 0$, the section commutes with all curvature values, which span $\mathfrak{su}(2)$, and hence $\theta = 0$. Independence and Proposition 3.2 give the basis assertion. The decay also follows directly from (126); translations have the better orders r^{-4} and r^{-5} . $\square$

For background tangents use the pair Coulomb connection

$$(131) \quad \Omega_q(v) = (d_q^{0,*} d_q^0)^{-1} d_q^{0,*} dq(v)$$

and its induced connection on the gluing frame bundle, given by evaluation at x . These are smooth over compact background charts. Split parameter tangents using this connection and the fixed geometric connections on the center and orthonormal-frame bundles. Thus at any one parameter a background direction can be represented by a horizontal variation of q and a parallel relative frame. This is a pointwise choice of tangent representatives, not an assertion that the horizontal distribution is integrable. Let $j_g : T_g \mathcal{G}^+ \rightarrow \mathcal{K}_X(q) \oplus \mathcal{K}_I^{\text{fr}}$ use A_q on these background directions and (128) on the normalized bubble directions. This uses the same A_q as before: the background kernel configurations are horizontal, so their L^2 Gram retraction annihilates gauge derivatives. In particular the term in (19) is still retained. The map j_g has a uniformly bounded inverse.

Proposition 5.6 (pregluing derivatives). *The normalized splicing derivatives satisfy*

$$(132) \quad R_g H_g dq_g = j_g + o(1),$$

with uniformly bounded inverse for j_g . In compatible local trivializations the tubular map

$$(133) \quad \mathcal{T}(g, h) = [q_g + h], \quad h \in \mathcal{C}_g^{\text{sl}},$$

has uniform second-derivative bounds. The tubular maps are continuous in the critical configuration norm, smooth on positive-scale strong Sobolev charts, and equivariant under the frame actions.

Proof. Fix N . In Lemma 3.5 take the transition annulus inside the pure-instanton region, with outer radius less than $r_0/(16N)$, where $r_0 = \sqrt{\lambda_0}$. Its radius ratios are fixed. After reducing the parameter chart it stays in that region; the spinor is zero there. All previous transport estimates remain valid. For each unit normalized bubble direction v , we claim that a smooth global infinitesimal gauge ζ_v gives

$$(134) \quad dq_g(v) = d_g^0 \zeta_v + \beta_I T_\lambda k_v + e_v, \quad \|e_v\|_{\mathbb{E}_g^1} \leq C_N \lambda^{1/3}.$$

We verify the critical divergence as well as the first-order norm.

For translation, take $\zeta_j = -\lambda(I_\lambda)_j$ in the regular core and cut it off to zero at radius comparable to r_0 , before the outer boundary of the inner chart. Equation (130) gives the core identity. On the cutoff annulus the gauge has size $O(\lambda/r_0)$; its first two spatial derivatives have sizes $O(\lambda/r_0^2)$ and $O(\lambda/r_0^3)$. The normalized center derivative of the transition logarithm is $O(\lambda/r_0)$ and gives the same connection bounds. The remaining annular errors therefore have value $O(\lambda/r_0^2)$ and covariant derivative $O(\lambda/r_0^3)$. Their first-order norm and critical divergence are $O(\lambda/r_0) = O(\sqrt{\lambda})$ by shrinking support.

For dilation, the angular transition is scale independent and the core variation is already $T_\lambda k_s$. At the connection cutoff the exterior instanton and its normalized scale derivative have size $O(\lambda^2/r_0^3)$, with spatial derivative $O(\lambda^2/r_0^4)$. Their annular errors have critical configuration norm $O(\lambda^2/r_0^2) = O(\lambda)$.

For a frame direction, the transition is $a^{-1}\rho$. The inner frame change identifying it with a^{-1} differentiates to $\sigma_{\text{reg}} = \chi a^{-1}\theta a$, while $\sigma_{\text{ext}} = \theta$. In the uncut instanton region put $\zeta_\theta = \sigma - h_\theta(z/\lambda)$. The potentially nonvanishing annular term cancels exactly:

$$d_I \sigma - d_I \zeta_\theta = T_\lambda k_\theta.$$

The inner expression is smooth at the origin. In the outer frame $\zeta_\theta = \lambda^2(\lambda^2 + r^2)^{-1}\theta$. Cut off this expression to zero at radius comparable to r_0 . Across the connection cutoff, the unmatched terms have value $O(\lambda^2/r_0^3)$ and spatial derivative $O(\lambda^2/r_0^4)$, hence critical configuration norm $O(\lambda)$. Background connection terms are smaller: in radial gauge they are $O(r_0)$ and multiply $\zeta_\theta = O(\lambda^2/r_0^2)$. The spinor vanishes on the support of this gauge.

These comparisons include the physical metric identifications. On the core the metric error is $O(r^2)$ with derivative $O(r)$. Using $r^2 \leq C_N \lambda$, the mode decay, and $\nabla_I k \in L^2 \cap L^\sharp$ bounds their first-order and critical-divergence errors by $O_N(\lambda)$. Normalized derivatives of the smooth coordinate and frame coefficients give additional values bounded by $C\lambda/(\lambda + r)$, with spatial derivative $C\lambda/(\lambda + r)^2$ on a fixed physical ball. The latter profile has L^2 norm $O(\lambda\sqrt{1 + |\log \lambda|})$ and $L^\sharp$ norm $O(\lambda(1 + |\log \lambda|))$, both $o(\lambda^{1/3})$. Radial monotonicity bounds the weighted supremum. At the spinor cutoff radius $R = \lambda^{1/3}$, scale differentiation gives value $O(1)$ and spatial derivative $O(R^{-1})$; center differentiation gives $O(\lambda/R)$ and $O(\lambda/R^2)$, respectively. These give first-order errors $O(R)$ and $O(\lambda)$; the spinor contribution to the gauge row is zero-order. This proves (134).

For a background direction represented horizontally as above, put $w = dq(v)$. Its pregluing derivative satisfies

$$(135) \quad dq_g(v) = \beta_X T_{X,g} w + e_v \pmod{\text{im } d_g^0}, \quad \|e_v\|_{\mathbb{E}_g^1} \leq C_N \lambda^{1/3}.$$

Indeed, the inner model is independent of the background. With the frame parallel for (131), the derivative of a background radial-gauge frame is zero at x and $O(r)$ nearby. Its cutoff extension gives a connection error $O(1)$ with spatial derivative $O(r_0^{-1})$ on the r_0 -annulus. The removed smooth background modes have the same bounds, giving $O(r_0)$ in the first-order norm and critical divergence. The removed spinor gives $O(R)$ in the first-order norm and $O(R^2)$ in the critical gauge row. Outside these balls the variation is exactly w . Equivariance of pregluing, with the induced action on relative frames, shows that changing the background representative adds a global gauge derivative. Without the induced frame connection a finite frame term would remain, so the tangent convention is needed here.

The same smooth-mode and escaping-tail estimates used in the proof of Proposition 3.4 give

$$(136) \quad \|d_g^{0,*}(\beta_I T_\lambda k)\|_{L^{\sharp,2}} = o(1), \quad \|d_g^{0,*}(\beta_X T_{X,g} w)\|_{L^{\sharp,2}} = o(1).$$

For bubble modes the cutoff gauge row is $O_N(\lambda)$ by the r^{-3} decay, and the metric terms have the bounds above. For background modes the cutoff row is $O_N(\sqrt{\lambda})$, the comparison coefficients tend to zero, and $d_g^{0,*}w = 0$. Thus Proposition 5.2 shows that H_g changes these cut-off modes by $o(1)$ in $\mathbb{E}_g^1$ and removes the exact gauge terms. The Gram retractions give

$$R_g(\beta_I T_\lambda k) = (0, k) + o(1), \quad R_g(\beta_X T_{X,g} w) = (r_X w, 0) + o(1) = (A_q v, 0) + o(1).$$

The diagonal and off-diagonal estimates are those in (71). This proves (132) uniformly over the compact data.

It remains to bound the tubular map in compatible quotient charts. Put $\ell_g = \text{pr}_{\mathbb{X}} \tilde{S}_g$. The identities $R_g \ell_g = 1$ and $d_g^{0,*} \ell_g = 0$ imply

$$(137) \quad \begin{aligned} d_g^{0,*} \partial_i \ell_g &= -(\partial_i d_g^{0,*}) \ell_g, \\ d_g^{0,*} \partial_j \partial_i \ell_g &= -(\partial_j \partial_i d_g^{0,*}) \ell_g - (\partial_i d_g^{0,*}) \partial_j \ell_g - (\partial_j d_g^{0,*}) \partial_i \ell_g. \end{aligned}$$

The differentiated gauge rows are bounded critical multipliers on L_1^2 . Together with Proposition 3.6, these equations upgrade the two derivatives of ℓ_g to $\mathbb{E}_g^1$ bounds. Difference quotients converge there because they converge in L_1^2 and their divergences converge in $L^{\sharp,2}$ by these identities. Consequently

$$(138) \quad Q_g^{\text{sl}} = H_g - \ell_g R_g H_g$$

is a projection onto $\mathcal{C}_g^{\text{sl}}$ with two uniform derivatives. The map $U_g = Q_g^{\text{sl}} Q_{g_0}^{\text{sl}} + (1 - Q_g^{\text{sl}})(1 - Q_{g_0}^{\text{sl}})$ is invertible on a fixed small chart and identifies the corresponding complements.

Equation (114) also bounds the first two derivatives of $q_g - q_{g_0}$ in $\mathbb{E}_{g_0}^1$: the connection divergence is a contraction of $\nabla_{A_g} a_\alpha$, and its spinor term is zero-order. If s_0 is the slice-representative map at q_{g_0} , the tubular map in that fixed slice is

$$(139) \quad (g, v) \mapsto s_0(q_g - q_{g_0} + U_g v), \quad v \in \mathcal{C}_{g_0}^{\text{sl}}.$$

Its argument stays in a fixed slice ball for small fixed parameter and complement radii. The chain rule and Proposition 5.4 prove the uniform second-derivative bound. Positive-scale strong smoothness follows from the same constructions and elliptic regularity. The definitions give continuity and equivariance. $\square$

5.4. Uniform parameter recovery. Local parameter charts are taken in a slightly larger background disk and scale interval; their artificial outer boundaries are fixed only after the following local construction.

Lemma 5.7 (uniform modulation). *There are constants $c_{\text{mod}}, r_{\text{mod}} > 0$ independent of small scale such that (133) is one-to-one for*

$$(140) \quad d_{\text{par}}(g, g') < c_{\text{mod}}, \quad \|h\|_{L_1^2} + \|h'\|_{L_1^2} < r_{\text{mod}}.$$

Its image contains a uniform neighborhood in the critical topology. The recovered parameter is characterized, in the Coulomb representative, by

$$(141) \quad R_g((A, \Phi) - q_g) = 0.$$

Proof. The differential along the zero section is

$$(142) \quad D\mathcal{T}_{(g,0)}(\dot{g}, \dot{h}) = H_g dq_g(\dot{g}) + \dot{h}.$$

Apply R_g and use (132) to recover $\dot{g}$; then recover $\dot{h}$ by subtraction. This gives a uniformly bounded inverse. Proposition 5.6 and the slice estimates allow the quantitative inverse-function theorem with a uniform radius. On the horizontal complement the critical and L_1^2 norms agree, which gives (140). $\square$

The correction in Proposition 4.2 is horizontal and lies in $\ker R_g$. Hence modulation recovers its input parameter exactly.

5.5. Global recovery of center, scale, and frame. For a framed charge-one ASD connection on $\mathbb{R}^4$, the model center and scale are

$$(143) \quad z[A] = \frac{1}{8\pi^2} \int_{\mathbb{R}^4} y |F_A|^2, \quad \sigma[A]^2 = \frac{1}{8\pi^2} \int_{\mathbb{R}^4} |y - z[A]|^2 |F_A|^2.$$

They identify translations and dilations of the charge-one instanton. The analytic scale λ is the dilation parameter in $I_\lambda(z) = \lambda^{-1}I(z/\lambda)$, so $\sigma[I_\lambda] = \lambda\sigma[I]$. For the corrected configurations we use only bounded local observables: a fixed fractional-energy radius, a first moment truncated in normalized coordinates, and convergence of the rescaled framed connection on compact sets. Proposition 4.2 implies that these observables tend to those of the input instanton. We do not apply the unbounded global moments in (143) directly to the corrected connection.

Lemma 5.8 (relative-frame recovery). *Let $q_{g_n} + t_n u_{g_n}$ and $q_{g'_n} + t'_n u_{g'_n}$ be gauge-equivalent pairs from Proposition 4.2, where $t_n, t'_n \in [0, 1]$ and both scales tend to zero. Suppose their background distance, center displacement divided by λ_n , and logarithmic scale ratio tend to zero. Use the same exterior background normalization and smooth local frame conventions. Then their relative frames tend to the same element of $\text{SO}(3)$. This assertion is uniform in t_n, t'_n .*

Proof. The local elliptic estimates following Proposition 4.2 give smooth convergence to the background on exterior compact sets and to the regular-gauge instanton on fixed rescaled compact sets. These conclusions remain true after multiplying the correction by any $t \in [0, 1]$; no equation for the interpolated pair is needed.

First align the second background, center, and scale with the first, leaving its relative frame unchanged in the chosen local conventions. Lemma 3.5 and Proposition 5.6 change the preglued pair by $o(1)$ in the critical configuration norm. Differences of the covariant derivatives and pair divergences are bounded zero-order critical multipliers. Thus the transported

corrections also remain $o(1)$ in that norm. These identifications preserve smooth endpoint convergence: they fix the exterior frame, are the identity toward the regular core, and the normalized center and scale changes tend to zero. We may therefore compare pairs with common q_n, x_n, λ_n , frames $\rho_{1,n}, \rho_{2,n}$, and errors $c_{i,n} = (a_{i,n}, \phi_{i,n})$ satisfying

$$\epsilon_n := \sum_{i=1}^2 \|c_{i,n}\|_{\text{crit}, g_{i,n}} \longrightarrow 0.$$

These errors need not remain exactly horizontal after alignment.

Pass to a subsequence with convergent frames and choose local $\text{SU}(2)$ lifts. A smooth determinant-one gauge v_n relating the pairs has subsequential smooth limits on all endpoint compact sets. Indeed its first-order connection equation and unitarity bound all its derivatives there. The exterior limit stabilizes the background pair and is the identity because its spinor is nonzero. The inner regular-gauge limit stabilizes I , and hence is $s \in \{1, -1\}$ by irreducibility. With $a(y) = y/|y|$, the two transitions are $a^{-1}\rho_i$, so the exterior matrix of the inner limiting gauge is the constant

$$(144) \quad c = \rho_1^{-1} a s a^{-1} \rho_2 = s \rho_1^{-1} \rho_2.$$

We show that $c = 1$.

Fix large R and small d , and set $\Omega_n = B(x_n, d) \setminus B(x_n, R\lambda_n)$. In its common exterior trivialization write the total connection forms as $b_i = A_i^{\text{pre}} + a_i$. The smooth background is in radial gauge. The preglued forms are radial-cutoff combinations of this background and $\rho_{i,n}^{-1} I_{\lambda_n}^{\text{ext}} \rho_{i,n}$. Formula (126) gives, for $r \geq R\lambda_n$,

$$|I_{\lambda_n}^{\text{ext}}| \leq C\lambda_n^2 r^{-3}, \quad |\partial I_{\lambda_n}^{\text{ext}}| \leq C\lambda_n^2 r^{-4}.$$

The flat divergence and radial contraction vanish. Gauss's lemma makes the contraction with the gradient of a radial cutoff vanish also for the physical metric. Since its normal-coordinate coefficients satisfy $g^{ij} - \delta^{ij} = O(r^2)$ and $\partial g = O(r)$, radial integration yields

$$(145) \quad \begin{aligned} \|A_i^{\text{pre}}\|_{L^4(\Omega_n) \cap L^{2\sharp}(\Omega_n)} &\leq C(d^2 + R^{-2}), \\ \|d^* A_i^{\text{pre}}\|_{L^\sharp(\Omega_n)} &\leq Cd^2 + C\lambda_n^2(1 + |\log \lambda_n|). \end{aligned}$$

Here d^* is the physical scalar codifferential on matrix-valued forms. Restricted critical norms use zero extension of the absolute coefficient; the divergence itself is taken before restriction. The bounds hold with the supremum over all points of X : dominate the exterior profiles by radially decreasing functions constant on $r < R\lambda_n$. Their weighted integrals are largest at the center, up to uniform coordinate constants. The weighted square integral of the potential is $O(R^{-4})$, while the metric divergence error $O(\lambda_n^2/r^2)$ contributes $O(\lambda_n^2(1 + |\log \lambda_n|))$. The background contributes $O(d^2)$.

Kato's inequality and Hardy's inequality bound $\|a_i\|_{L^4 \cap L^{2\sharp}}$ by $C\epsilon_n$. Expressing the ordinary divergence through the pair divergence adds contractions of $A_i^{\text{pre}} a_i$ and $\Phi_i^{\text{pre}} \phi_i$. The preglued spinor is uniformly bounded, with restricted $L^{2\sharp}$ norm at most Cd . The critical product inequality gives

$$(146) \quad \|d^* a_i\|_{L^\sharp(\Omega_n)} \leq C\epsilon_n(1 + d + d^2 + R^{-2}).$$

This is where control of the critical divergence, in addition to the first-order norm, enters the comparison.

The gauge equation is $dv_n = v_n b_2 - b_1 v_n$. A fixed determinant connection contributes a scalar matrix and cancels from this equation. Unitarity gives $\|dv_n\|_{L^{2\sharp}(\Omega_n)} \leq C\mu_n$, where

$\mu_n = d^2 + R^{-2} + \epsilon_n$. Applying d^* to the equation expresses Δv_n as $v_n d^* b_2 - (d^* b_1) v_n$ and contractions of dv_n with b_1, b_2 . Consequently,

$$(147) \quad \|\Delta v_n\|_{L^\sharp(\Omega_n)} \leq C(d^2 + \lambda_n^2(1 + |\log \lambda_n|) + \epsilon_n(1 + d + d^2 + R^{-2}) + \mu_n^2).$$

Extend v_n as a matrix-valued function by setting it equal to c on the inner ball and to 1 outside the outer ball, using linear cutoffs on $R\lambda_n < r < 2R\lambda_n$ and $d/2 < r < d$. This extension need not take values in $\text{SU}(2)$. For fixed R, d , smooth endpoint convergence makes its additional Laplacian terms $o(1)$ in $L^\sharp$: an inner cutoff error has size $o(1)/(R\lambda_n)^2$ on an annulus of radius $R\lambda_n$, and the outer error has the analogous fixed-scale bound. Subtract 1 and extend by zero to obtain a global function w_n with values in a trivial matrix bundle. It satisfies $\|w_n\|_2 \leq Cd^2$. The scalar case of [Fee01, Lemma 5.3(2)] gives

$$(148) \quad \|w_n\|_{C^0} \leq C(\|\Delta w_n\|_{L^\sharp} + \|w_n\|_2).$$

The metric in this estimate is the fixed physical metric throughout. Taking $n \rightarrow \infty$ in (147) and (148), with R, d fixed, gives

$$|c - 1| \leq C(d^2 + (d^2 + R^{-2})^2).$$

Let $d \downarrow 0$ and then $R \rightarrow \infty$. Thus $c = 1$, and (144) identifies the two $\text{SO}(3)$ frames. Every subsequence of frames has the same conclusion, which proves the lemma. All estimates are uniform in the interpolation parameters. $\square$

Proposition 5.9 (separation of parameters). *Suppose g_n and g'_n have scales tending to zero and $\gamma(g_n) = \gamma(g'_n)$. Then their background points and ideal centers agree in the limit. For positive scales, the normalized center displacement tends to zero, the scale ratio tends to one, and the relative-frame distance in $\text{SO}(3)$ tends to zero. Consequently the map is one-to-one after decreasing the maximum scale.*

Proof. To prove injectivity after decreasing the maximum scale, argue by contradiction with a sequence of failed pairs whose maximum scales tend to zero. Thus $\lambda_n, \lambda'_n \rightarrow 0$. The assertion needed here is this small-scale separation; a local inverse at fixed positive scale would not by itself prove global injectivity. Uhlenbeck convergence first identifies the two background limits and ideal points. The fixed background slices then give $d_N(q_n, q'_n) \rightarrow 0$.

Rescale the common corrected configuration about x_n by λ_n . The first representation converges on compact sets to the centered unit-scale instanton I . In these coordinates, the second bubble has relative translation and scale

$$(149) \quad z_n = \lambda_n^{-1} \exp_{x_n}^{-1}(x'_n), \quad t_n = \lambda'_n / \lambda_n.$$

The curvature difference from either preglued connection tends to zero in L^2 , so their curvature measures have the same limit in these coordinates. Each preglued curvature measure differs by $o(1)$ in total variation from the sum of its background measure and its localized instanton measure. Indeed the connection-cutoff errors and the omitted instanton tail have L^2 norm $O_N(\lambda)$, the removed background curvature has vanishing norm, and the metric comparison on the shrinking ball tends to the Euclidean one. The same bounds hold with λ'_n for the second description. A translated and dilated instanton has density $Ct_n^4/(t_n^2 + |y - z_n|^2)^4$. If $|z_n| \rightarrow \infty$, its energy on any fixed compact set tends to zero, uniformly in t_n : the density there is at most $C/(|z_n| - K)^4$ on $|y| \leq K$. If $t_n \rightarrow \infty$ with bounded z_n , the density also tends to zero. If $t_n \rightarrow 0$ with bounded z_n , a subsequence of its curvature measures tends to a point mass. Each alternative contradicts the smooth unit-scale limit. Thus, after a subsequence, $z_n \rightarrow z \in \mathbb{R}^4$ and $t_n \rightarrow t \in (0, \infty)$.

The two rescaled limits are gauge equivalent, and the second is the translate and dilate of I by (z, t) . Applying (143) to these exact limiting instantons gives $z = 0$ and $t = 1$. Lemma 5.8, with the same exterior background normalization, then identifies their relative $\mathrm{SO}(3)$ frames modulo exactly the central and diagonal actions already divided out. Consequently

$$(150) \quad d_{\mathrm{par}}(g_n, g'_n) \longrightarrow 0.$$

For large n , the two parameters lie within the uniform modulation radius. Put the common configuration in the Coulomb slice and apply Lemma 5.7; uniqueness gives literal equality. At zero scale the cone quotient has already identified all relative frames. $\square$

Because the compactified parameter space over the compact background is compact after the outer boundary is fixed, sequential separation also rules out coincidences between parameters that are not initially close. We have therefore proved the following statement.

Corollary 5.10 (compactified embedding). *After decreasing ε and δ , γ is an equivariant topological embedding of the compactified virtual gluing space. On the positive-scale stratum it is a smooth embedding.*

Proof. Separation and uniform modulation give injectivity for a uniform small scale bound. Continuity at zero scale and compactness of the domain then give a topological embedding into the Hausdorff Uhlenbeck space. At positive scale the tubular coordinates identify the corrected map with the smooth graph $g \mapsto (g, u_g)$, so it is a smooth embedding. $\square$

Finally write $\gamma(g) = q_g + u_g$ in the chosen Coulomb slice. For $0 \leq t \leq 1$, set

$$(151) \quad \gamma_t(g) = q_g + t u_g.$$

The correction remains horizontal and in $\ker R_g$ after multiplication by t , and $\|t u_g\|_{\mathrm{crit}, g} \leq C\lambda^{1/3}$. Lemma 5.8 supplies frame recovery uniformly in t ; Lemma 5.7 supplies the uniform local uniqueness. The parameter-separation argument therefore applies uniformly in t . Hence every γ_t is a compactified embedding. The construction is equivariant, γ_0 is the splicing map, and $\gamma_1 = \gamma$.

6. REVERSE GLUING AND NEIGHBORHOOD SURJECTIVITY

Continuity and injectivity do not imply that every nearby monopole is obtained by gluing. We prove the converse by first putting an arbitrary one-bubble sequence close to a preglued pair and then applying the same complement and inverse used in the construction.

Proposition 6.1 (strong splicing approximation). *Suppose that irreducible monopoles $[A_n, \Phi_n]$ converge in the Uhlenbeck topology to*

$$([B, \Psi], x) \in M_{\mathfrak{s}} \times X$$

with exactly one unit of instanton charge lost at x . There are coarse gluing data

$$g_n^0 = ([B, \Psi], x_n, \lambda_n, \rho_n), \quad x_n \rightarrow x, \quad \lambda_n \downarrow 0,$$

standard gluing bundle identifications, and determinant-one gauges for which

$$(152) \quad \|(A_n, \Phi_n) - q_{\lambda_n}(g_n^0)\|_{\mathrm{crit}, g_n^0} \longrightarrow 0.$$

The relative frames ρ_n use the quotient convention in (23); no auxiliary tangent or bundle frame remains in the datum.

6.1. Extraction of the bubble. The monopole Weitzenböck formula gives a uniform L^∞ bound for Φ_n on the fixed compact perturbation family. Choose x_n and λ_n by a fixed fractional-energy centering and radius convention in a normal-coordinate ball about x . Pull back by $y \mapsto \exp_{x_n}(\lambda_n y)$ and choose Uhlenbeck gauges. The rescaled curvature equation has the form

$$(153) \quad F_{A_n^{\text{res}}}^+ = \lambda_n^2 \mathcal{Q}(\Phi_n^{\text{res}}) + \lambda_n^2 \mathcal{P}_n,$$

where $\mathcal{P}_n$ is bounded on each fixed normalized ball. The right side tends to zero. Uhlenbeck compactness and removal of the singularity at infinity therefore give a finite-energy ASD connection I on S^4 . The lost charge is one, so I has charge one. The centering and radius conventions make it centered and of unit scale; the framed charge-one classification identifies the remaining parameter with a framed rotation $\rho_n \in \text{SO}(3)$. We choose this rotation by comparing the rescaled bubble gauge with the outer background gauge through the annular patching normalization below; this is equivalent to the framed classification and prevents a second, unrecorded frame from entering the gluing datum.

After a subsequence,

$$(154) \quad \begin{aligned} A_n^{\text{res}} &\longrightarrow I && \text{strongly in } L_1^2 \text{ on compact subsets of } \mathbb{R}^4, \\ (A_n, \Phi_n) &\longrightarrow (B, \Psi) && \text{strongly in } L_1^2 \text{ on compact subsets of } X \setminus \{x\}. \end{aligned}$$

The conformally rescaled spinor tends to zero on the bubble side.

Lemma 6.2 (one-bubble no neck). *For the centers and scales above,*

$$(155) \quad \lim_{R \rightarrow \infty} \lim_{d \downarrow 0} \limsup_{n \rightarrow \infty} \int_{B(x_n, d) \setminus B(x_n, R\lambda_n)} |F_{A_n}|^2 = 0.$$

Proof. The curvature measures converge to $|F_B|^2 dV + 8\pi^2 \delta_x$. For fixed d , the exterior accounts for the background energy. The rescaled ball $B(0, R)$ accounts for the corresponding part of the charge-one instanton's energy. The remaining background energy tends to zero as $d \downarrow 0$, and the instanton tail tends to zero as $R \rightarrow \infty$. Subtracting these contributions proves the assertion. The same conclusion holds on any fixed-ratio enlargement of the annulus. $\square$

6.2. Local regularity and annular gauges. We first obtain gauges on the complete neck with small local norms. Summability will follow from the curvature equation in a further gauge. All annuli and overlaps below have fixed shapes after dilation, and local estimates are applied on slightly enlarged sets.

Lemma 6.3 (local neck regularity). *For the genuine monopoles under consideration, let r be a sufficiently small coordinate-annulus radius and normalize by $z = ry$. On a fixed normalized ball with sufficiently small curvature energy, there is a determinant-one gauge in which the trace-free connection form a and the spinor $\psi = r\Phi$ satisfy, on a smaller ball,*

$$(156) \quad \|a\|_{C^k} \leq C_k(\|F_A\|_{L^2} + r^2), \quad \|\psi\|_{C^k} \leq C_k r.$$

Here k is any fixed finite integer. The constants are uniform in the center, radius, and fixed compact perturbation family.

Proof. Use the Euclidean coordinate metric in the ball gauge theorem [Uhl82, Theorem 2.1]. Curvature norms for it and the normalized geometric metric are uniformly comparable. The resulting gauge has $d_\delta^* a = 0$ and $\|a\|_{W^{1,2}} \leq CE$, where $E = \|F_A\|_{L^2}$. The geometric metric

differs from δ by $O(r^2)$ with any fixed number of normalized derivatives. The fixed determinant and spin connections are smooth coefficients. Constant dilation of the monopole equations, with $\psi = r\Phi$, gives $|\psi| \leq Cr$ and

$$(157) \quad Q_\delta a + K_r \nabla a + B_a a = s_r, \quad Q_\delta = d_\delta^* \oplus d_\delta^+, \quad \|K_r\|_{C^k} \leq C_k r^2, \quad |B_a| \leq C|a|, \quad \|s_r\|_{L^\infty} \leq Cr^2.$$

The term $B_a a$ is the quadratic curvature term, viewed as linear in one factor. The spinor and smooth determinant terms give s_r .

Choose a finite chain of nested balls, with the number depending only on k . For an interior cutoff χ , extend $v = \chi a$ and the cutoff coefficients to $\mathbb{R}^4$. The right side of its equation is in $L^2 \cap L^3$ with norm at most $C(E + r^2)$: the additional terms are cutoff derivatives times $a \in L^4$. The extended B_a has L^4 norm $O(E)$. The constant-coefficient inverse Q_δ^{-1} maps L^s into $\dot{W}^{1,s}$ for $s = 2, 3$. Sobolev multiplication therefore bounds $Q_\delta^{-1}(K_r \nabla + B_a)$ on either space by $C_s(r^2 + E)$. Choose the thresholds to make both norms less than one half. The same Neumann series converges in both spaces. Uniqueness in $\dot{W}^{1,2}$ identifies it with χa , proving

$$\|a\|_{W^{1,3}} + \|a\|_{L^{12}} \leq C(E + r^2)$$

on the first smaller ball.

The interior Dirac estimate now gives $\|\psi\|_{W^{1,12}} \leq Cr$, because the connection term is $a\psi$ and $|\psi| \leq Cr$. The curvature equation has its quadratic term in L^6 , so the interior estimate for (d_δ^*, d_g^+) gives $\|a\|_{W^{1,6}} \leq C(E + r^2)$. These operators are uniformly elliptic in the normalized metrics. Since $W^{1,6}$ is an algebra in dimension four, differentiation of the two equations and interior estimates on the remaining nested balls prove (156). The spinor estimates retain the factor r , and their quadratic curvature sources retain r^2 . This argument uses constant dilation on each ball; it does not identify that spinor normalization with a variable conformal weight. $\square$

Lemma 6.4 (patching annular gauges). *Suppose a chain of overlapping normalized annuli has local connection forms with $W^{1,p}$ norms ϵ_j , where $2 < p < 4$ and $\sup_j \epsilon_j$ is sufficiently small. There is one gauge in which the connection form on the j th slab has norm bounded by $C(\epsilon_{j-1} + \epsilon_j + \epsilon_{j+1})$, allowing a fixed number of adjacent terms for enlarged slabs. Thus both the ℓ^1 and ℓ^∞ bounds are independent of chain length. The outer frame can be fixed, with one constant rotation retained at the inner end. The same assertion holds in any fixed higher regularity supplied by Lemma 6.3.*

Proof. For adjacent forms a, b , a transition v satisfies

$$(158) \quad dv = vb - av.$$

Put $q = 4p/(4 - p) > 4$. Unitarity and Sobolev embedding give $\|dv\|_{L^q} \leq C(\|a\|_{W^{1,p}} + \|b\|_{W^{1,p}})$. Hence v is continuous. Differentiating the equation gives $\nabla^2 v$ as a sum of $v\nabla b$, $(\nabla a)v$, $(dv)b$, and $a(dv)$. Since $q/2 \geq p$, these terms are bounded in L^p . Poincaré and Morrey estimates on the fixed connected overlap give a constant $c \in \text{SU}(2)$ such that

$$(159) \quad \|v - c\|_{W^{2,p}} + \|v - c\|_{C^0} \leq C(\|a\|_{W^{1,p}} + \|b\|_{W^{1,p}}).$$

The right side is small enough to take the logarithm of vc^{-1} . Its $W^{2,p}$ norm has the same bound. No embedding $W^{2,2} \hookrightarrow C^0$ is used.

If $e_{j+1} = e_j v_j$ are the local frames, put $s_1 = 1$ and $s_{j+1} = c_j^{-1} s_j$. The frames $e'_j = e_j s_j$ have transitions

$$v'_j = s_j^{-1} (v_j c_j^{-1}) s_j$$

close to 1 with the same bounds. Products of constant unitary matrices cause no norm growth. On disjoint interpolation regions use the frame $e'_j \exp(\chi_j \log v'_j)$, where χ_j runs from zero to one with fixed normalized derivative bounds. It agrees with the adjacent frames near its ends. Differentiating the gauge action gives the asserted local estimate. Each nonconstant correction is applied only on its own overlap. The only accumulated transformation is the constant s_j .

To obtain one small frame on each normalized annulus from the ball gauges of Lemma 6.3, use a fixed finite good cover of $S^3 \times (1, 2)$. The transition equation gives C^{k+1} bounds modulo constants on its overlaps. Normalize those constants on a spanning tree of the nerve. The nerve is simply connected; fix fillings by triangles for the finitely many remaining edge loops. The cocycle identity on a triple overlap bounds each constant triangle product by $1 + O(\epsilon)$. The fixed fillings therefore make all normalized transitions $1 + O(\epsilon)$. Using a determinant trivialization, view the local frames as unit sections of a quaternionic line. Their partition-of-unity average is nonzero, since in any local frame it is $1 + O(\epsilon)$. Normalize this average to obtain a global unit section and hence an $SU(2)$ frame. The finite construction preserves the derivative bounds. Apply the preceding chain argument to these annular frames. $\square$

6.3. Coulomb gauge on spheres and summable decay. Write $t = \log r$ on the neck. Apply the preceding lemmas on a fixed-ratio enlargement, such as $R\lambda_n/4 < r < 4d$. The no-neck energy gives, on $R\lambda_n < r < d$, one smooth gauge with

$$(160) \quad \mathcal{A} = \alpha_0 dt + a_0, \quad \sup_j \|\mathcal{A}\|_{C^k(D_j^+)} \leq C\epsilon,$$

for a fixed sufficiently large k , say $k = 8$. Here D_j^+ are fixed enlarged cylinder slabs, and ϵ tends to zero in the iterated limit of (155), with an additional $O(d^2)$ term. We now improve the supremum bound to a sum bound.

Lemma 6.5 (sphere Coulomb gauge). *Under (160), a further determinant-one gauge gives $\mathcal{A} = \alpha dt + a$ with*

$$(161) \quad d_{S^3}^* a = 0, \quad \int_{S^3} \alpha = 0.$$

The resulting connection has uniformly small C^2 slab norms. The gauge consists of a uniformly small transformation and a path of spatially constant rotations, normalized at the outer end. Its derivative bounds are independent of cylinder length.

Proof. At each t , solve

$$d_{S^3}^*(u^{-1}a_0u + u^{-1}d_{S^3}u) = 0, \quad u = e^\xi, \quad \int_{S^3} \xi = 0.$$

The derivative at zero is the scalar Laplacian on mean-zero adjoint functions on the fixed round sphere. Apply the implicit-function theorem in high Sobolev spaces. Its inverse and differentiated equations give the required spatial and time derivative bounds; the initial C^8 control permits these derivatives and traces. All estimates are on the same sphere and have no length parameter. Let β be the temporal part after this gauge and let $\bar{\beta}$ denote its spherical average. Solve

$$h' = -\bar{\beta}h, \quad h(t_+) = 1.$$

Gauge by the spatially constant h . The temporal part becomes $h^{-1}(\beta - \bar{\beta})h$, while the tangential Coulomb condition is preserved. Unitarity bounds h and the equation bounds its derivatives locally from those of $\bar{\beta}$. The endpoint $h(t_-)$ is retained as a relative rotation. $\square$

In this gauge, the curvature equation, expressed using the round product metric and the orientation $dt \wedge \text{vol}_{S^3}$, is

$$(162) \quad \partial_t a - d_{S^3} \alpha + *_{S^3} d_{S^3} a = f - [\alpha, a] - *_{S^3} (a \wedge a).$$

The forcing satisfies, for any fixed $2 < p < 4$,

$$(163) \quad \|f\|_{W^{1,p}(D_j^+)} \leq C r_j^2, \quad \sum_j \|f\|_{W^{1,p}(D_j^+)} \leq C d^2.$$

Indeed, the normalized monopole source and its first derivative are $O(r_j^2)$ by Lemma 6.3. In normal polar coordinates $r^{-2}g$ differs from the round product metric by $O(r^2)$, with the same derivative bounds. The difference of the self-duality projections applied to curvature therefore contributes $O(\epsilon r_j^2)$ with one derivative, using the already established local C^2 connection bounds. Smooth determinant and perturbation coefficients obey the same estimates. The gauges in Lemma 6.5 preserve these bounds under conjugation. Finally sum the geometric series in r_j^2 .

Lemma 6.6 (summable neck estimate). *Let e_-, e_+ be the $W^{1,p}$ norms of the connection on fixed enlarged endpoint slabs. In the gauge (161),*

$$(164) \quad \sum_j \|\alpha dt + a\|_{W^{1,p}(D_j)} \leq C(e_- + e_+ + d^2),$$

with C independent of the neck length.

Proof. Write $N = f - [\alpha, a] - *_{S^3} (a \wedge a)$. Applying $d_{S^3}^*$ to (162) gives $-\Delta_{S^3} \alpha = d_{S^3}^* N$. The mean-zero inverse composed with $d_{S^3}^*$ maps L^s to $W^{1,s}$ on the sphere. Absorbing the term with coefficient a gives, for $s = 2, p$,

$$(165) \quad \|\alpha(t)\|_{W^{1,s}(S^3)} \leq C \|f(t)\|_{L^s(S^3)} + C \epsilon \|a(t)\|_{L^s(S^3)}.$$

Let P be projection onto coexact one-forms. There are no harmonic one-forms on S^3 , so $Pa = a$. Projection of the curvature equation gives

$$(166) \quad (\partial_t + L)a = PN, \quad L = *_{S^3} d_{S^3}|_{\ker d_{S^3}^*}.$$

The spectrum of L avoids $(-2, 2)$. Let P_+ and P_- be its positive and negative spectral projections, and let t_-, t_+ be the two ends of the cylinder. Variation of constants gives

$$\begin{aligned} a(t) &= e^{-(t-t_-)L} P_+ a(t_-) + e^{(t_+-t)L} P_- a(t_+) \\ &\quad + \int_{t_-}^t e^{-(t-s)L} P_+ PN(s) ds - \int_t^{t_+} e^{-(t-s)L} P_- PN(s) ds. \end{aligned}$$

Each propagator decays in L^2 at least as $e^{-2|t-s|}$ in its indicated time direction. The endpoint traces satisfy $b_{\pm} := \|a(t_{\pm})\|_2 \leq C e_{\pm}$ by the trace estimate on the fixed enlarged endpoint slabs. If b_j and F_j are the slab L^2 norms of a and f , respectively, this formula and (165) give

$$(167) \quad b_j \leq C \left(\theta^{j-j_-} b_- + \theta^{j_+-j} b_+ + \sum_k \theta^{|j-k|} (F_k + \epsilon b_k) \right)$$

for a fixed $0 < \theta < 1$. The convolution is bounded on ℓ^1 independently of length. Summing and absorbing gives a summed L^2 bound. Fixed-slab interior estimates for the elliptic system $((\partial_t + *d)a, d^*a)$ then give the corresponding summed $W^{1,2}$ bound.

For $2 < p < 4$, the slab embedding $W^{1,2} \hookrightarrow L^4$ controls the summed L^p norm of a . The Hodge projection P is bounded on L^p , and (165) bounds PN by $C\|f\|_{L^p} + C\epsilon\|a\|_{L^p}$. Apply the fixed-slab interior estimate at exponent p . Enlarged endpoint slabs account for the boundary terms, yielding

$$\sum_j \|a\|_{W^{1,p}(D_j)} \leq C \left(e_- + e_+ + \sum_j \|f\|_{L^p(D_j^+)} \right).$$

Finally differentiate the scalar equation for α in t . The mean remains zero. Absorb the term $(\partial_t \alpha)a$; the other nonlinear terms have coefficients α or a , already small in C^0 . Thus

$$\|\partial_t \alpha(t)\|_{W^{1,p}(S^3)} \leq C \|\partial_t f(t)\|_{L^p(S^3)} + C\epsilon \|\partial_t a(t)\|_{L^p(S^3)}.$$

Integration, (165), and (163) prove (164). $\square$

6.4. Critical comparison and the endpoint gauges. On a physical annulus U_j of radius r_j , let

$$\delta_j = r_j^{1-4/p} \|c\|_{L^p(U_j)} + r_j^{2-4/p} \|\nabla c\|_{L^p(U_j)}.$$

This is equivalent to the $W^{1,p}$ norm of its pulled-back one-form on the normalized annulus. For $p > 2$, Hölder's inequality gives

$$(168) \quad \sup_x \int_{U_j} \frac{|v(z)|}{|x-z|^2} dz \leq C r_j^{2-4/p} \|v\|_{L^p(U_j)}.$$

For x within distance $O(r_j)$ of U_j , integrate the kernel to the power $p' < 2$; for more distant x , bound it by $C r_j^{-2}$. Thus the supremum includes every center. Applying the analogous estimate to $|c|^2$, and using the normalized embedding $W^{1,p} \hookrightarrow L^{4p/(4-p)}$, shows

$$(169) \quad \|c\|_{L^4} + \|\nabla c\|_{L^2} + \|c\|_{L^{2\sharp}} + \|\nabla c\|_{L^{\sharp,2}} \leq C \sum_j \delta_j.$$

These are norms of the indicated functions on the annulus, extended by zero as measurable functions for the weighted estimates. No derivative of a zero extension is taken. Smooth metric comparability and its lower-order connection terms give the same physical estimates.

The spliced background-instanton connection has normalized annular norms bounded by $C(r_j^2 + (\lambda_n/r_j)^2)$ in its exterior trivialization. Their sum on $R\lambda_n < r < d$ is at most $C(d^2 + R^{-2})$, including the fixed-ratio cutoff annuli. Together with (164), this gives

$$(170) \quad \|A_n - A_n^{\text{spl}}\|_{L^4} + \|\nabla_{A_n^{\text{spl}}}(A_n - A_n^{\text{spl}})\|_{L^2} + \|d_{A_n^{\text{spl}}}^*(A_n - A_n^{\text{spl}})\|_{L^{\sharp,2}} \leq C(e_- + e_+ + d^2 + R^{-2}) + o_n(1).$$

Here the gauges at the endpoints are chosen as follows.

At the outer end, the sequence in its background convergence gauge has normalized connection norm $O(d^2) + o_n(1)$ in a radial frame. At the inner end, its extracted instanton in exterior gauge has norm $O(R^{-2}) + o_n(1)$. Compare these gauges with the cylinder gauge using (158). The nonconstant transition parts are small in $W^{2,p}$ and C^0 , bounded by the endpoint errors and the corresponding model norms. Fix the outer constant and retain the inner one as ρ_n . Interpolate only the small nonconstant parts by their logarithms on fixed-ratio endpoint annuli. The regular-to-exterior instanton transition is kept intact: it is the charge-one clutching map. The interpolated gauge agrees with the original background convergence gauge outside and with the extracted bubble gauge, up to ρ_n , inside. It therefore

defines one determinant-one bundle identification of the prescribed splicing type. Its additional connection errors obey (169), proving (170). Changing ordinary to model-covariant derivatives uses $L^4 \cdot L^4 \rightarrow L^2$ and $L^{2\sharp} \cdot L^{2\sharp} \rightarrow L^\sharp$.

Proof of Proposition 6.1. The exterior and rescaled convergence in (154) improves, after gauge, to smooth convergence on fixed compact sets. Cover each such set by finitely many small-energy balls. The interior bootstrap in Lemma 6.3 gives uniform higher bounds in local Coulomb gauges; the transition equation patches their limits, which the original convergence identifies with the background or instanton. On the neck use the gauges and estimate (170). The endpoint terms tend to zero in the iterated limit by (160). The fixed-domain exterior and rescaled core differences tend to zero in the critical norms by smooth convergence and (168).

For the spinor, the uniform pointwise bound gives

$$\|\Phi_n\|_{L^4(B_d)} + \|\Phi_n\|_{L^{2\sharp}(B_d)} \leq Cd.$$

Choose η supported in B_{2d} and equal to one on B_d . Since $D_{A_n}\Phi_n = -\rho(\vartheta)\Phi_n$, the Weitzenböck identity for $\eta\Phi_n$ gives

$$(171) \quad \|\nabla_{A_n}\Phi_n\|_{L^2(B_d)}^2 \leq C \int_{B_{2d}} (|d\eta|^2 + 1)|\Phi_n|^2 \leq Cd^2.$$

Only self-dual curvature acts on the positive spinor. Its trace-free part is bounded by the monopole equation and its determinant part is fixed, so no bound on the full curvature is needed here. The preglued spinor is cut off at radius $\lambda_n^{1/3}$, outside the modified connection region. Its derivative is the cutoff times the background derivative plus the cutoff derivative times the background spinor; its $L^2(B_d)$ norm is bounded by $C(d^2 + \lambda_n^{1/3})$. Changing the derivative of Φ_n to the spliced connection adds at most $Cd\|A_n - A_n^{\text{spl}}\|_{L^4}$. The spinor contribution to the pair divergence has $L^{\sharp,2}(B_d)$ norm $O(d^2)$, since both spinors are bounded. These estimates and (170) give the full critical pair comparison. On the exterior the spinors converge smoothly.

Choose a slow diagonal $R_n \rightarrow \infty$, $d_n \downarrow 0$ after the fixed- R, d estimates, retaining $R_n\lambda_n \ll \sqrt{\lambda_n} \ll \lambda_n^{1/3} \ll d_n$. All pregluing cutoffs then lie inside the comparison annulus. The resulting global determinant-one identifications and relative frames give (152). $\square$

6.5. Exact modulation and complemented uniqueness. Apply Lemma 5.7 to the configurations in Proposition 6.1. After determinant-one gauge, there are unique exact gluing data $g_n = (q_n, \hat{x}_n, \hat{\lambda}_n, \hat{\rho}_n)$ and differences u_n such that

$$(172) \quad (A_n, \Phi_n) = q_{\hat{\lambda}_n}(g_n) + u_n, \quad d_{q_{\hat{\lambda}_n}}^{0,*} u_n = 0, \quad R_{g_n} u_n = 0, \quad \|u_n\|_{L_1^2} \rightarrow 0.$$

The exact parameters are close to the coarse ones in (62); in particular, q_n tends to the reducible background and $b(q_n) \rightarrow 0$.

Set

$$(173) \quad H_n = (u_n, -b(q_n), 0).$$

The complement condition depends only on u_n , so H_n lies in $\mathbb{H}_{\hat{\lambda}_n, N}^{\text{cmp}}$. Since (A_n, Φ_n) is a monopole and is in the Coulomb slice,

$$(174) \quad \hat{\mathcal{F}}_{g_n}(H_n) = 0.$$

Let $e_n = \hat{\mathcal{F}}_{g_n}(0)$. The inverse and quadratic estimates give

$$\|H_n\| \leq C\|e_n\| + C\|H_n\|^2 \leq C\hat{\lambda}_n^{1/3} + C\|H_n\|^2.$$

The qualitative convergence in (172) absorbs the quadratic term, and hence

$$(175) \quad \|H_n\| \leq 2C\hat{\lambda}_n^{1/3}.$$

Thus H_n lies in the same contraction ball as the correction in Proposition 4.2. Uniqueness of that fixed point gives

$$(176) \quad (u_n, -b(q_n), 0) = (u_{g_n}, \eta_{s,g_n}, \eta_{i,g_n}).$$

It follows that $[A_n, \Phi_n] = \gamma(g_n)$ and $\chi(g_n) = 0$.

Proposition 6.7 (neighborhood surjectivity). *There is an Uhlenbeck-open neighborhood $\mathcal{U}$ of $M_s \times X$ such that*

$$(177) \quad \mathcal{U} \cap \overline{\mathcal{M}}_v^{*,0} = \mathcal{U} \cap \gamma(\chi^{-1}(0)).$$

Consequently, γ restricts to a homeomorphism from $\chi^{-1}(0)$ onto an Uhlenbeck-open neighborhood of the ideal stratum, after restricting the virtual neighborhood over its outer boundary.

Proof. At positive scale, the augmented equation and injectivity of J_g identify the obstruction zero set with the monopoles in the image. At zero scale, $\chi^{-1}(0) = b^{-1}(0) \times X$ is the nearby lower-level monopole space. We first check that its middle-stratum points belong to the Uhlenbeck closure of the positive-scale moduli space. We work in the interior of the background disk, which suffices near $M_s \times X$.

Fix $q_0 \in b^{-1}(0) - M_s$ and a center x . Transversality of b gives a local disk D through q_0 on which b is a diffeomorphism onto a neighborhood of zero in Ξ_{q_0} , after trivializing Ξ . Trivialize the relative-frame bundle over this disk. Proposition 4.3 gives $\chi_s - b = O(\lambda^{1/3})$ in C^1 in the background variable, uniformly for $\rho \in \text{SO}(3)$. The implicit-function theorem therefore gives a smooth choice $q(\lambda, \rho) \in D$ with $\chi_s(q(\lambda, \rho), x, \lambda, \rho) = 0$ and $q(\lambda, \rho) \rightarrow q_0$ uniformly in ρ . For each small positive λ , the remaining section over this graph takes values in

$$\text{SU}(2) \times_{\{\pm 1\}} \mathbb{C} \longrightarrow \text{SO}(3).$$

This complex line is nontrivial: its first Chern class generates $H^2(\text{SO}(3); \mathbb{Z}) = \mathbb{Z}/2$. It consequently admits no nowhere-zero section. Choose a zero ρ_λ of the restricted χ_i . The corresponding positive-scale monopoles converge to (q_0, x) , proving the required closure assertion. If $r_\Xi = 0$, take $D = \{q_0\}$ in this argument. The lowest stratum already belongs to the compactification by (1). Thus the inclusion from right to left holds at every scale.

Conversely, nearby zero-scale ideal monopoles are in the image by the background finite-dimensional reduction. If the reverse inclusion failed in every neighborhood, there would therefore be genuine monopoles outside $\gamma(\chi^{-1}(0))$ converging to the compact ideal stratum. A subsequence converges to one point of that stratum, and the argument above places every sufficiently late member in $\gamma(\chi^{-1}(0))$, a contradiction. The inverse is continuous because γ is a compactified embedding by Corollary 5.10. $\square$

7. THE OBSTRUCTION CONE AND ORIENTATIONS

7.1. The compactified section. The background summand $\overline{\Upsilon}^s$ is the pullback of the fixed complex bundle $\Xi \rightarrow M_s$ to $N_{t,s}$. The charge-one Dirac operator has the kernel and cokernel in (21). At positive scale its obstruction line is

$$(0, \delta] \times \text{SU}(2) \times_{\{\pm 1\}} \mathbb{C} \longrightarrow (0, \delta] \times \text{SO}(3).$$

Its compactification is the cone map (27); the whole fiber, not merely its norm, is identified with the vertex at zero scale. There are no partition-overlap maps at level one.

At positive scale, define χ by (100); at zero scale, set

$$(178) \quad \chi(q, x, 0) = (b(q), 0).$$

The contraction is equivariant, because the nonlinear map, cutoffs, transports, and inverse are equivariant and its fixed point is unique. We use the diagonal presentation before taking the effective residual circle; it has weight two on the normal and background-obstruction factors. Write ν for minus the first Chern class of the line associated to the effective circle character of the background obstruction on the virtual link quotient; equivalently the weight-one associated line has class $-\nu$ in this convention. Since Ξ was chosen globally trivial, after passage to the effective action its Euler class is

$$(179) \quad e(\bar{\Upsilon}^s/S^1) = (-\nu)^{r\Xi}.$$

The instanton line retains the nontrivial $SU(2)/\{\pm 1\}$ framing twist, so its Euler class is computed from its tensor square in Section 8 rather than copied from (179).

Proposition 7.1 (compactified obstruction section). *The two components of χ are smooth at positive scale. The background component extends continuously over the compactification and smoothly on each zero-scale stratum. The instanton component is a continuous section of the cone pseudo-bundle. With the metric induced by $J_{s,g} \oplus J_{i,g}$, the norm of χ is continuous along the section.*

Proof. Positive-scale smoothness follows from Proposition 4.2. At zero scale, its estimate gives

$$\|\eta_{s,g}\| + \|\eta_{i,g}\| \leq C\lambda^{1/3}.$$

Thus χ_s tends to $b(q)$ and χ_i tends to the cone vertex. For $g_n \rightarrow (q, x, 0)$, the Gram estimate in Lemma 3.3 gives

$$\|J_{s,g_n}\chi_s(g_n) + J_{i,g_n}\chi_i(g_n)\|_{L^2}^2 = \|b(q)\|^2 + o(1),$$

which is the asserted norm continuity. This does not extend the ambient instanton line as a constant-rank bundle; only the section tends to its collapsed fiber. $\square$

We next compare the global reduction with the local small-eigenvalue reduction of [FL99, Section 8]. There are three distinct finite-dimensional reductions. On a relatively compact positive-scale chart U^+ on which a cutoff μ avoids the spectrum, let

$$(180) \quad E_{\text{glob},g} = \Xi_q \oplus K_I, \quad E_{\text{loc},g} = \ker d_q^{1,*} \oplus K_I, \quad E_{\text{spec},g} = \text{Ran } \Pi_{q,g,\mu}.$$

The published cut-and-project and synthesis estimates [FL99, Propositions 8.19 and 8.26] identify E_{loc} with E_{spec} on a constant-rank chart: their two composites are $I + o(1)$. They also transport the ordered orientation local system of $\ker d_q^{1,*} \oplus K_I$ to E_{spec} , compatibly on overlaps. There is generally no isomorphism between E_{glob} and E_{loc} , since the global augmentation may contain extra stabilizing directions.

Proposition 7.2 (stable comparison of reductions). *On every positive-scale constant-rank germ of the monopole zero set, the global and spectral reductions admit a common equivariant finite-dimensional enlargement. Their zero loci agree, the comparisons satisfy the cocycle condition up to equivariant homotopy on a common germ, and the induced determinant orientation and relative Euler–Thom pairing agree. No equality of the two ranks and no spectral bundle at zero scale is asserted.*

Proof. For a smooth Fredholm section F with two surjective augmentations $i_j : E_j \rightarrow Y$, form

$$\mathcal{V}_{01} = \{(x, e_0, e_1) : F(x) - i_0 e_0 - i_1 e_1 = 0\}.$$

The two reductions embed by setting the other coefficient equal to zero. At a point of the first reduction, projection to e_1 identifies its normal bundle in $\mathcal{V}_{01}$ with E_1 : surjectivity of $DF \oplus (-i_0)$ supplies a lift of every prescribed e_1 . The analogous statement holds with the indices reversed. Thus both obstruction zero loci are canonically $F^{-1}(0)$.

Apply this construction with $E_0 = E_{\text{glob}}$ and $E_1 = E_{\text{spec}}$. The first augmented derivative is surjective at the pregluing by Proposition 3.4, and remains so at the small correction by the quadratic estimate and openness. On $E_{\text{spec}}^\perp$, the second has right inverse

$$d_{q_g}^{1,*} (d_{q_g}^1 d_{q_g}^{1,*})^{-1},$$

bounded by the spectral gap at μ . Positive-scale smoothness of the inverse and the Riesz formula for $\Pi_{q_g, \mu}$ give smooth tubular maps on a common germ. On overlaps, order all spectral factors by chart label and use literal zero-coordinate embeddings; associativity of direct sum gives the cocycle before tubular choices, and nearby invariant tubular choices are homotopic through their graph maps.

Along the first embedding,

$$\det T\mathcal{V}_{01} \cong \det T\mathcal{V}_0 \otimes \det E_1.$$

The same E_1 occurs in the common obstruction bundle, so its determinant line cancels its dual. Along the other embedding the fixed order contributes the Koszul factor $(-1)^{\text{rank}_{\mathbb{R}} E_0 \text{rank}_{\mathbb{R}} E_1}$; it is $+1$ because $E_0 = \Xi \oplus K_I$ has even real rank. If E_1 is not globally oriented, its orientation local system is carried by both factors and cancels canonically. The identical cancellation holds in relative equivariant cohomology: the Thom class of the added normal E_1 is the Euler factor of the added obstruction E_1 . Hence stabilization changes neither the oriented zero locus nor the relative Euler–Thom pairing. $\square$

7.2. The exact transversality statement. At a positive-scale zero, the extended equation is

$$(181) \quad \mathfrak{S}(\gamma(g)) = J_g \chi(g).$$

Let Q_g be projection onto the obstruction image. The contraction parametrizes the local virtual manifold $(1 - Q_g)\mathfrak{S} = 0$, and the complemented inverse identifies its tangent space with $D\gamma_g(T_g \mathcal{G}^+)$. Differentiating (181) at a zero gives

$$(182) \quad D\mathfrak{S}_{\gamma(g)} D\gamma_g = J_g D\chi_g.$$

Proposition 7.3 (stratumwise transversality). *For a generic perturbation at which the irreducible monopole section is transverse, the total section $\chi_s \oplus \chi_i$ is transverse on $\mathcal{G}^+$. On $(N_{t,s} - M_s) \times X$, the surviving background component $\chi_s = b$ is transverse. No transversality holds or is asserted on $M_s \times X$ when $r_\Xi > 0$.*

Proof. Given $e \in \Upsilon_g$ at a positive-scale zero, surjectivity of $D\mathfrak{S}_{\gamma(g)}$ supplies v with $D\mathfrak{S}_{\gamma(g)} v = J_g e$. Its projected equation component vanishes, so v is tangent to the virtual manifold and equals $D\gamma_g \dot{g}$ for some $\dot{g}$. Equation (182) and injectivity of J_g give $D\chi_g \dot{g} = e$.

On the middle stratum, the instanton fiber has collapsed and the background finite-dimensional reduction makes b transverse; the X factor is passive. On the lowest stratum b is identically zero. An identically zero section of a positive-rank bundle is not transverse. This is why [FL02, Theorem 3.8 and Section 5.1] asserts precisely the first two conclusions and disclaims the third. $\square$

Accordingly, the all-strata formulation in [FL18, Hypothesis 7.8.1(2)] is not asserted literally for the fine decomposition (25). The relative Euler construction uses the positive-scale and middle-stratum transversality in Proposition 7.3; it does not require transversality on the lowest stratum.

7.3. Orientation under instanton addition. For a real Fredholm operator F , use

$$\det F = \Lambda^{\max} \ker F \otimes (\Lambda^{\max} \operatorname{coker} F)^*.$$

Direct-sum factors are ordered as written, and complex vector spaces carry their complex orientations. For $q \in N(\varepsilon)$, write $p = \pi_N(q) \in M_5$. Orient $T_p M_5$ using the homology orientation Ω . Orient the centered framed instanton parameters by

$$(183) \quad (\partial_\lambda, i, j, k) \quad \text{on} \quad \mathbb{R} \oplus \mathfrak{so}(3) = \mathbb{H}.$$

Write $O^{\text{asd}}(\Omega, c_1(L))$ for the monopole orientation obtained from the ASD orientation by turning off the spinor and using the complex Dirac orientation, as in [FL01b, Lemma 2.2 and Definition 2.3]. We compare the specific kernel map S_g with the instanton-addition map of [Don87, Proposition (3.20)]. The latter is constructed by projecting cutoff curvature-contraction modes and background kernel fields. The following argument includes finite stabilizations along the homotopy that removes the spinor couplings.

Proposition 7.4 (instanton-addition orientation). *The synthesis S_g induces an orientation-preserving isomorphism*

$$(184) \quad \det \widehat{\mathcal{D}}_q^X \otimes \det \mathcal{K}_I^{\text{fr}} \longrightarrow \det \widehat{\mathcal{D}}_{q_g},$$

where the monopole determinant lines have orientation $O^{\text{asd}}(\Omega, c_1(L))$, the finite obstruction spaces have their complex orientations, and the bubble has (183) followed by its four translations. This comparison is compatible with the frame and residual-circle actions.

Proof. We first fix the rotation sign. In the regular quaternionic gauge put $v_\theta(y) = \theta y$ for $\theta \in \operatorname{Im} \mathbb{H}$. Left rotations preserve $I = \operatorname{Im}(\bar{y} dy)/(1 + |y|^2)$, and $\iota_{v_\theta} I = h_\theta = \bar{y} \theta y/(1 + |y|^2)$. Covariant Cartan's formula gives the exact identity

$$(185) \quad 0 = \mathcal{L}_{v_\theta} I = \iota_{v_\theta} F_I + d_I h_\theta, \quad -\iota_{\theta y} F_I = k_\theta.$$

In the exterior framing, pullback by $y \mapsto e^{-t\theta} y$ acts as $I \mapsto e^{-t\theta} I e^{t\theta}$. This agrees with the relative-frame convention used for the modes above. The negative dilation and translation flows give $k_s = -\iota_y F_I$ and $k_j = -\iota_{e_j} F_I$, respectively, with the based gauge corrections already included in their Coulomb representatives.

For $y = x^0 + x^1 i + x^2 j + x^3 k$, the positive self-dual basis is

$$\begin{aligned} \omega_1 &= dx^0 \wedge dx^1 + dx^2 \wedge dx^3, & \omega_2 &= dx^0 \wedge dx^2 - dx^1 \wedge dx^3, \\ \omega_3 &= dx^0 \wedge dx^3 + dx^1 \wedge dx^2. \end{aligned}$$

Indeed $-\operatorname{Im}(dy \wedge d\bar{y}) = 2(i\omega_1 + j\omega_2 + k\omega_3)$. With complex coordinates $z_1 = x^0 + ix^1$, $z_2 = x^2 + ix^3$, the real $(0, 2)$ plane has complex orientation (ω_2, ω_3) after the Kähler form ω_1 . These are the conventions of [Don87, Section 3(c), p. 413]. Thus the oriented curvature modes for $\mathbb{R} \oplus \Lambda^+ \oplus \mathbb{R}^4$ are precisely $(k_s, k_i, k_j, k_k, k_0, \dots, k_3)$, where the subscripts i, j, k denote frame directions and $0, \dots, 3$ denote translations. All changes from normalized scale and center coordinates multiply basis vectors by positive numbers. Interchanging the two blocks of four directions also has positive determinant.

We next establish compatibility with arbitrary finite stabilizations. Let D_b be a compact smooth background family obtained by varying the zeroth-order monopole coefficients, and choose smooth target columns $J_b : W \rightarrow C^\infty$ such that

$$(186) \quad B_b(u, w) = D_b u - J_b w$$

is surjective. The columns need not be independent. Localize them as $\beta_X T_X^Y J_b$ and retain the complex bubble Dirac column. The parametrix and weak estimates of Propositions 3.4 and 3.7 still apply, with constants uniform on this family. A smooth target column in the real gauge row has the same cutoff estimates as one in the other rows.

For dependent columns, change the background retraction to the positive Gram inverse of the full augmented kernel, using $\|u\|_2^2 + |w|^2$ and applying its duals to $(\beta_X T_X^{-1} u, w)$. Keep the compact bubble duals. This retains pure auxiliary kernel vectors, which configuration integrals alone would lose. The preliminary synthesis is $(v, w) \mapsto (\beta_X T_X v, w, 0)$ on the background kernel and $k \mapsto (\beta_I T_\lambda k, 0, 0)$ on the bubble. The weighted norm is now $\|u\|_2^2 + |w|^2 + \lambda^2 |c_I|^2$. The Gram lower bound and both mode errors in (88) are unchanged. The index increase is eight: the ASD block contributes eight, the real Dirac index decreases by two, and its complex bubble column adds two. The corrected lift and orthogonal projection consequently give the same orientation isomorphism

$$(187) \quad \mathcal{G}_{J,g} : \ker B_b \oplus \mathcal{K}_I^{\text{fr}} \longrightarrow \ker B_{J,g}.$$

Local frames of these kernels suffice, and the maps vary continuously at positive scale.

Suppose that B_b is already surjective and add columns J'_b on W' . The two exact sequences, on the background and glued sides, are

$$(188) \quad \begin{aligned} 0 &\longrightarrow \ker B_b \longrightarrow \ker(B_b - J'_b) \xrightarrow{\text{pr}_{W'}} W' \longrightarrow 0, \\ 0 &\longrightarrow \ker B_{J,g} \longrightarrow \ker B_{J \oplus J',g} \xrightarrow{\text{pr}_{W'}} W' \longrightarrow 0. \end{aligned}$$

For a background right inverse P_b , the first sequence splits by $(k, w') \mapsto (k + P_b J'_b w', w')$. For the enlarged glued problem let T_g^+ be its preliminary synthesis and use the old right inverse with zero new coordinate: $\hat{P}_g f = (P_g f, 0)$. Its corrected preliminary lift

$$(1 - \hat{P}_g B_{J \oplus J',g}) T_g^+$$

restricts to the old lift and induces the identity on W' . It is an isomorphism by the two exact sequences. The weak error bound shows that it has the same orientation as orthogonal projection of T_g^+ , just as in (86). Moving W' past the eight-dimensional bubble kernel has sign $(-1)^{8 \dim W'} = 1$. This proves compatibility with the determinant stabilization sequences and independence of the columns by common enlargement. In this comparison the actual negative maps $-J_b$ and $-J'_b$ are retained in both sequences. Common real auxiliary factors cancel between the background and glued problems; no parity assumption on $\dim W$ or $\dim W'$ is needed.

Apply this construction to the spinor homotopy

$$(189) \quad D_t = \mathcal{D}_{A,t\Phi}, \quad D_{g,t} = \mathcal{D}_{A_g,t\Phi_g}, \quad 0 \leq t \leq 1.$$

At each t choose smooth representatives spanning the background cokernel. They give surjectivity on a neighborhood of t . A finite cover of $[0, 1]$ and the union of these columns give one finite stabilization for the whole path. Include the original Ξ columns at $t = 1$, and separate real ASD and complex Dirac columns at $t = 0$. Redundancy is allowed by the preceding argument. The construction works on neighborhoods in the background and center parameters; a finite cover makes the bounds uniform over the compact family. Thus the localized glued

family is surjective for sufficiently small scale throughout the path. The intermediate pairs need not solve the equations. In particular, real gauge cokernels at zero spinor are included in the stabilization.

The determinant comparison varies continuously along this homotopy. At $t = 0$, common enlargement permits replacement by the separated ASD and Dirac columns; at $t = 1$ it permits cancellation back to the original augmentation. At zero spinor choose Hermitian complements in the Dirac blocks. Their columns, localizations, inverses, and projections are complex linear, so their determinant comparison is positive. Only the real ASD comparison remains.

For that block, the map in [Don87, Section 3(d), Proposition (3.20)] is the orthogonal projection of background cutoff fields and contractions of the anti-self-dual curvature with cutoff conformal-affine fields. For our normalized parameters (s, θ, ξ) , the latter field on the core is $sz + \theta z + \lambda \xi$ in physical coordinates. Its curvature contraction is exactly $T_\lambda k$ there. On a transition annulus $r \asymp \sqrt{\lambda}$ the curvature is $O_N(1)$, while the dilation and rotation fields are $O_N(\sqrt{\lambda})$ and translations are $O(\lambda)$. The resulting L^2 error is $O_N(\lambda^{3/2})$. Discarded mode tails have the same bound by r^{-3} decay; metric projection errors are smaller by the $O(r^2)$ metric discrepancy. Different background cutoffs contribute $O_N(\lambda)$ in L^2 . These errors tend to zero in the weighted operator norm of Proposition 3.7.

Flattening cutoffs and column cutoffs may also be matched to those in the source. Alternatively, the difference of flattened connections is $O_N(\sqrt{\lambda})$ on radius $O_N(\sqrt{\lambda})$, hence $O_N(\lambda)$ in L^4 ; a difference of smooth cutoff columns is $O_N(\lambda^{3/2})$ in $L^{4/3}$. Their linear interpolation is therefore a small perturbation of the uniformly surjective augmented family, with the same mode estimates. It preserves the orientation comparison. On the ASD block there is no bubble auxiliary coordinate, so the Hilbert projection used above is the ordinary configuration L^2 projection with the background stabilization coordinates retained.

The source uses gauge row $-d_A^*$. Apply the fixed target isomorphism $C(f, \omega) = (-f, \omega)$ to both its background and glued operators and to their finite columns. Kernels and the projected modes are unchanged, while cokernels and determinant sequences are transported by C . The induced square of determinant addition maps commutes, since on the stabilized kernels both horizontal maps are the same projected synthesis. Both standard orientations are propagated from a fixed split reference; any sign between their reference normalizations is consequently carried equally to the background and glued sides and cancels. In particular, one does not discard a real constant-gauge factor separately on one side. This gives the $+d_A^*$ convention used in [FL01b, Section 2.2]. The mode identification and Proposition 3.7 now identify (187), in orientation, with the instanton-addition map that defines the standard ASD orientation from a split reference bundle and Ω .

Finally, the orientation O^{asd} is defined by the same spinor homotopy and the complex Dirac orientation [FL01b, Lemma 2.2, Equation (2.5), and Definition 2.3]. Transport back to $t = 1$ and cancel the common auxiliary columns. At a reducible background, [FL01b, Lemma 2.6] identifies $O^{\text{asd}}(\Omega, c_1(L))$ with the Seiberg–Witten tangent orientation and complex normal orientation. The original complex augmentation gives the stated background kernel orientation. Continuity and the background derivative comparison extend this identification over the small disk $N(\varepsilon)$. All original maps are equivariant; independence of the extra local columns makes these local orientation comparisons agree on overlaps and under the frame and circle actions. This proves the proposition. $\square$

In particular the orientation of the kernel identification

$$(190) \quad \ker \widehat{\mathcal{D}}_{q_g} \cong \ker \widehat{\mathcal{D}}_q^X \oplus \mathcal{K}_I^{\text{fr}}$$

is represented by the synthesis in (64).

The ordered model kernels and finite-dimensional stabilizations give

$$(191) \quad \begin{aligned} \det \widehat{\mathcal{D}}_q^X &= \det N_p \otimes \det T_p M_s, \\ \det \mathcal{K}_I^{\text{fr}} &= \det T_I M_1^s(S^4, \delta), \\ \det \widehat{\mathcal{D}}_{q_g} &= \det \mathcal{D}_{q_g} \otimes \det \Xi_p \otimes \det K_I. \end{aligned}$$

The last identity is the determinant exact sequence for adding the finite-dimensional domain $\Xi_p \oplus K_I$ with the ordered map $-(J_{s,\lambda} \oplus J_{i,\lambda})$. Its minus sign has positive determinant because the real stabilization dimension is $2r_\Xi + 2$. Combining (184)–(191) and putting the obstruction determinants on the dual side yields

$$(192) \quad \begin{aligned} \det \mathcal{D}_{q_g} &\cong \det N_p \otimes \det T_p M_s \otimes \det T_I M_1^{s,\natural}(S^4, \delta) \otimes \det T_x X \\ &\otimes (\det \Xi_p)^* \otimes (\det K_I)^*. \end{aligned}$$

Here the centered bubble factor contains scale and relative rotation; its four translations are identified orientation-preservingly with $T_x X$ by the chosen oriented tangent frame. Moving the four-dimensional translation factor past the four-dimensional scale–rotation factor contributes no sign.

The splitting (192) has no orientation monodromy around a loop in M_s . The tangent bundle is oriented by Ω , the bundles N and Ξ have their complex orientations, the instanton modes have the fixed quaternionic orientation, TX is oriented, and K_I is a complex line. Possible spectral flow of an associated self-adjoint family changes neither this real orientation nor the global stabilization.

The tangent- and adjoint-frame determinants introduced before taking the diagonal $\text{SO}(4) \times \text{SO}(3)$ quotient cancel between the ordered tangent exact sequences for the framed parameter space and for the orbit directions. Both group dimensions are moved past identical factors, so the two Koszul signs cancel. The remaining relative-frame factor is the $\text{SO}(3)$ already contained in $M_1^{s,\natural}$.

There is also no orientation monodromy around this relative-frame factor. The generator of $\pi_1(\text{SO}(3))$ lifts to a path in $\text{SU}(2)$ from 1 to -1 . Equivariance of Proposition 7.4 transports the determinant along this lift. At the endpoint, -1 acts trivially on adjoint connection directions, by complex multiplication by -1 on each complex spinor or obstruction factor, and by -1 on the four-dimensional quaternionic parameter space. Each real determinant is $+1$. Hence the orientation descends around the nontrivial loop in $\text{SO}(3)$.

The path $q_g + tu_g$ transports the determinant to the corrected operator. The following proposition identifies this transport with the horizontal tangent of the corrected chart. Comparison with the published obstruction reduction uses the common enlargement in Proposition 7.2; the additional complex factors cancel with their duals. The normalization discrepancy in the spinor-bilinear term is removed by a zero-order homotopy through positive coefficients. The ordered factors in (192) are those of [FL02, Section 3.9].

Proposition 7.5 (orientation of the corrected chart). *At a positive-scale zero of χ , there is a canonical orientation isomorphism*

$$(193) \quad \det \mathcal{D}_{\gamma(g)} \cong \det T_g \mathcal{G}^+ \otimes (\det \Upsilon_g)^*.$$

Under this isomorphism, $D\gamma$ identifies the transverse-zero-locus orientation of $\chi^{-1}(0)$ with the orientation $O^{\text{asd}}(\Omega, c_1(L))$ of the actual monopole moduli space.

Proof. Put $q_t = q_g + tu_g$ and keep J_g and R_g fixed. On augmented variables define

$$(194) \quad B_t(v, \eta) = (d_{q_t}^{0,*} v, D\mathfrak{S}_{q_t} v - J_g \eta), \quad \mathcal{F}_t = (B_t, R_g \text{pr}_{\mathbb{X}}).$$

The quadratic estimates give $B_t - B_0 = O(\lambda^{1/3})$ from the first-order augmented space to $\mathbb{Z} \oplus \mathbb{Y}$, uniformly for $t \in [0, 1]$. The graph completions therefore coincide, and (82) and a Neumann series make every $\mathcal{F}_t$ invertible. The maps

$$\tilde{S}_t k = \mathcal{F}_t^{-1}(0, k)$$

identify the full kernels continuously, with $R_g \text{pr}_{\mathbb{X}} \tilde{S}_t = 1$. They give the ordered determinant transport along this path; intermediate pairs need not solve the equations.

At a zero of χ , differentiating (181) removes the derivative of J_g . Gauge equivariance gives $D\mathfrak{S}_{\gamma(g)} d_{\gamma(g)}^0 = 0$, so

$$(195) \quad \Theta_g \dot{g} = (H_{\gamma(g)} D\gamma_g \dot{g}, D\chi_g \dot{g}) \in \ker B_1.$$

Lemma 5.3, Proposition 4.3, and the uniformly bounded pregluing derivative give

$$(196) \quad \begin{aligned} R_g \text{pr}_{\mathbb{X}} \Theta_g &= R_g H_g dq_g + O(\lambda^{1/3}) \\ &= j_g + o(1). \end{aligned}$$

After applying j_g^{-1} this map has positive determinant. Thus Θ_g is an isomorphism and identifies the parameter orientation with the augmented kernel orientation transported from $t = 0$. The horizontal projection in (195) is essential; R_g need not vanish on gauge directions.

Proposition 7.4 identifies the orientation at $t = 0$ with the first four factors of (192). These are the determinant line of the tangent bundle of the virtual domain, while $\Upsilon_g = \Xi_p \oplus K_I$. The stabilization exact sequence therefore gives (193) with its stated orientation. The ordering $K_I \oplus \Xi_p$ used in one source formula may be interchanged with $\Xi_p \oplus K_I$ without sign because both have even real dimension. Similarly, the analytic stabilization is $-J_g$, but $\dim_{\mathbb{R}} \Upsilon_g = 2r_{\Xi} + 2$ is even.

Proposition 7.3 gives the exact sequence

$$0 \longrightarrow T_g \chi^{-1}(0) \longrightarrow T_g \mathcal{G}^+ \xrightarrow{D\chi_g} \Upsilon_g \longrightarrow 0.$$

Its determinant is the right side of (193) with the same ordered factors. The claim follows. $\square$

7.4. The residual circle and the link boundary. Let $\mathcal{Z}^+ = \chi^{-1}(0) \cap \mathcal{G}^+$. On the free circle locus define

$$(197) \quad \lambda_{\mathcal{Z}} = v_{S^1} \wedge \tilde{\lambda}_{\mathcal{Z}/S^1}.$$

The weight-two presentation changes the positive speed of an orbit but not its orientation. Let $\vec{r}$ point outward from the deleted neighborhood of the ideal stratum. The boundary of its complement has

$$\lambda_{\mathcal{Z}/S^1} = -\vec{r} \wedge \lambda_L.$$

Together with (197), this proves

$$(198) \quad \lambda_{\mathcal{M}} = -v_{S^1} \wedge \vec{r} \wedge \tilde{\lambda}_L,$$

which is the convention in [FL02, Equation (3.81)]. The sign can also be read in one complex fiber: moving the positive orbit vector $i\vec{r}$ past the outward radial vector contributes the displayed minus sign.

The compactified virtual link has a background-normal face and an instanton-scale face. Their common edge receives opposite orientations, because exchanging the two outward normals contributes a minus sign. Thus the faces form an oriented relative cycle. This statement concerns the smooth positive-scale zero set and the relative Euler cycle; it does not reinterpret the lowest stratum as a transverse zero set.

Finally, if w is an integral lift of $w_2(\mathfrak{t}')$, the orientation comparison of [FL01b] is

$$(199) \quad O^{\text{asd}}(\Omega, w) = (-1)^{(w - c_1(L))^2/4} O^{\text{asd}}(\Omega, c_1(L)).$$

Write $o_{\nu}(w, \mathfrak{s}) = \frac{1}{4}(w + c_1(\mathfrak{s}) - c_1(\mathfrak{t}'))^2$ for this integral exponent. The circle and boundary constructions are linear in the ambient orientation, so the same factor relates the corresponding link orientations. This completes the orientation assertion in Theorem 2.1.

8. THE LEVEL-ONE PAIRING

In this section write $\delta_{\text{gl}} > 0$ for the fixed small upper bound on the gluing scale; δ will denote the insertion degree.

We now apply the family neighborhood theorem to the published level-one pairing. It identifies the actual and virtual links, allows geometric representatives to be pulled back, and carries the universal bundle through an equivariant homotopy to the splicing map. The obstruction section represents the smooth zero locus by a relative Euler class, and the orientation comparison preserves signed intersections. Corollary 5.10, Proposition 6.7, Proposition 7.1, and Proposition 7.5 supply these inputs. The dimension argument of [FL02, Lemmas 3.9 and 5.4] puts the counted intersection in the positive-scale smooth locus and away from the common edge. The universal-bundle calculation, Thom and Segre reduction, charge-one fiber evaluations, and Jacobi identity are retained as published topological inputs. The proved orientation comparison permits their application to signed intersections on the actual link.

We retain the notation of that paper for the pairing. Let $\mathfrak{t}'$ be a spin^u structure with $w_2(\mathfrak{t}')$ good in the sense of [FL02, Definition 2.3], and suppose

$$(200) \quad (c_1(\mathfrak{t}') - c_1(\mathfrak{s}))^2 = p_1(\mathfrak{t}') + 4.$$

Let $h \in H_2(X; \mathbb{R})$ and let $x \in H_0(X; \mathbb{Z})$ be the positive generator. Write Q_X for the intersection form on $H_2(X; \mathbb{R})$. For a monomial z in h and x , let $\overline{V}(z)$ denote the closure of a transverse geometric representative of the Donaldson class $\mu_p(z)$; let $\overline{W}$ represent the circle class μ_c ; and let $\overline{L}_{\mathfrak{t}', \mathfrak{s}}$ be the standard oriented link of $M_{\mathfrak{s}} \times X$ in $\overline{\mathcal{M}}_{\mathfrak{t}'}^{*,0}/S^1$. The symbol $\#$ below denotes the signed count of a transverse zero-dimensional intersection. Let $\mu_{\mathfrak{s}} \in H^2(M_{\mathfrak{s}}; \mathbb{Z})$ be the first Chern class of the Seiberg–Witten base-point fibration. With the orientation determined by the chosen homology orientation, set

$$(201) \quad \text{SW}_X(\mathfrak{s}) = \left\langle \mu_{\mathfrak{s}}^d, [M_{\mathfrak{s}}] \right\rangle.$$

Also write

$$(202) \quad d_a(\mathfrak{t}') = -2p_1(\mathfrak{t}') - \frac{3}{2}(\chi(X) + \sigma(X)),$$

and set

$$(203) \quad \beta = c_1(\mathfrak{s}) - c_1(\mathfrak{t}'), \quad B_h = \langle \beta, h \rangle.$$

Write $c_1^2(X) = 2\chi(X) + 3\sigma(X)$. For the Jacobi polynomials we use the normalization

$$(204) \quad P_d^{A,B}(\zeta) = 2^{-d} \sum_{v=0}^d \binom{d+A}{v} \binom{d+B}{d-v} (\zeta-1)^{d-v} (\zeta+1)^v.$$

8.1. Pullbacks and Euler classes. Let $\mathcal{G}^{\text{vir},*}$ be the locus in the virtual gluing space on which the effective circle acts freely. With the associated-line convention fixed before (179),

$$(205) \quad \nu = c_1(\mathcal{G}^{\text{vir},*} \longrightarrow \mathcal{G}^{\text{vir},*}/S^1).$$

The characteristic classes below are first taken on this locus and then extended in the relative cohomology used for the virtual link. The line defining μ_c on the monopole configuration quotient has weight -2 , while the action on the gluing domain before effectivization has twice the angular parameter. Hence

$$(206) \quad \gamma^* \mu_c = -\nu.$$

Away from the incidence diagonal, the pulled-back universal $SO(3)$ bundle splits as $\mathbb{R} \oplus \mathbb{L}^{\text{vir}}$, where

$$(207) \quad c_1(\mathbb{L}^{\text{vir}}) = 2\pi_s^* \mu_s - \nu + \pi_{X,2}^* c_1(L), \quad c_1(L) = -\beta.$$

The equivariant isotopy (151) identifies the corrected and splicing pullbacks. Insertion of one instanton changes the Pontrjagin class by the incidence diagonal, so

$$(208) \quad (\gamma \times \text{id}_X)^* p_1(\mathbb{F}_{\mathfrak{t}}) = (2\pi_s^* \mu_s - \nu + \pi_{X,2}^* c_1(L))^2 - 4(\pi_X \times \text{id}_X)^* \text{PD}[\Delta].$$

The coefficient -4 is fixed by a chain meeting the diagonal once. Since $\mu_p(\alpha) = -\frac{1}{4}p_1(\mathbb{F}_{\mathfrak{t}})/\alpha$, slant product with a point and a surface gives

$$(209) \quad \begin{aligned} \gamma^* \mu_p(x) &= -\frac{1}{4}(2\mu_s - \nu)^2 + \pi_X^* \text{PD}[x], \\ \gamma^* \mu_p(h) &= \frac{1}{2}\langle \beta, h \rangle (2\mu_s - \nu) + \pi_X^* \text{PD}[h]. \end{aligned}$$

These are the formulas of [FL02, Lemma 4.10]; the second summand in each line is the diagonal correction.

The background Euler class is (179). For the instanton factor, the square of the framing line loses the central twist:

$$(\text{SU}(2) \times_{\{\pm 1\}} \mathbb{C})^{\otimes 2} \cong \text{SO}(3) \times \mathbb{C}.$$

Accounting for the central and residual circle weights gives

$$(210) \quad c_1((\Upsilon^i/S^1)^{\otimes 2}) = \pi_X^* c_1(\mathfrak{t}') - \nu.$$

Thus, in rational cohomology,

$$(211) \quad e(\Upsilon^i/S^1) = \frac{1}{2}(\pi_X^* c_1(\mathfrak{t}') - \nu).$$

The factor $1/2$ comes from taking the tensor square of the framed associated line.

8.2. Relative Euler reduction. Let δ, m, η be nonnegative integers with $2m \leq \delta$. Put $n = \delta - 2m$ and $z = h^n x^m$; the exponent η records the circle-class insertion. Here δ is the degree parameter for the Donaldson insertions. On the positive-scale virtual link, the section determines a relative Euler class. The geometric-representative cocycle is supported away from the lower-level zero set. The homotopy

$$(212) \quad \chi_s \oplus \chi_i \longmapsto \chi_s \oplus t\chi_i, \quad 0 \leq t \leq 1,$$

does not vanish off $\chi_s^{-1}(0)$. At $t = 0$ the class factors as

$$(213) \quad e(\Upsilon/S^1, \chi_s) = e(\overline{\Upsilon}^s/S^1, \chi_s) \smile e(\Upsilon^i/S^1).$$

This is precisely where one needs continuity of the background section and the cone topology, but not transversality of the lowest stratum. The signed actual-link count is therefore

$$(214) \quad \left\langle \bar{\mu}_p(z) \smile \bar{\mu}_c^\eta \smile \bar{e}_s \smile \bar{e}_i, [\bar{L}^{\text{vir}}] \right\rangle.$$

Write

$$H = \pi_X^* \text{PD}[h], \quad X_0 = \pi_X^* \text{PD}[x], \quad c = \pi_X^* c_1(\mathfrak{t}').$$

Substitution of (206), (209), (179), and (211) into (214) gives

$$(215) \quad \begin{aligned} \mathcal{P} = & \left\langle \left[\frac{1}{2} B_h (2\mu_s - \nu) + H \right]^n \right. \\ & \smile \left[-\frac{1}{4} (2\mu_s - \nu)^2 + X_0 \right]^m (-\nu)^{\eta+r_\Xi} \\ & \left. \smile \frac{1}{2} (c - \nu), [\bar{L}^{\text{vir}}] \right\rangle. \end{aligned}$$

The dimension relation is

$$(216) \quad \delta + \eta = d + r_N - r_\Xi + 2.$$

The equivariant Thom division of [FL02, Propositions 5.20–5.21] expresses this pairing over

$$M_s \times \partial \overline{\text{Gl}}_t(\delta_{\text{gl}})/S^1$$

as a finite sum involving the Segre classes of N . The normal-bundle calculation [FL02, Lemma 2.4] gives

$$(217) \quad s_j(N) = S_j \mu_s^j,$$

where S_j is the binomial coefficient in [FL02, Equation (6.14)]. On this product base, [FL02, Lemma 5.23] gives

$$(218) \quad \nu = \nu_t + 2\mu_s.$$

Every nonzero term then has base factor

$$(219) \quad \left\langle \mu_s^d, [M_s] \right\rangle = \text{SW}_X(\mathfrak{s}).$$

In particular, positive-dimensional base classes have not been discarded; the Segre calculation combines all of them into the single pairing in (219).

Put

$$(220) \quad A_J = \eta - d + 1, \quad B_J = \frac{1}{2}(2\delta - d_a(\mathfrak{t}') - d_s) - \frac{1}{4}(\chi(X) + \sigma(X)),$$

and set

$$(221) \quad J_d = P_d^{A_J, B_J}(0), \quad J'_d = P_d^{A_J-1, B_J+1}(0),$$

using (204). The finite Segre sum is $2^d J_d$, except in the term containing Hc , where it is $2^d J'_d$; this is the Jacobi identity of [FL02, Lemma 6.2].

Lemma 8.1 (published charge-one link integrals). *Write $\nu_t = \nu_t$ for the effective circle class on the charge-one instanton link in (218). With the orientation in (198),*

$$(222) \quad \begin{aligned} \langle \nu_t X_0, [\partial \overline{\text{Gl}}/S^1] \rangle &= 2, & \langle \nu_t^2 H, [\partial \overline{\text{Gl}}/S^1] \rangle &= -4B_h, \\ \langle \nu_t Hc, [\partial \overline{\text{Gl}}/S^1] \rangle &= 2\langle c_1(\mathfrak{t}'), h \rangle, & \langle \nu_t^3, [\partial \overline{\text{Gl}}/S^1] \rangle &= 6\beta^2 + 2c_1^2(X), \\ \langle \nu_t^2 c, [\partial \overline{\text{Gl}}/S^1] \rangle &= -4\beta \cdot c_1(\mathfrak{t}'). \end{aligned}$$

Proof. These numerical evaluations are cohomological inputs from [FL02, Lemma 5.24 and Equation (5.77)]; they are not reproved here. We recall enough of that calculation to fix their normalization. On a finite cover of X , the framed charge-one gluing bundle is covered with degree -2 by the boundary of a rank-two Hermitian bundle F . The induced map of projectivizations has degree -1 . For the usual hyperplane class $\xi = c_1(\mathcal{O}_{\mathbb{P}(F)}(1))$, the circle class used in that source is $-\xi$, so ν_t pulls back to -2ξ . The projective-bundle pushforward

$$\pi_*(\xi^{1+j}) = s_j(F)$$

therefore gives $\langle \nu_t X_0 \rangle = (-1)(-2) = 2$ and, from the published Segre classes of F , the second and fourth values in (222). Taking $h = \text{PD}(c)$ in the second value gives the fifth. Finally, Poincare duality in Equation (5.77) gives

$$\langle \nu_t Hc \rangle = \langle \nu_t X_0 \rangle \langle c_1(\mathfrak{t}'), h \rangle,$$

which is the third value. Since $c_1(L) = -\beta$, these conventions agree with (206) and (198). $\square$

Combining the base reduction with Lemma 8.1 gives the following polynomial form, which does not divide by a Jacobi value and therefore remains defined when $J_d = 0$.

Corollary 8.2 (family level-one pairing). *Assume the hypotheses of Theorem 2.1. Suppose in addition that $w_2(\mathfrak{t}')$ is good in the sense of [FL02, Definition 2.3]. Let δ, m, η be nonnegative integers satisfying*

$$0 \leq m \leq \lfloor \delta/2 \rfloor, \quad 2(\delta + \eta) = \dim(\mathcal{M}_\nu/S^1) - 1,$$

put $n = \delta - 2m$ and use the notation in (203) and (220)–(221). Then

$$\begin{aligned} & \#(\overline{V}(h^{\delta-2m}x^m) \cap \overline{W}^\eta \cap \overline{L}_{\nu,s}) \\ &= (-1)^{m+1+d} 2^{d-\delta} \left\langle \mu_{\mathfrak{s}}^d, [M_{\mathfrak{s}}] \right\rangle \\ & \quad \times [J_d(a_0 B_h^n + a_1 B_h^{n-2} Q_X(h, h)) + 2n J'_d B_h^{n-1} \langle c_1(\mathfrak{t}'), h \rangle], \end{aligned}$$

where terms with negative powers of B_h are omitted and

$$a_0 = 3\beta^2 + c_1^2(X) + 2\beta \cdot c_1(\mathfrak{t}') + 4n - 4m, \quad a_1 = 4 \binom{n}{2}.$$

Proof. Equation (215), equivariant Thom division, (217), and (218) reduce the base degree to (219). The Jacobi identity gives $2^d J_d$ and $2^d J'_d$ in the two indicated groups of terms. The ν_t^3 and $\nu_t^2 c$ evaluations in Lemma 8.1 give $3\beta^2 + c_1^2(X)$ and $2\beta \cdot c_1(\mathfrak{t}')$. The $\nu_t^2 H$ evaluation gives $4n$, while $\nu_t X_0$ gives both $-4m$ and the $Q_X(h, h)$ term, using $H^2 = Q_X(h, h)X_0$. The $\nu_t Hc$ evaluation supplies the term with J'_d . Collecting these terms gives a_0 and a_1 . The sign and power of two are $(-1)^{m+1+d} 2^{d-\delta}$, as in [FL02, Equations (6.15)–(6.18)]; Proposition 7.5 introduces no further sign. This proves the formula. $\square$

Corollary 8.3 (zero-dimensional specialization). *Under the hypotheses of Corollary 8.2, if $d_{\mathfrak{s}} = 0$, then $J_0 = J'_0 = 1$ and*

$$\begin{aligned} & \#(\overline{V}(h^{\delta-2m}x^m) \cap \overline{W}^\eta \cap \overline{L}_{\nu,s}) \\ &= (-1)^{m+1} 2^{-\delta} \text{SW}_X(\mathfrak{s}) (a_0 B_h^n + b_0 B_h^{n-1} \langle c_1(\mathfrak{t}'), h \rangle + a_1 B_h^{n-2} Q_X(h, h)), \end{aligned}$$

where $b_0 = 2n$ and a_0, a_1 are as in Corollary 8.2.

8.3. Blow-up and the degree consequence. For completeness, write

$$c(X) = -\frac{1}{4}(7\chi(X) + 11\sigma(X)).$$

Our series conventions are

$$\mathcal{D}_X^w(h) = D_X^w((1 + x/2)e^h), \quad \text{SW}_X^w(h) = \sum_K (-1)^{(w^2 + w \cdot K)/2} \text{SW}_X(K) e^{\langle K, h \rangle},$$

where K runs over the Seiberg–Witten basic classes and $\text{SW}_X(K) = \sum_{c_1(\mathfrak{s})=K} \text{SW}_X(\mathfrak{s})$, so that no injectivity of the first Chern class map is assumed. Congruences are congruences of formal power series by total degree in h ; a congruence modulo h^k imposes no condition when $k \leq 0$.

The blow-up specialization below preserves a_0 . On $\tilde{X} = X \# \overline{\mathbb{CP}}^2$, let e be the exceptional class and

$$c_1(\mathfrak{s}^\pm) = c_1(\mathfrak{s}) \pm \text{PD}[e], \quad \text{SW}_{\tilde{X}}(\mathfrak{s}^\pm) = \text{SW}_X(\mathfrak{s}).$$

For $\tilde{\mathfrak{t}}$ one has $c_1(\tilde{\mathfrak{t}}) = c_1(\mathfrak{t})$ and $p_1(\tilde{\mathfrak{t}}) = p_1(\mathfrak{t}) - 1$. The exceptional evaluations are ∓ 1 , while (199) gives exponents $\sigma_{\tilde{\mathfrak{t}}}(w + \text{PD}[e], \mathfrak{s}^+) = \sigma_{\mathfrak{t}}(w, \mathfrak{s}) - 1$ and $\sigma_{\tilde{\mathfrak{t}}}(w + \text{PD}[e], \mathfrak{s}^-) = \sigma_{\mathfrak{t}}(w, \mathfrak{s})$. The two exceptional contributions therefore add after orientation weighting. Their factor two converts $2^{-(\delta+1)}$ to $2^{-\delta}$.

Algebraically,

$$\begin{aligned} & 3(\beta^2 - 1) + (c_1^2(X) - 1) + 2\beta \cdot c_1(\mathfrak{t}') + 4(\delta + 1 - 2m) - 4m \\ &= 3\beta^2 + c_1^2(X) + 2\beta \cdot c_1(\mathfrak{t}') + 4(\delta - 2m) - 4m. \end{aligned}$$

Together with $Q_{\tilde{X}}(e, h) = 0$ and $e^2 = -1$, this is the $k = 0$ case of [FL02, Proposition 6.5].

Recall that abundance means that the orthogonal complement of the Seiberg–Witten basic classes contains a hyperbolic sublattice. Effectiveness is the link-pairing hypothesis of [FL02, Definitions 1.2–1.3]; in particular, it requires vanishing of the relevant link pairings at every level when the Seiberg–Witten invariant vanishes. This hypothesis is retained in the following application.

Corollary 8.4 (degree consequence). *Assume the hypotheses of Corollary 8.2 for every level-one stratum entering the comparison, including those on the blow-up used below. Suppose also that X is abundant, effective, and of Seiberg–Witten simple type. Let B be its set of Seiberg–Witten basic classes, and choose $\Lambda, w \in H^2(X; \mathbb{Z})$ satisfying*

$$(223) \quad \Lambda \in B^\perp, \quad \Lambda^2 = 4 - (\chi(X) + \sigma(X)), \quad w - \Lambda \equiv w_2(X) \pmod{2},$$

as in [FL02, Theorem 1.1]. Then

$$(224) \quad \begin{aligned} \mathcal{D}_X^w(h) &\equiv 0 \equiv \text{SW}_X^w(h) && \pmod{h^{c(X)-2}}, \\ \mathcal{D}_X^w(h) &\equiv 2^{2-c(X)} e^{Q_X(h, h)/2} \text{SW}_X^w(h) && \pmod{h^{c(X)+2}}. \end{aligned}$$

The first coefficient identity below applies when $c(X) \geq 2$ and the second when $c(X) \geq 0$, with terms of negative exponent omitted:

$$\begin{aligned}
 D_X^w(h^{c(X)-2}x) &= 2^{3-c(X)} \sum_{\mathfrak{s}} (-1)^{(w^2+w \cdot c_1(\mathfrak{s}))/2} \text{SW}_X(\mathfrak{s}) \langle c_1(\mathfrak{s}), h \rangle^{c(X)-2}, \\
 D_X^w(h^{c(X)}) &= 2^{2-c(X)} \sum_{\mathfrak{s}} (-1)^{(w^2+w \cdot c_1(\mathfrak{s}))/2} \text{SW}_X(\mathfrak{s}) \\
 (225) \quad &\times \left(\langle c_1(\mathfrak{s}), h \rangle^{c(X)} + \binom{c(X)}{2} \langle c_1(\mathfrak{s}), h \rangle^{c(X)-2} Q_X(h, h) \right).
 \end{aligned}$$

Proof. Use the choices (223) in the comparison of [FL02, Section 6.3]. Its degree restrictions put the strata with nonzero Seiberg–Witten invariant in levels zero and one; effectiveness makes the remaining link pairings vanish, including those at higher levels. The published top-level pairing [FL01b], Corollary 8.2, and the Donaldson and Seiberg–Witten blow-up formulas therefore give the same comparison. Together with the Seiberg–Witten moment vanishing in [FL02, Equation (6.51)], this gives (224).

For clarity, set $c = c(X)$ and write $S_j(h)$ for the signed Seiberg–Witten sum of $\langle c_1(\mathfrak{s}), h \rangle^j$ appearing in (225). The characteristic conditions imply $w^2 \equiv \sigma(X) \pmod{4}$, and the Donaldson dimension rule is

$$D_X^w(h^{\delta-2m}x^m) = 0 \quad \text{unless} \quad \delta \equiv c \pmod{4}$$

[FL02, Equation (6.53)]. Thus the coefficients of degrees $c-2$ and c in $\mathcal{D}_X^w$ are respectively $D_X^w(h^{c-2}x)/(2(c-2)!)$ and $D_X^w(h^c)/c!$, whenever the indicated degrees are nonnegative. Since $S_j = 0$ for $j < c-2$, the corresponding coefficients on the right of (224) are

$$2^{2-c} \frac{S_{c-2}}{(c-2)!}, \quad 2^{2-c} \left(\frac{S_c}{c!} + \frac{Q_X(h, h) S_{c-2}}{2(c-2)!} \right),$$

where terms with negative indices are omitted. Multiplication by the factorials gives (225), agreeing with [FL02, Equations (6.57)–(6.58)]. $\square$

As a normalization test, for a K3 surface one has $c(X) = 2$ and only the zero basic class, with Seiberg–Witten invariant one in the standard homology orientation. For admissible (Λ, w) as in (223), the sign of the zero basic class is positive because $w^2 \equiv 0 \pmod{4}$. Formula (225) becomes

$$(226) \quad D_{K3}^w(x) = 2, \quad D_{K3}^w(h^2) = Q_{K3}(h, h),$$

the constant and quadratic coefficients of $e^{Q_{K3}(h, h)/2}$ in the convention with insertion $1+x/2$. These coefficients agree with the standard K3 normalization.

9. SCOPE AND FURTHER PROBLEMS

The compactified embedding and reverse uniqueness give the neighborhood in Theorem 2.1. Critical strong approximation for arbitrary one-bubble sequences follows from uniform local neck gauges and summability of the coexact tangential modes. The constant temporal mode is retained through its endpoint frame.

The weak inverse compares the kernel map with projected cutoff modes at the L^2 scale of the bubble. Finite stabilization through zero spinor reduces its orientation to ASD instanton addition and the complex Dirac orientation. Proposition 7.5 carries this comparison to the corrected chart and its link boundary. The pairing applications retain the named topological assumptions; they do not give an unconditional Donaldson–Seiberg–Witten comparison.

The discussion concerns one charge-one bubble over a compact regular nonzero-spinor background. Nonregular bases, zero-spinor solutions, collision strata, and levels two and higher require additional arguments. At level two, a neighborhood of $M_s \times \text{Sym}^2(X)$ must accommodate both separated charge-one bubbles and concentration on the diagonal. Extending reverse gluing to these regimes requires compatible charge-two models and uniform approximation and parameter recovery as the bubbling description changes. The topological overlap constructions in [FL18] provide part of the existing framework. Establishing the corresponding analytic neighborhood statements remains a further problem. An extension to every relevant level, together with the other inputs to the cobordism formula, would address the gluing hypothesis retained in the conditional deduction of Witten’s conjecture in [FL19b].

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